FIVE-STOREY VACATION VILLA

SHAMANDEEL, QUESNA, MENOUFIA, EGYPT · PRIVATE CLIENT

SHEET	TITLE	LEVEL	SCALE	REV
ARC-001	Cover Sheet & Drawing Index	the whole set	NTS	**P21**
ARC-002	General Notes, Symbols & Compliance	the whole set	NTS	**P21**
ARC-000	Master Deliverables List	the whole set	NTS	**P21**
A-000	Site Plan — Detached Villa	+0.00	1:75	**P21**
A-100	Ground Floor Plan — Detached Villa	+0.00	1:50	**P21**
A-101	First Floor Plan — Detached Villa	+3.30	1:50	**P21**
A-102	Second Floor Plan — Detached Villa	+6.60	1:50	**P21**
A-103	Third Floor Plan — Detached Villa	+9.90	1:50	**P21**
A-104	Penthouse Plan — Detached Villa	+13.20	1:50	**P21**
ARC-106	Roof Plan	+16.45 / +13.20	1:50	**P21**
A-200	Section A–A — Detached Villa	looking west	1:50	**P21**
ARC-301	Section B–B — Transverse	looking north	1:50	**P21**
A-400	South Elevation — Scheme A “Wave”	street frontage	1:50	**P21**
A-401	Side & Rear Elevations — Scheme A “Wave”	east, north and west	1:75	**P21**
A-402	Street Elevation in Context	scheme A, as surveyed	1:100	**P21**
A-430	Axonometric — Three Balcony Options	from the south-west	1:250	**P21**
ARC-500	Door & Window Schedule	all levels	1:100	**P21**
ARC-400	Reflected Ceiling Plan — Ground Floor	+0.00	1:50	**P21**
ARC-401	Reflected Ceiling Plan — First Floor	+3.30	1:50	**P21**
ARC-402	Reflected Ceiling Plan — Second Floor	+6.60	1:50	**P21**
ARC-403	Reflected Ceiling Plan — Third Floor	+9.90	1:50	**P21**
ARC-404	Reflected Ceiling Plan — Penthouse Floor	+13.20	1:50	**P21**
ARC-700	External Works & Landscape	-0.50 garden, +0.00 villa	1:75	**P21**
STR-001	Structural Notes & Design Criteria	the whole frame	NTS	**P21**
STR-100	Excavation, Grading & Shoring	-0.50 garden, +0.00 villa	1:75	**P21**
STR-101	Foundation Layout	formation by GEO-001	1:50	**P21**
STR-200	Column Schedule & Grid	all levels	1:50	**P21**
STR-301	Slab Framing Layout at +0.00	+0.00	1:50	**P21**
STR-302	Slab Framing Layout at +3.30	+3.30	1:50	**P21**
STR-303	Slab Framing Layout at +6.60	+6.60	1:50	**P21**
STR-304	Slab Framing Layout at +9.90	+9.90	1:50	**P21**
STR-305	Slab Framing Layout at +13.20	+13.20	1:50	**P21**
STR-306	Slab Framing Layout at +16.45	+16.45	1:50	**P21**
STR-400	Retaining & Boundary Walls	-0.50 to +0.00	1:75	**P21**
STR-500	Cantilever Balcony Structure	+13.20 and the three floors below	1:20	**P21**
EL-100	Power & Small Power Layout — Ground Floor	+0.00	1:50	**P21**
EL-101	Power & Small Power Layout — First Floor	+3.30	1:50	**P21**
EL-102	Power & Small Power Layout — Second Floor	+6.60	1:50	**P21**
EL-103	Power & Small Power Layout — Third Floor	+9.90	1:50	**P21**
EL-104	Power & Small Power Layout — Penthouse Floor	+13.20	1:50	**P21**
EL-200	Lighting & Switching Layout — Ground Floor	+0.00	1:50	**P21**
EL-201	Lighting & Switching Layout — First Floor	+3.30	1:50	**P21**
EL-202	Lighting & Switching Layout — Second Floor	+6.60	1:50	**P21**
EL-203	Lighting & Switching Layout — Third Floor	+9.90	1:50	**P21**
EL-204	Lighting & Switching Layout — Penthouse Floor	+13.20	1:50	**P21**
EL-300	Low Voltage, Data & Security Layout — Ground Floor	+0.00	1:50	**P21**
EL-301	Low Voltage, Data & Security Layout — First Floor	+3.30	1:50	**P21**
EL-302	Low Voltage, Data & Security Layout — Second Floor	+6.60	1:50	**P21**
EL-303	Low Voltage, Data & Security Layout — Third Floor	+9.90	1:50	**P21**
EL-304	Low Voltage, Data & Security Layout — Penthouse Floor	+13.20	1:50	**P21**
ME-100	HVAC Layout — Ground Floor	+0.00	1:50	**P21**
ME-101	HVAC Layout — First Floor	+3.30	1:50	**P21**
ME-102	HVAC Layout — Second Floor	+6.60	1:50	**P21**
ME-103	HVAC Layout — Third Floor	+9.90	1:50	**P21**
ME-104	HVAC Layout — Penthouse Floor	+13.20	1:50	**P21**
PL-100	Water Supply Layout — Ground Floor	+0.00	1:50	**P21**
PL-101	Water Supply Layout — First Floor	+3.30	1:50	**P21**
PL-102	Water Supply Layout — Second Floor	+6.60	1:50	**P21**
PL-103	Water Supply Layout — Third Floor	+9.90	1:50	**P21**
PL-104	Water Supply Layout — Penthouse Floor	+13.20	1:50	**P21**
PL-200	Drainage Layout — Ground Floor	+0.00	1:50	**P21**
PL-201	Drainage Layout — First Floor	+3.30	1:50	**P21**
PL-202	Drainage Layout — Second Floor	+6.60	1:50	**P21**
PL-203	Drainage Layout — Third Floor	+9.90	1:50	**P21**
PL-204	Drainage Layout — Penthouse Floor	+13.20	1:50	**P21**
EL-400	Electrical Single Line Diagram	the whole building	schematic	**P21**
ME-200	HVAC Schematic & Plant Arrangement	the whole building	schematic	**P21**
PL-300	Drainage & Water Riser Diagram	the whole building	schematic	**P21**
A-02	Interior Cutaway — Ground Floor	+0.00	1:75	**P21**
A-03	Interior Cutaway — First Floor	+3.30	1:75	**P21**
A-04	Interior Cutaway — Second Floor	+6.60	1:75	**P21**
A-05	Interior Cutaway — Third Floor	+9.90	1:75	**P21**
A-06	Interior Cutaway — Penthouse Floor	+13.20	1:75	**P21**
A-300	Typical Apartment — Enlarged Plan	+3.30 / +6.60	1:25	**P4**
A-301	Third Floor Apartment — Enlarged Plan	+9.90	1:25	**P4**
A-302	Central Kitchen & Laundry — Enlarged Plan	+0.00	1:25	**P4**
A-410	South Elevation — Scheme B “Arcade”	street frontage	1:50	**P21**
A-411	Side & Rear Elevations — Scheme B “Arcade”	east, north and west	1:75	**P21**
A-420	South Elevation — Scheme C “Facet”	street frontage	1:50	**P21**
A-421	Side & Rear Elevations — Scheme C “Facet”	east, north and west	1:75	**P21**

SHEETS IN GREY ARE BUILT BUT NOT PART OF THE ISSUE

COVER SHEET & DRAWING INDEX

PROJECT DATA · DRAWING INDEX · REVISION P21

PROJECT DATA

PLOT	264.5 m²
FOOTPRINT	128.2 m²
SITE COVERAGE	48.5 %
LEVELS	5
FLOOR TO FLOOR	3.30 m
TOP OF BUILDING	+17.05
EXTERNAL WORKS	136.2 m²
SHEETS IN THE ISSUE	73
SHEETS BUILT	80
CODES	ECP 201 / ECP 203

GENERAL NOTES

1. The index is built, not listed: every row is a sheet this project can actually draw, with the title and revision that sheet carries. An index typed beside the drawings goes wrong within a revision or two, and it is the one document a reader trusts when deciding a drawing does not exist.

2. Rows in grey are built but outside the issue: the A-300 enlargements, which are of apartments the villa no longer has, and the scheme B and C elevations, which the client did not choose.

3. The prefix is part of the number. A-402 is the street elevation and ARC-402 the second floor's ceiling plan.

4. Deliverables beyond the drawings — the structural, services and quantities packages — are registered on ARC-000 with their status.

5. Not for construction. Issued for the owner's record, costing and construction information.

FIVE-STOREY VACATION VILLA SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

COVER SHEET & DRAWING INDEX

DRAWING No	REVISION	SCALE	DATE
ARC-001	**P21**	**NTS**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

GENERAL NOTES & COMPLIANCE MATRIX

SHAMANDEEL, QUESNA, MENOUFIA, EGYPT · ECP 201 — Egyptian Code for Loads · ECP 203 — Egyptian Code for Reinforced Concrete

1. Figured dimensions govern. Do not scale from these drawings.

2. Dimensions are in millimetres unless a metre value is shown with two decimals. Levels are in metres to the project datum, which is the ground floor finished floor level, +0.00.

3. Dimensions to structural faces unless noted. Opening sizes are structural; frame and tolerance allowances are the supplier's.

4. Every discrepancy between drawings is to be referred before any work is set out or any material ordered.

5. Boundaries, levels and the neighbours' heights are as described by the client and await a measured survey, SUR-001.

6. Foundations, framing and every member size are the structural engineer's and are not set on any architectural sheet. The STR series is issued as layout for that design.

7. Services layouts are coordination only. Cable, duct and pipe sizes, loads and capacities are the services engineer's.

8. Materials, finishes, waterproofing build-ups and performance specifications are not set by these drawings.

9. Not for construction.

CODE COMPLIANCE MATRIX

ITEM	AS DRAWN	REQUIRED	CLAUSE	COMPLIES
Site coverage	**48.5 %**			
Setback, west / north / east	**1.00 m each**			
Height, top of building	**+17.05**			
Storeys	**5**			
Floor to floor	**3.30 m**			
Clear height	**3.00 m**			
Stair, private	**18R at 183 × 260**			
Stair, flight width	**0.90 m**			
Stair, well	**2.00 × 3.30 m**			
Balustrade and parapet	**1.10 m**			
Window sill	**0.90 m above floor**			
Balcony projection	**1.95 m maximum**			
Overlooking, east	**single storey 1.00 m away**			
Overlooking, west	**3 storeys, blind**			
Overlooking, north	**1 storey**			

GENERAL NOTES, SYMBOLS & COMPLIANCE

GENERAL NOTES · SYMBOLS · COMPLIANCE

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 Finished floor level datum

GENERAL NOTES

1. The matrix carries our figure for every item a house is normally tested against, measured from the geometry and exact.

2. The required value, the clause and the verdict are left open on purpose. They belong to ECP 201 and ECP 203 and to whoever signs the compliance statement; a column of limits invented here would be worse than an empty one, because it would be believed.

3. The symbols panel is the same key every sheet in the set carries, so a reader who has this sheet can read any of them.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

GENERAL NOTES, SYMBOLS & COMPLIANCE

DRAWING No	REVISION	SCALE	DATE
ARC-002	**P21**	**NTS**	**2026-10-07**

Layout for the architect of record — the compliance matrix is completed and signed against ECP 201 and ECP 203. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

MASTER DELIVERABLES LIST

81 DELIVERABLES · 5 LEVELS, 6 SLABS · STATUS DERIVED AT BUILD TIME

CODE	CL	STATUS	DELIVERABLE	SCALE
PACKAGE 0 — INFORMATION REQUIRED FROM OTHERS (3)				
SUR-001	X	· not started	Measured topographic and boundary survey	1:200
GEO-001	X	! blocked	Soil investigation and foundation recommendation	-
CLI-001	D	+ issued	Client instructions and decisions record	-
PACKAGE A — PRESENTATION AND ESTIMATION (8)				
A-01	D	+ issued	Exterior views - axonometric, three schemes	1:250
A-02	D	+ issued	Interior cutaway axonometric - ground floor, +0.00	1:75
A-03	D	+ issued	Interior cutaway axonometric - first floor, +3.30	1:75
A-04	D	+ issued	Interior cutaway axonometric - second floor, +6.60	1:75
A-05	D	+ issued	Interior cutaway axonometric - third floor, +9.90	1:75
A-06	D	+ issued	Interior cutaway axonometric - penthouse floor, +13.20	1:75
A-07	D	+ issued	Architectural design brief	-
A-08	X	+ issued	Cost estimation report - elemental	-
PACKAGE B — ARCHITECTURE (21)				
ARC-000	D	+ issued	Master deliverables list and document register	-
ARC-001	D	+ issued	Cover sheet, project data and drawing index	-
ARC-002	L	+ issued	General notes, symbols and code compliance matrix	-
ARC-100	D	+ issued	Context and master site plan	1:75
ARC-101	D	+ issued	Ground floor plan, +0.00	1:50
ARC-102	D	+ issued	First floor plan, +3.30	1:50
ARC-103	D	+ issued	Second floor plan, +6.60	1:50
ARC-104	D	+ issued	Third floor plan, +9.90	1:50
ARC-105	D	+ issued	Penthouse floor plan, +13.20	1:50
ARC-106	D	+ issued	Roof plan over the penthouse, +16.45	1:50
ARC-200	D	+ issued	Facade elevations - all four faces, scheme A	1:50 / 1:75
ARC-300	D	+ issued	Longitudinal section A-A	1:50
ARC-301	D	+ issued	Transverse section B-B	1:50
ARC-400	D	+ issued	Reflected ceiling plan - ground floor, +0.00	1:50
ARC-401	D	+ issued	Reflected ceiling plan - first floor, +3.30	1:50
ARC-402	D	+ issued	Reflected ceiling plan - second floor, +6.60	1:50
ARC-403	D	+ issued	Reflected ceiling plan - third floor, +9.90	1:50
ARC-404	D	+ issued	Reflected ceiling plan - penthouse floor, +13.20	1:50
ARC-500	D	+ issued	Door and window schedule	1:100
ARC-600	L	· not started	Construction details	1:5 / 1:10
ARC-700	D	+ issued	External works and landscape	1:75
PACKAGE C — STRUCTURE (12)				
STR-001	L	+ issued	Structural notes, materials and design criteria	-
STR-100	L	+ issued	Excavation, grading and shoring	1:75
STR-101	L	+ issued	Foundation layout	1:50
STR-200	L	+ issued	Column schedule and grid	1:50
STR-301	L	+ issued	Slab framing layout at +0.00	1:50
STR-302	L	+ issued	Slab framing layout at +3.30	1:50
STR-303	L	+ issued	Slab framing layout at +6.60	1:50
STR-304	L	+ issued	Slab framing layout at +9.90	1:50
STR-305	L	+ issued	Slab framing layout at +13.20	1:50
STR-306	L	+ issued	Slab framing layout at +16.45	1:50
STR-400	L	+ issued	Retaining and boundary walls	1:75
STR-500	L	+ issued	Cantilever balcony structure	1:20
PACKAGE D — BUILDING SERVICES (33)				
EL-100	L	+ issued	Power and small power layout - ground floor, +0.00	1:50
EL-101	L	+ issued	Power and small power layout - first floor, +3.30	1:50
EL-102	L	+ issued	Power and small power layout - second floor, +6.60	1:50
EL-103	L	+ issued	Power and small power layout - third floor, +9.90	1:50
EL-104	L	+ issued	Power and small power layout - penthouse floor, +13.20	1:50
EL-200	L	+ issued	Lighting and switching layout - ground floor, +0.00	1:50
EL-201	L	+ issued	Lighting and switching layout - first floor, +3.30	1:50
EL-202	L	+ issued	Lighting and switching layout - second floor, +6.60	1:50
EL-203	L	+ issued	Lighting and switching layout - third floor, +9.90	1:50
EL-204	L	+ issued	Lighting and switching layout - penthouse floor, +13.20	1:50
EL-300	L	+ issued	Low voltage, data and security layout - ground floor, +0.00	1:50
EL-301	L	+ issued	Low voltage, data and security layout - first floor, +3.30	1:50
EL-302	L	+ issued	Low voltage, data and security layout - second floor, +6.60	1:50
EL-303	L	+ issued	Low voltage, data and security layout - third floor, +9.90	1:50
EL-304	L	+ issued	Low voltage, data and security layout - penthouse floor, +13.20	1:50
ME-100	L	+ issued	HVAC layout - ground floor, +0.00	1:50
ME-101	L	+ issued	HVAC layout - first floor, +3.30	1:50
ME-102	L	+ issued	HVAC layout - second floor, +6.60	1:50
ME-103	L	+ issued	HVAC layout - third floor, +9.90	1:50
ME-104	L	+ issued	HVAC layout - penthouse floor, +13.20	1:50
PL-100	L	+ issued	Water supply layout - ground floor, +0.00	1:50
PL-101	L	+ issued	Water supply layout - first floor, +3.30	1:50
PL-102	L	+ issued	Water supply layout - second floor, +6.60	1:50
PL-103	L	+ issued	Water supply layout - third floor, +9.90	1:50
PL-104	L	+ issued	Water supply layout - penthouse floor, +13.20	1:50
PL-200	L	+ issued	Drainage layout - ground floor, +0.00	1:50
PL-201	L	+ issued	Drainage layout - first floor, +3.30	1:50
PL-202	L	+ issued	Drainage layout - second floor, +6.60	1:50
PL-203	L	+ issued	Drainage layout - third floor, +9.90	1:50
PL-204	L	+ issued	Drainage layout - penthouse floor, +13.20	1:50
EL-400	L	+ issued	Electrical single line diagram	Schematic
ME-200	L	+ issued	HVAC schematic and plant arrangement	Schematic
PL-300	L	+ issued	Drainage and water riser diagram	Schematic
PACKAGE E — QUANTITIES AND COST CONTROL (4)				
QS-001	D	+ issued	Bill of quantities - itemised, unpriced, by trade	-
QS-002	D	+ issued	Bill of materials	-
QS-003	L	+ issued	Material requisitions and specifications	-
QS-004	X	+ issued	Owner's cost baseline	-

+ ISSUED > IN PROGRESS ! BLOCKED · NOT STARTED

MASTER DELIVERABLES LIST

DOCUMENT CONTROL · PACKAGES 0 AND A TO E · STATUS DERIVED

THE SET AT A GLANCE

DELIVERABLES	81
ISSUED	78
IN PROGRESS	0
BLOCKED	1
NOT STARTED	2
LEVELS	5
STRUCTURAL SLABS	6
D — DERIVED	29
L — LAYOUT FOR OTHERS	48
X — EXTERNAL	4

GENERAL NOTES

1. Nothing is issued as engineering that has not been engineered. Class D is generated from geometry.py and is ours to issue. Class L is layout and coordination only, sized and certified by others. Class X cannot be produced here at all.

2. So the structural package carries foundation layouts on the real grid and column coordinates, and no reinforcement; the services package carries coordination layouts, and no load calculations.

3. Status is derived at build time from the issued set, never typed. A row naming a sheet nobody draws, or a document nobody writes, fails the checks rather than printing here.

4. Five levels, 6 structural slabs: the slab at +13.20 is the penthouse floor and the roof over the third floor at once.

5. Deliverable codes are the client's. Where one resolves to sheets already issued, the existing number is the cross reference — the agreed set is not renumbered. The prefix is part of the number: A-402 is the street elevation and ARC-402 the second floor's ceiling plan, and they are different sheets.

6. The exterior is settled: scheme A, adopted at P18. Blocked rows now wait on the measured survey and the plot's soil figures, neither of which can be settled from the drawings.

7. For the owner's own record, costing and construction information. Not a tender issue, and not for construction.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

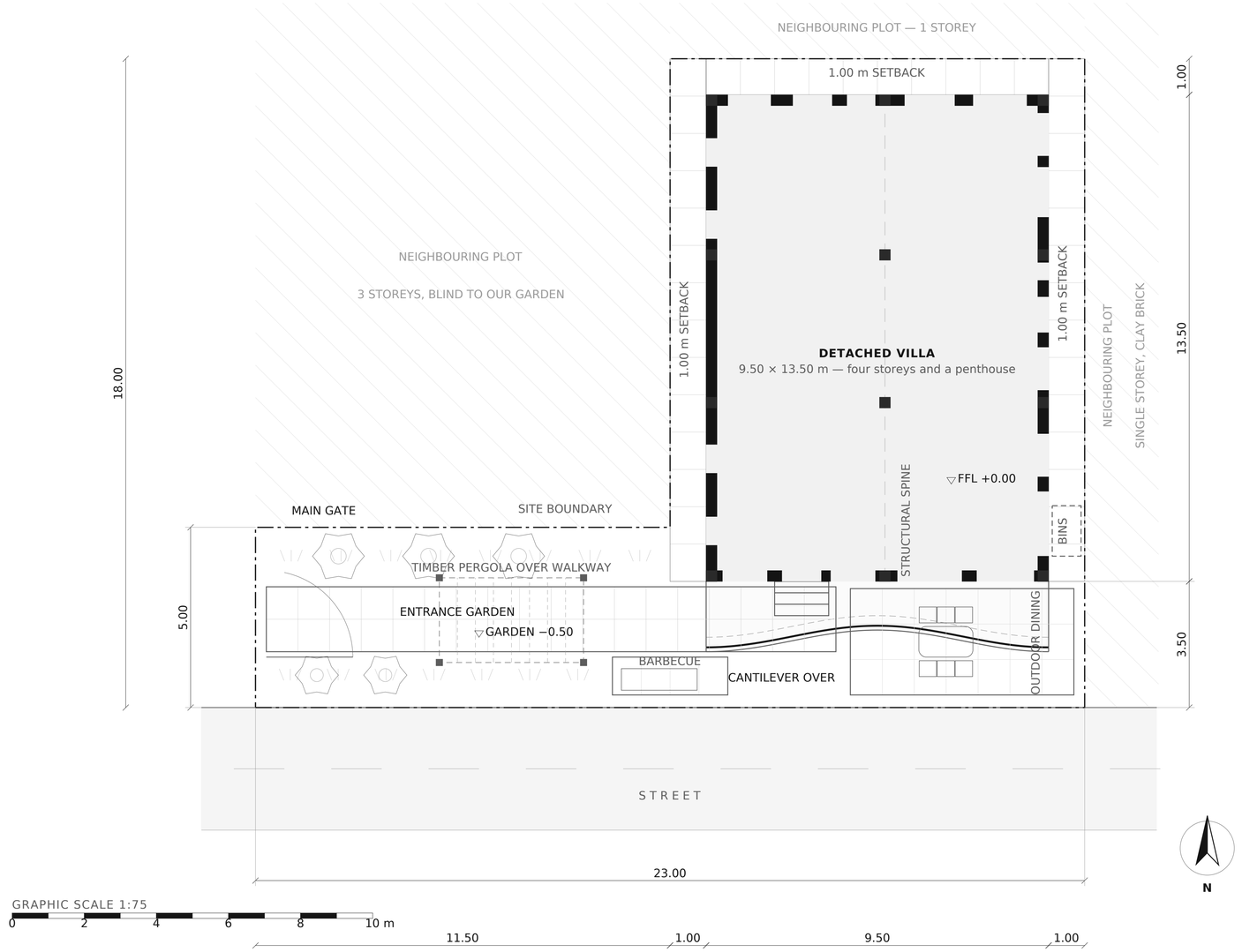
MASTER DELIVERABLES LIST

DRAWING No	REVISION	SCALE	DATE
ARC-000	**P21**	**NTS**	**2026-10-07**

Document control — every status on this sheet is derived at build time from the issued set.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SITE PLAN — DETACHED VILLA

LEVEL +0.00 · 1:75 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

AREA SCHEDULE

PLOT, SOUTH BAND 23.00 × 5.00	115.0 m²
PLOT, EAST PART 11.50 × 13.00	149.5 m²
TOTAL	**264.5 m²**
VILLA FOOTPRINT 9.50 × 13.50 M	128.2 m²
SITE COVERAGE	48.5 %
ENTRANCE GARDEN, WEST	57.5 m²
FORECOURT, IN FRONT	40.3 m²
SETBACKS, THREE SIDES	38.5 m²
GROSS INTERNAL AREA, FIVE STOREYS	505.2 m²

GENERAL NOTES

1. Final draft — layout agreed. The villa is detached: set back 1.00 m from the west, north and east boundaries, the three sides a neighbour stands on. The south is not set back — it faces the street across our own forecourt.

2. Footprint 9.50 × 13.50 m = 128.2 m² on a 264.5 m² plot, 48.5 % coverage against 63.0 % for the attached scheme at P4. Detaching costs 22 % of each floor and buys an open aspect on all four sides and no party wall.

3. The setbacks are our own land, so each is a 1.00 m paved side passage rather than a dead strip, reached from the forecourt on the east and from the entrance garden on the west.

4. The cantilever oversails the forecourt by 1.95 m at each end and draws in to 1.35 m at the waist, leaving at least 1.55 m clear to the street, and covers the front door and its three risers. Nothing projects over a boundary, but the overhang is subject to the local projection rule; confirm before construction.

5. The client's survey sketch recorded a 0.50 m step in the north boundary and a 5.50 m offset between the two blocks' north walls; the two cannot both be true. Resolved in favour of a straight boundary at 18.00. To be confirmed by measured survey before any work starts on site.

6. Structure is an RC frame on the grid shown, 300 mm columns on wall centrelines so none stands in a room. The A-300 enlargements are the last sheets still on the old 11.50 × 14.50 footprint.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

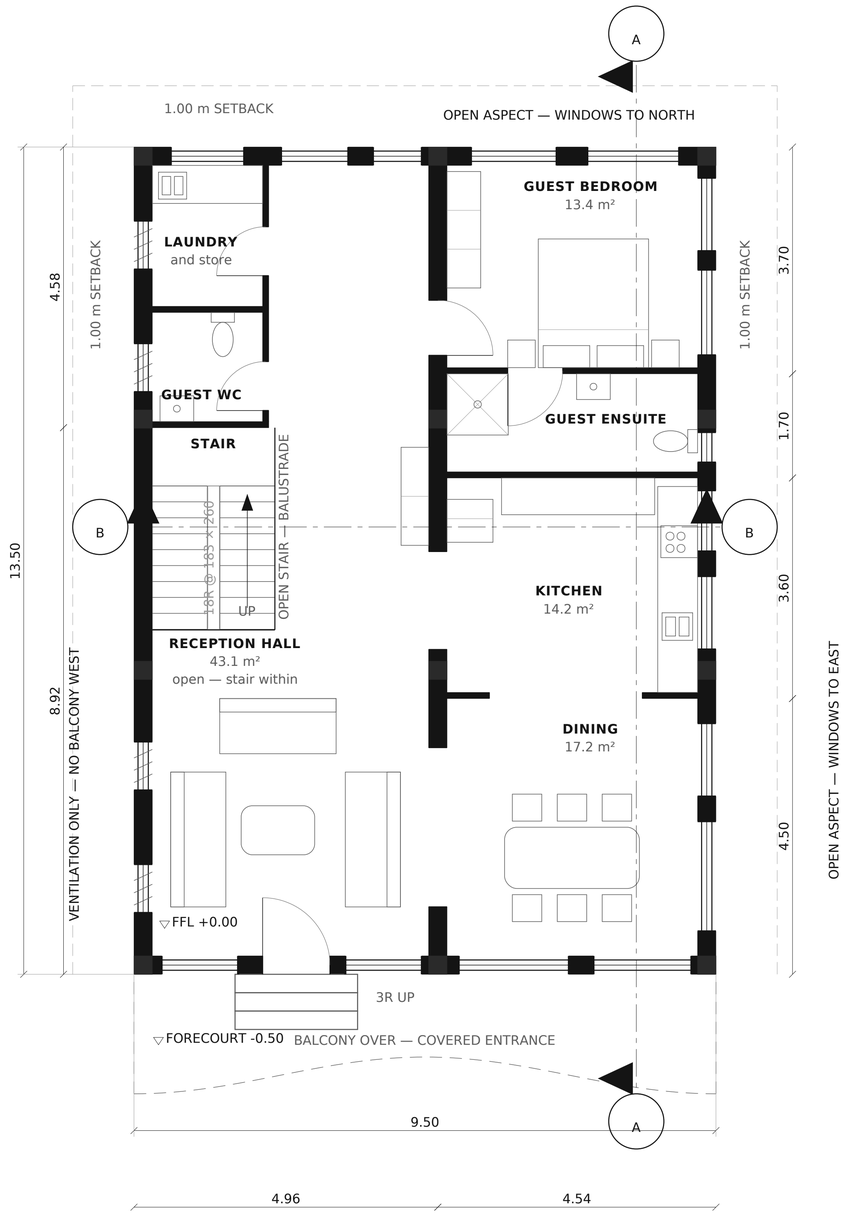
SITE PLAN — DETACHED VILLA

DRAWING No	REVISION	SCALE	DATE
A-000	**P21**	**1:75 @ A1**	**2026-10-07**

Final draft — layout agreed with the client; not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted





GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

GROUND FLOOR PLAN — DETACHED VILLA

LEVEL +0.00 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

AREA SCHEDULE

RECEPTION HALL	43.1 m²
STAIR	6.6 m²
GUEST WC	3.2 m²
LAUNDRY	4.1 m²
DINING	17.2 m²
KITCHEN	14.2 m²
GUEST ENSUITE	6.5 m²
GUEST BEDROOM	13.4 m²
TOTAL	**108.3 m²**
GROSS INTERNAL AREA, THIS FLOOR	114.8 m²
BEDROOMS	1 — the guest suite; 8 above
FLOOR ABOVE	+3.30

GENERAL NOTES

1. Final draft — layout agreed. The floor you receive people on: the family living is upstairs where the balconies are, so this one takes arrival, eating and a guest suite for anyone who cannot manage the stair.

2. Hall, dining and kitchen are one open sequence. The spine is cased open to both at 2.60 and 1.60 m, and the kitchen serves the dining through a 2.50 m opening rather than a door. Only the guest suite and the two service rooms are closed.

3. The reception hall is the whole west strip at 43.1 m², with the stair standing in it. It turns the corner behind the stair and runs on to the north wall, so the guest bedroom opens off the hall and there is no circulation room anywhere on the floor.

4. Guest WC, laundry and store are one block against the west wall, the only part of this floor that is walled off. East and north carry ordinary windows, the west louvred ventilation only.

5. The front door is in the hall's south wall, under the cantilever, which projects about 1.75 m there — so the entrance is covered without a porch being added to the elevation. Three risers up from the forecourt at −0.50.

6. Bedrooms are numbered 1 to 11 through the house, south to north and west before east on each floor; the ground floor's room is the guest bedroom and carries no number.

7. The A-300 enlargements are the last sheets still on the old 11.50 × 14.50 footprint.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

GROUND FLOOR PLAN — DETACHED VILLA

DRAWING No	REVISION	SCALE	DATE
A-100	**P21**	**1:50 @ A1**	**2026-10-07**

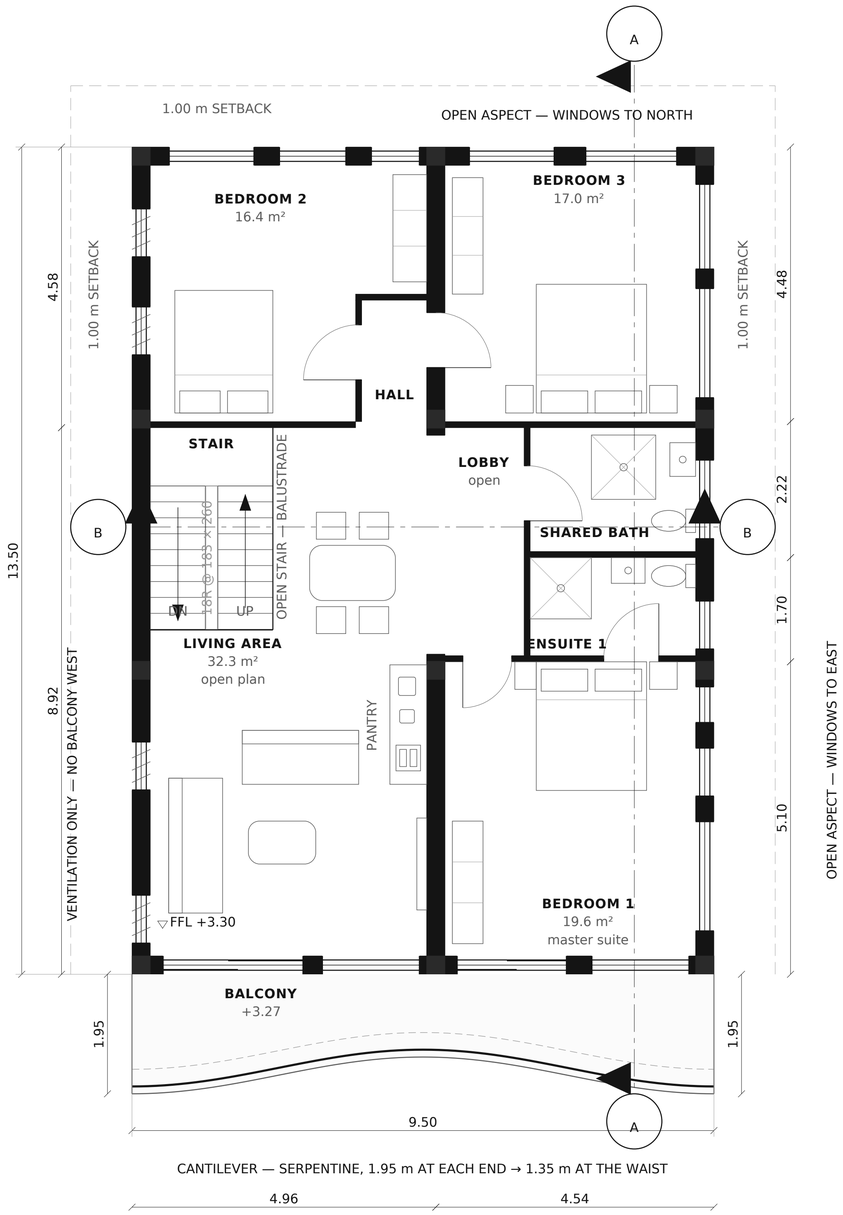
Final draft — layout agreed with the client; not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

FIRST FLOOR PLAN — DETACHED VILLA

LEVEL +3.30 · 1:50 @ A1 · METRES







KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

AREA SCHEDULE

LIVING AREA	32.3 m²
STAIR	6.6 m²
BEDROOM 1 — MASTER	19.6 m²
BEDROOM 1 ENSUITE	4.2 m²
LOBBY	4.8 m²
SHARED BATH	5.4 m²
BEDROOM 2	16.4 m²
HALL	2.0 m²
BEDROOM 3	17.0 m²
TOTAL	**108.3 m²**
GROSS INTERNAL AREA, THIS FLOOR	114.8 m²
BALCONY, EXTERNAL	15.7 m²
BEDROOMS	3 — the master, bedrooms 2 and 3
FLOOR ABOVE / BELOW	+6.60 / +0.00

GENERAL NOTES

1. Final draft — layout agreed. The living floor: a living area and bedrooms 1 to 3. The living area is 32.3 m² with the stair standing in it and the pantry open along the spine, opening through a sliding door onto the cantilever.

2. Bedrooms are numbered 1 to 11 through the house, south to north and west before east on each floor; the ground floor's room is the guest bedroom and carries no number.

3. Bedroom 1 is the master at 19.6 m²: its north wall sits 1.30 m further north than the suite below it, and the shared bath gives up the same depth at 5.4 m².

4. The pantry is a counter against the solid stretch of spine, not the walled room the third floor has. The lobby has no wall either: the spine is opened 3.69 m between the columns, a downstand beam rather than a wall.

5. The west strip is otherwise the third floor's and the east strip the second's, so the wet rooms stack and the stair lines through. East and north carry ordinary windows, the west louvred ventilation only. The hall to the north bedrooms stops at its last door; the dead end beyond goes to bedroom 2, 16.4 m².

6. The cantilever runs on all three upper floors and the penthouse — the same serpentine, 1.95 m deep at each end drawing in to 1.35 m at the waist, 15.7 m² — with sliding doors from the living area and bedroom 1.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

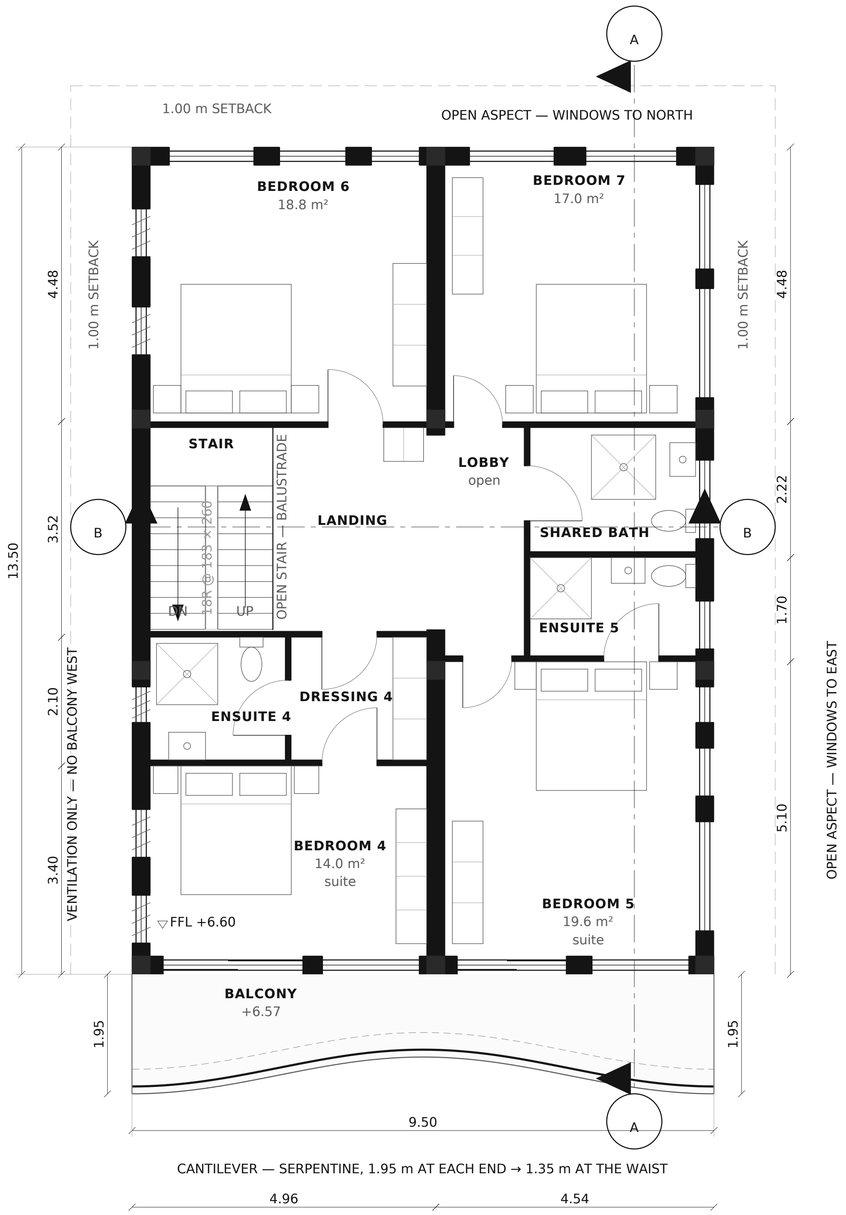
FIRST FLOOR PLAN — DETACHED VILLA

DRAWING No	REVISION	SCALE	DATE
A-101	**P21**	**1:50 @ A1**	**2026-10-07**

Final draft — layout agreed with the client; not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted







SECOND FLOOR PLAN — DETACHED VILLA

LEVEL +6.60 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

AREA SCHEDULE

BEDROOM 4 — SUITE	14.0 m²
BEDROOM 4 ENSUITE	4.4 m²
BEDROOM 4 DRESSING	4.3 m²
BEDROOM 5 — SUITE	19.6 m²
BEDROOM 5 ENSUITE	4.2 m²
LOBBY	4.8 m²
STAIR	6.6 m²
LANDING	7.9 m²
SHARED BATH	5.4 m²
BEDROOM 6	18.8 m²
BEDROOM 7	17.0 m²
TOTAL	**107.0 m²**
GROSS INTERNAL AREA, THIS FLOOR	114.8 m²
BALCONY, EXTERNAL	15.7 m²
BEDROOMS	4 — two of them suites
FLOOR ABOVE / BELOW	+9.90 / +3.30

GENERAL NOTES

1. Final draft — layout agreed. A bedroom floor: bedrooms 4 to 7, two of them suites, on the detached envelope.

2. Bedrooms are numbered 1 to 11 through the house, south to north and west before east on each floor; the ground floor's room is the guest bedroom and carries no number.

3. Bedroom 4 takes the south-west with its ensuite and dressing behind; bedroom 5 the south-east, its ensuite off the bedroom.

4. The east strip is the first floor's, line for line — shared bath 5.4 m², lobby 4.8 m², bedroom 5 at 19.6 m² as its wall moves 1.30 m north. Every wet room and east window stacks, one drainage stack each, straight to the ground.

5. The lobby has no door: the spine is opened its whole length between the columns, so lobby and landing read as one space. It serves bedroom 5, the shared bath and bedroom 7. Bedroom 6 has the old hall — the whole north-west quarter, 18.8 m².

6. The cantilever is carried here too — the same serpentine, 1.95 m at each end, 1.35 m at the waist, 15.7 m². Both suites open onto it; a screen between them is worth considering.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

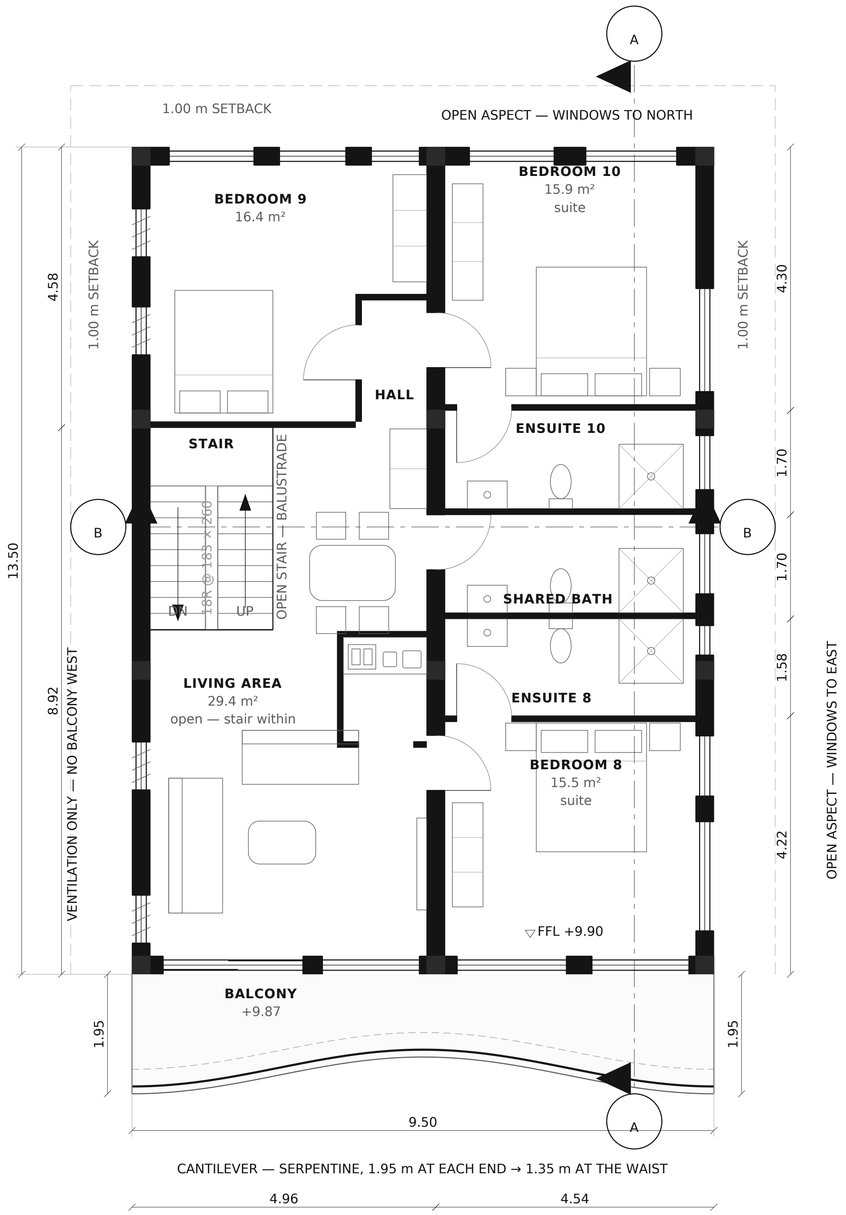
SECOND FLOOR PLAN — DETACHED VILLA

DRAWING No	REVISION	SCALE	DATE
A-102	**P21**	**1:50 @ A1**	**2026-10-07**

Final draft — layout agreed with the client; not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted





GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

THIRD FLOOR PLAN — DETACHED VILLA

LEVEL +9.90 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

AREA SCHEDULE

LIVING AREA	29.4 m²
STAIR	6.6 m²
HALL	2.0 m²
BEDROOM 8 — SUITE	15.5 m²
BEDROOM 8 ENSUITE	6.5 m²
SHARED BATH	6.5 m²
BEDROOM 10 ENSUITE	6.5 m²
BEDROOM 10 — SUITE	15.9 m²
BEDROOM 9	16.4 m²
TOTAL	**105.3 m²**
GROSS INTERNAL AREA, THIS FLOOR	114.8 m²
BALCONY, EXTERNAL	15.7 m²
ROOF OVER, TOP OF SLAB	+13.20

GENERAL NOTES

1. Final draft — layout agreed. The building is set back 1.00 m from the west, north and east boundaries; the south faces the street across our own forecourt. Detaching costs 22% of the floor — 114.8 m² against 147.3 — and removes the party wall, so one structural spine divides the plate.

2. Bedrooms are numbered 1 to 11 through the house, south to north and west before east on each floor; the ground floor's room is the guest bedroom and carries no number.

3. The west strip south of bedroom 9 is one room: no wall divides the living area from the stair standing in it. The stair is a private villa stair — 900 mm flights, 18R at 183 × 260 — needing no enclosure, no smoke vent and no lift.

4. The south cantilever is reinforced concrete: a serpentine of one full period, 1.95 m deep at both ends and drawn in to 1.35 m at the waist, which is deep enough to sit out on along the whole frontage.

5. East and north carry ordinary windows. The east neighbour is a single storey in clay brick 1.00 m away, so they clear its roof by 1.20 m at the first floor. The west is louvred only — that neighbour is 3 storeys, and blind toward us.

6. Bedroom 8 and the shared bath open off the living area; bedrooms 9 and 10 off the hall, whose dead end goes to bedroom 9. Bedroom 10 has no balcony: it takes the north wall and two windows.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

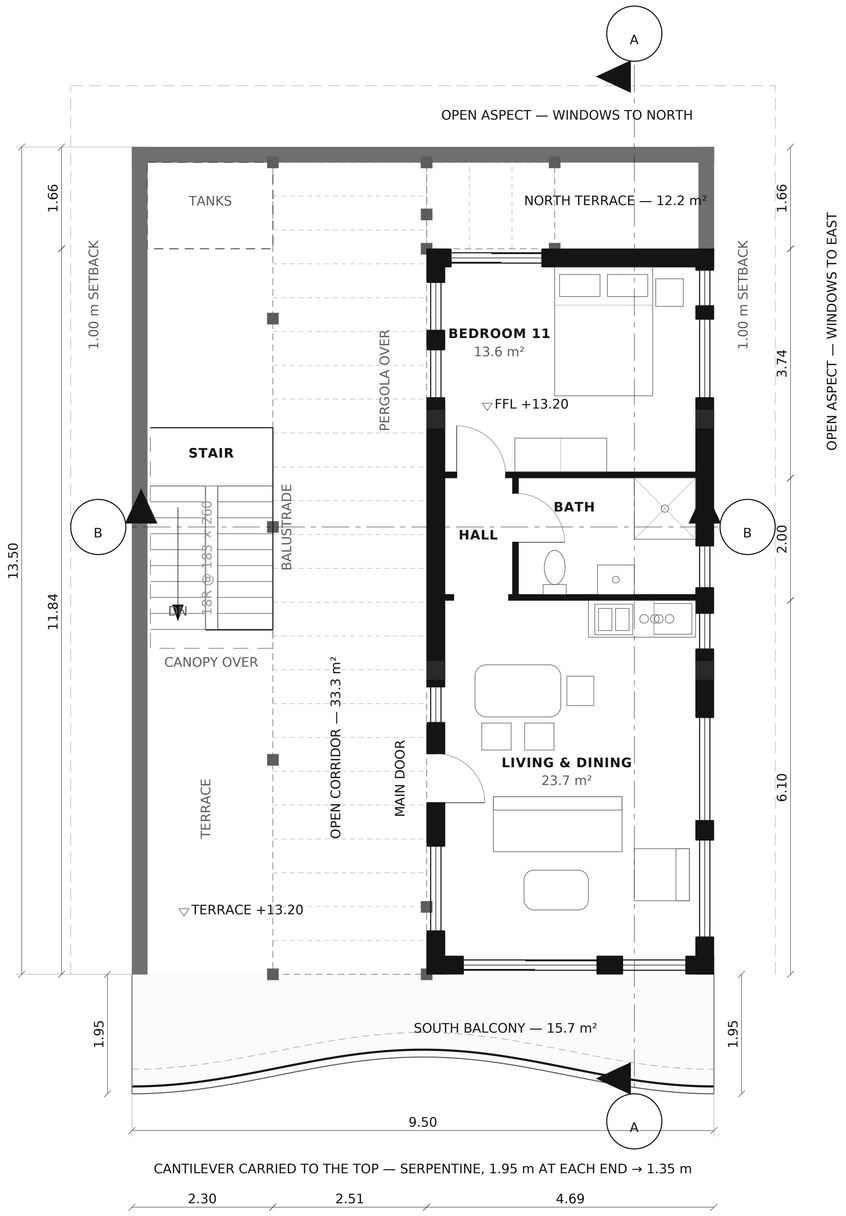
THIRD FLOOR PLAN — DETACHED VILLA

DRAWING No	REVISION	SCALE	DATE
A-103	**P21**	**1:50 @ A1**	**2026-10-07**

Final draft — layout agreed with the client; not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

PENTHOUSE PLAN — DETACHED VILLA

LEVEL +13.20 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

AREA SCHEDULE

LIVING AND DINING	23.7 m²
HALL	2.1 m²
BATH	5.4 m²
BEDROOM 11	13.6 m²
TOTAL	**44.8 m²**
GROSS INTERNAL AREA, FLAT	46.0 m²
SOUTH BALCONY, ON THE CANTILEVER	15.7 m²
SOUTH TERRACE, WEST OF THE CORRIDOR	11.5 m²
NORTH TERRACE	12.2 m²
OPEN CORRIDOR	33.3 m²
TOP OF PENTHOUSE SLAB	+16.45
FLOOR BELOW	+9.90

GENERAL NOTES

1. Bedrooms are numbered 1 to 11 through the house, south to north and west before east on each floor; the ground floor's room is the guest bedroom and carries no number. This is bedroom 11.

2. Final draft — layout agreed. The two open areas are joined outside the flat, down the west, by a corridor 2.51 m clear against the 2.00 m asked for, pergola over its whole length. The front door is off it.

3. 46.0 m² indoors; 15.7 m² of balcony on the cantilever, 11.5 m² of terrace by the stair, 12.2 m² north and 33.3 m² of corridor — all four run into one another.

4. The corridor passes east of the stair: the well is 2.00 m wide and hard against the west parapet, its half landing 1.65 m below this floor, so it stays open. Its east side is the flat's west wall, on the spine at x 17.46 — the one line on this side of the roof with a beam under it.

5. That leaves the flat 4.09 m clear, which will not take two bedrooms side by side: stacking them needs 12.06 m of the 13.50 m depth and leaves 1.44 m for both terraces. The second bedroom is what pays for the corridor.

6. The living room runs south to the building's face, so it opens straight onto the cantilever; y 3.65 is a grid line, so that wall sits on the columns and over the wall below. At +16.45 the penthouse is a fourth storey for most planning codes.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

PENTHOUSE PLAN — DETACHED VILLA

DRAWING No	REVISION	SCALE	DATE
A-104	**P21**	**1:50 @ A1**	**2026-10-07**

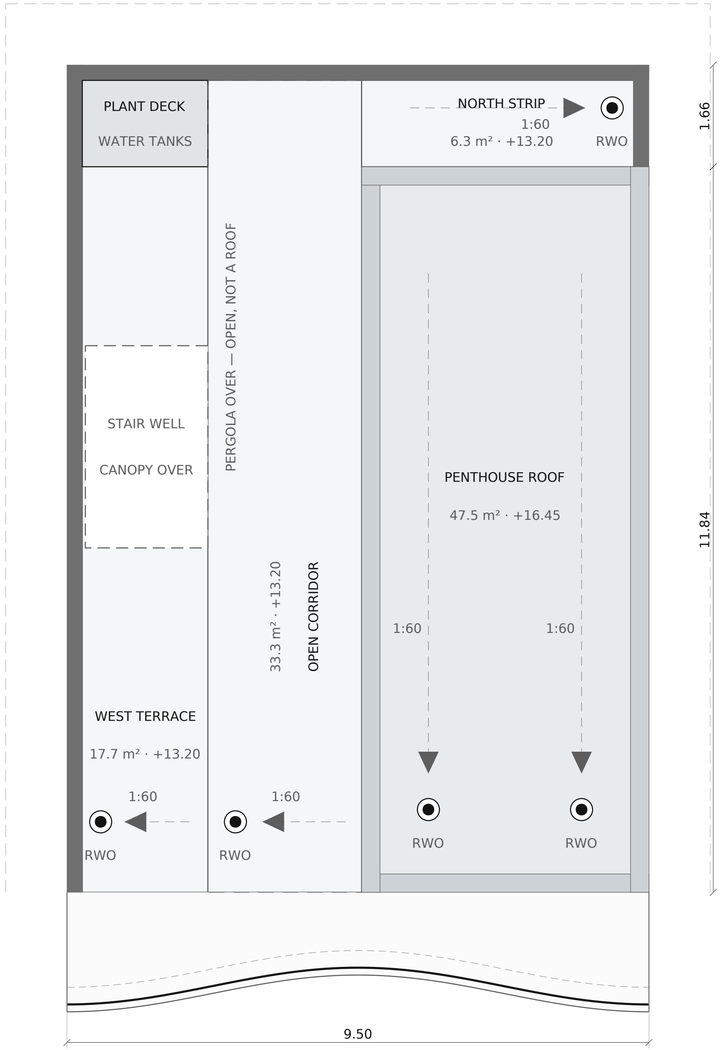
Final draft — layout agreed with the client; not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

ROOF PLAN

LEVEL +16.45 / +13.20 · 1:50 @ A1 · METRES



BALCONY +13.20 — 15.7 m², FALLS TO THE EDGE

GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE ROOF

ROOF OVER PENTHOUSE	+16.45
TERRACES AND BALCONY	+13.20
PARAPET	250 mm × 1100 high
WATERPROOFING UPSTAND	150 mm minimum
FALLS	1:60, indicative
OUTLETS	5, indicative — see PL-200
BALCONY	15.7 m², falls to the edge
PLANT DECK	water tanks, north-west

GENERAL NOTES

1. The surfaces that shed water, which no other sheet draws: the roof over the penthouse at +16.45, the terraces and the open corridor at +13.20, the balcony, and the plant deck.

2. Outlines, parapets, the plant deck and the levels are measured from the geometry and are exact. The falls and the outlet positions are architectural intent, drawn so the drainage package has something to coordinate against.

3. PL-200 and PL-300 set the outlet positions, the pipe sizes, the downpipe routes and the overflow provision. Nothing here is a drainage design.

4. Waterproofing turned up 150 mm minimum at every parapet, upstand and threshold and dressed into the outlet. Membrane and insulation build-up to the specification.

5. The pergola over the corridor is open and is not a roof: the corridor under it is an external surface and drains as one. The stair well is open to the sky under its canopy.

6. The areas here are drainage zones, not the use areas on A-104, and they divide the same surfaces differently — hence "north strip" rather than A-104's north terrace, which also counts the slab north of the stair. The two differ by 0.17 m²: A-104 deducts the full 2.05 m strip for the stair well, which is 2.00 m wide.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

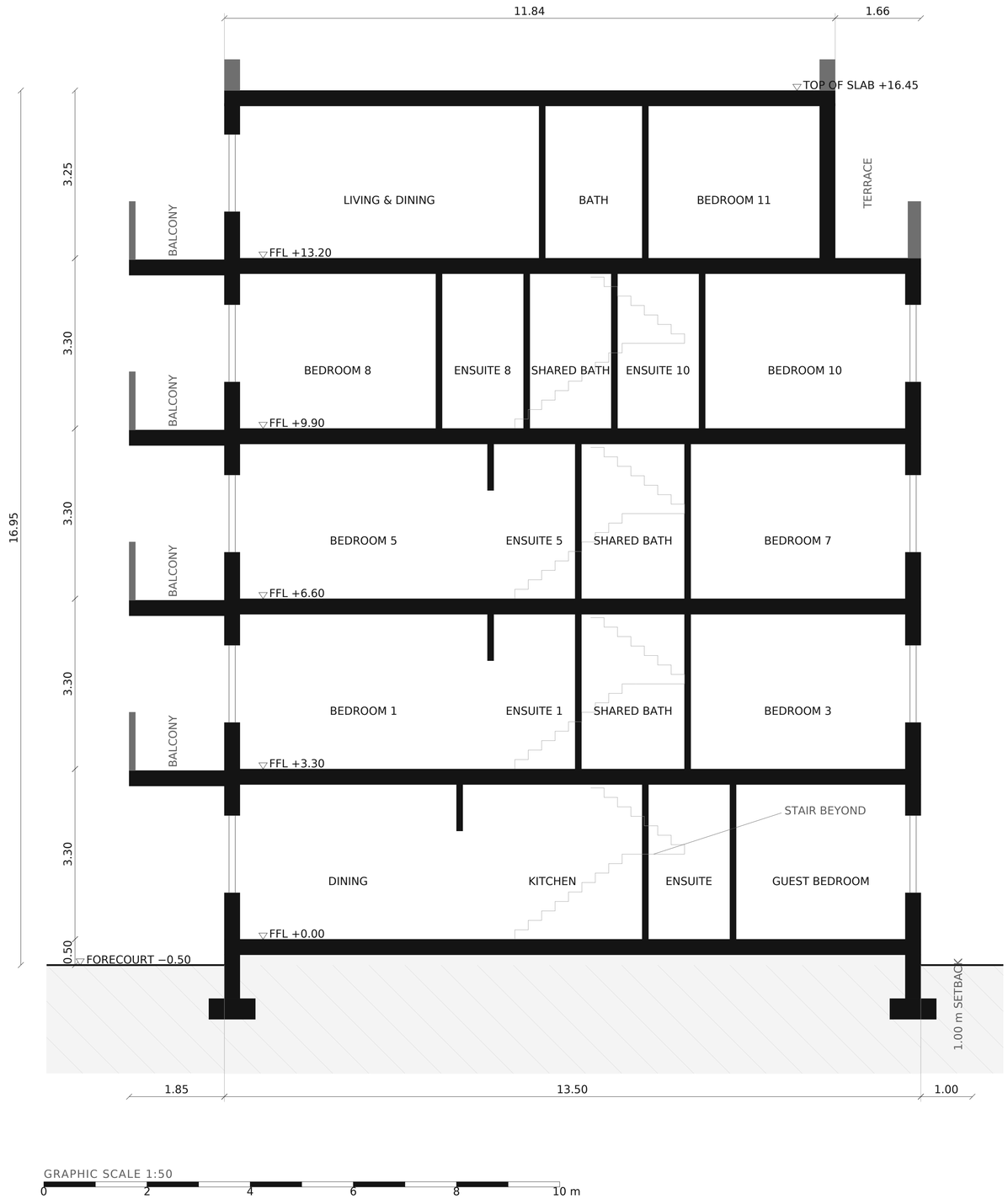
ROOF PLAN

DRAWING No	REVISION	SCALE	DATE
ARC-106	**P21**	**1:50 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SECTION A–A — DETACHED VILLA

LEVEL looking west · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

LEVEL DATUMS

FORECOURT AND GARDEN	-0.50
GROUND FLOOR	+0.00
FIRST FLOOR	+3.30
SECOND FLOOR	+6.60
THIRD FLOOR	+9.90
PENTHOUSE FLOOR	+13.20
PENTHOUSE, TOP OF SLAB	+16.45
CLEAR CEILING HEIGHT	3.00 m
STRUCTURAL ZONE	0.30 m
OVERALL, GARDEN TO TOP	16.95 m

GENERAL NOTES

1. Final draft — layout agreed. Looking west, cut on the line marked A–A on every plan, at x 20.70. That line is not arbitrary: it is the one north-south cut that passes through a room at every level and through every wet room in the east strip, and it clears all three column lines, so the slabs are seen spanning rather than a column standing in the cut.

2. The wet rooms on the first and second floors sit one over the other, wall for wall — 8.60, 10.30 and 12.42 on both — so each drops a single drainage stack straight to the ground. The third floor divides the strip on its own lines and is shown as it is.

3. Flat slab on the column grid, 300 mm structural zone, giving 3.00 m clear at every level and 3.30 m floor to floor; the penthouse is 3.25 m floor to top of slab.

4. The south cantilever is carried on all three upper floors and the penthouse. It projects 1.85 m at this cut, against 1.95 m at each end of the curve and 1.35 m at its waist, with a glass balustrade at the slab edge.

5. The penthouse stands clear of the north edge, leaving the roof terrace beyond its north wall. The corridor that joins the two terraces runs west of this cut and does not appear; see A-104.

6. The stair is drawn in thin line beyond the cut, where it is seen through the open spine. Foundations are indicative only and subject to a ground investigation and the engineer's design.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

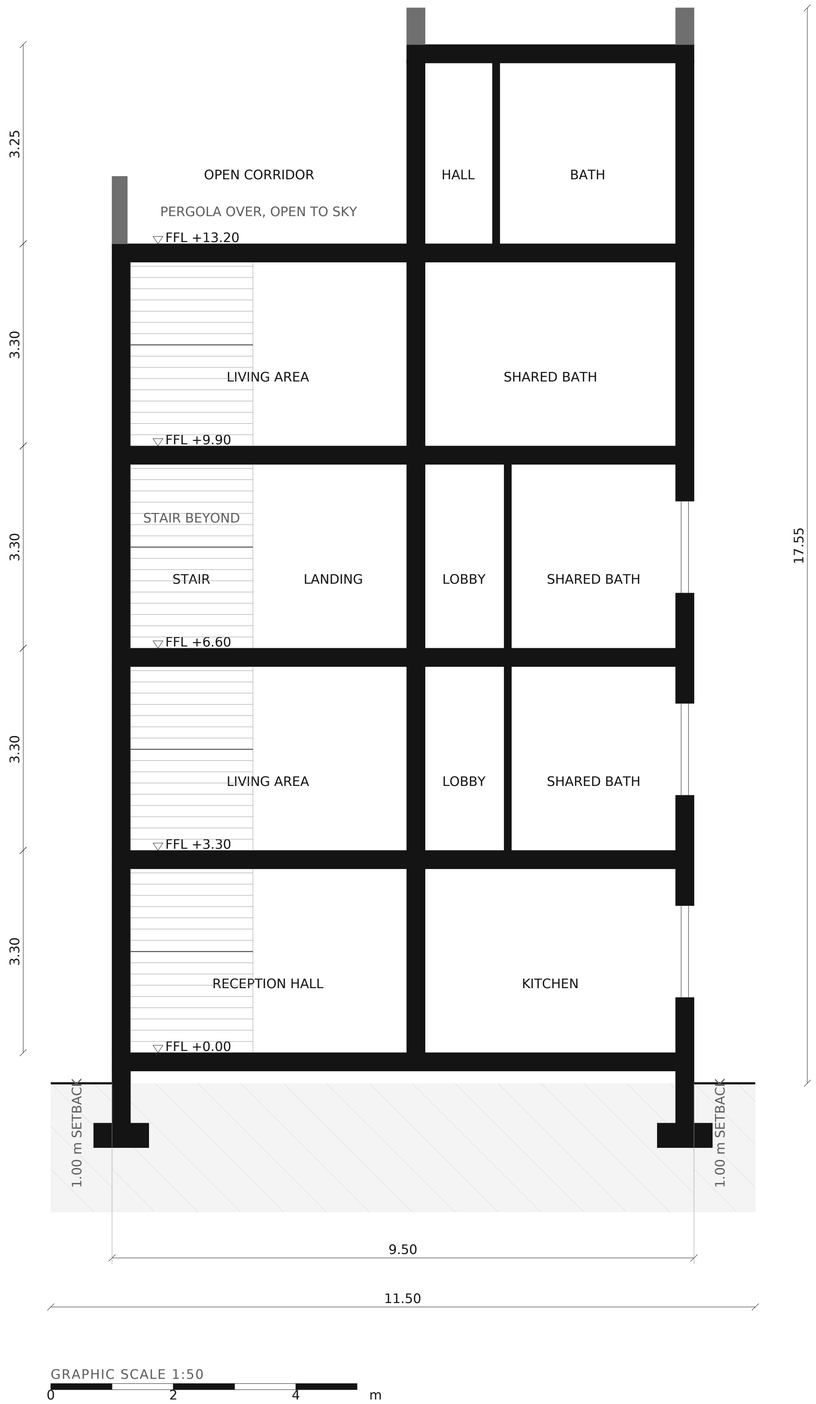
SECTION A–A — DETACHED VILLA

DRAWING No	REVISION	SCALE	DATE
A-200	**P21**	**1:50 @ A1**	**2026-10-07**

Final draft — layout agreed with the client; not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SECTION B–B — TRANSVERSE

LEVEL looking north · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE CUT

SECTION LINE	y = 10.80, looking north
WIDTH BETWEEN SETBACKS	9.50 m
STRUCTURAL SPINE	x 17.31 – 17.61
FLOOR TO FLOOR	3.30 m
CLEAR HEIGHT	3.00 m
STRUCTURAL ZONE	300 mm
TOP OF BUILDING	+17.05
EAST GLAZED AT THIS CUT	levels 0, 1 and 2

GENERAL NOTES

1. Cut at y = 10.80, inside the stair well and clear of every cross wall, looking north. A-A cuts the other way and shows the depth; this is the only drawing of the width.

2. What the cut passes through was read off the plans by the probe in checks.py, not written from memory, and whether the east wall is glazed at the cut is worked out from the plan's own opening list each time the sheet is built.

3. The east wall is glazed on the lower three floors and solid on the third, which is the plans' arrangement and not a drafting choice. It can be glazed at all because the neighbour a metre east is a single storey.

4. The stair is beyond the cut, not in it: its flights run north-south, so this section crosses them end-on and they are drawn in thin line as seen through the open well.

5. At the penthouse the west half of the plate is the open corridor, outside the flat and open to the sky under its pergola. Foundations are indicative pending GEO-001.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

SECTION B–B — TRANSVERSE

DRAWING No	REVISION	SCALE	DATE
ARC-301	**P21**	**1:50 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



THE BALCONY CURVE IN PLAN — IT CANNOT SHOW IN ELEVATION

1.95 m DEEPEST · 1.35 m SHALLOWEST · 15.7 m² A FLOOR



GRAPHIC SCALE 1:100

0 2 4 m

SOUTH ELEVATION — SCHEME A “WAVE”

LEVEL street frontage · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

SCHEME A — WAVE

WALLS	warm limestone render
SLAB NOSING	rounded, 180 mm
COVE LIGHT	in the underside of the nosing
BALUSTRADE	clear glass, 120 mm inboard
FRAMES	bronze anodised
BALCONY CURVE	serpentine
PROJECTION	1.95 / 1.35 m
BALCONY AREA	15.7 m² a floor
TOP OF BUILDING	+17.05

GENERAL NOTES

1. The exterior only. The plans, site plan and section are agreed and re-issued to the adopted curve; nothing here changes the rest of them: same five levels, same 9.50 × 13.50 m footprint, same 1.00 m setbacks, same openings. Only the skin differs between the schemes.

2. Every opening drawn here is read from the plan it belongs to rather than redrawn, and checks.py fails the build if an elevation ever shows one the plan does not have.

3. Scheme A takes its material from the contemporary reference: warm limestone render, a rounded slab nosing with a cove light tucked into its underside, clear glass balustrades and planting along the balcony edge. The four south bays read as glazed ribbons between the slabs.

4. The balcony curve is a serpentine — one full period across the frontage, 1.95 m deep at each end drawn in to 1.35 m at the waist. At 1.35 m is usable the whole way along; the deep ends carry the dining and the furniture.

5. This is the adopted curve, and the plans are drawn to it. It cost area to take: 15.7 m² a floor against the 17.4 m² of the curve it replaced, 7 m² over the four balcony levels. A-000 and A-100 to A-104 are re-issued to it.

6. The balconies do not wrap the corner as the reference does: the 1.00 m setbacks forbid anything projecting west, north or east, so the cantilever stays on the street frontage.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

SOUTH ELEVATION — SCHEME A “WAVE”

DRAWING No	REVISION	SCALE	DATE
A-400	**P21**	**1:50 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SIDE & REAR ELEVATIONS — SCHEME A “WAVE”

LEVEL east, north and west · 1:75 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

SCHEME A — WAVE

WALLS	warm limestone render
SLAB NOSING	rounded, 180 mm
COVE LIGHT	in the underside of the nosing
BALUSTRADE	clear glass, 120 mm inboard
FRAMES	bronze anodised
BALCONY CURVE	serpentine
PROJECTION	1.95 / 1.35 m
BALCONY AREA	15.7 m² a floor
TOP OF BUILDING	+17.05

GENERAL NOTES

1. The exterior only. The plans, site plan and section are agreed and re-issued to the adopted curve; nothing here changes the rest of them: same five levels, same 9.50 × 13.50 m footprint, same 1.00 m setbacks, same openings. Only the skin differs between the schemes.

2. Every opening drawn here is read from the plan it belongs to rather than redrawn, and checks.py fails the build if an elevation ever shows one the plan does not have.

3. The east and north boundaries are open and carry ordinary windows on every floor. The west is the blind elevation: louvred ventilation only, drawn at high level, and nothing projects.

4. The penthouse stands back 4.81 m from the west and 1.66 m from the north, so on those two faces it is drawn in thin line behind the roof terrace parapet.

5. The balcony appears on the street frontage only and so does not show on any of these three faces.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

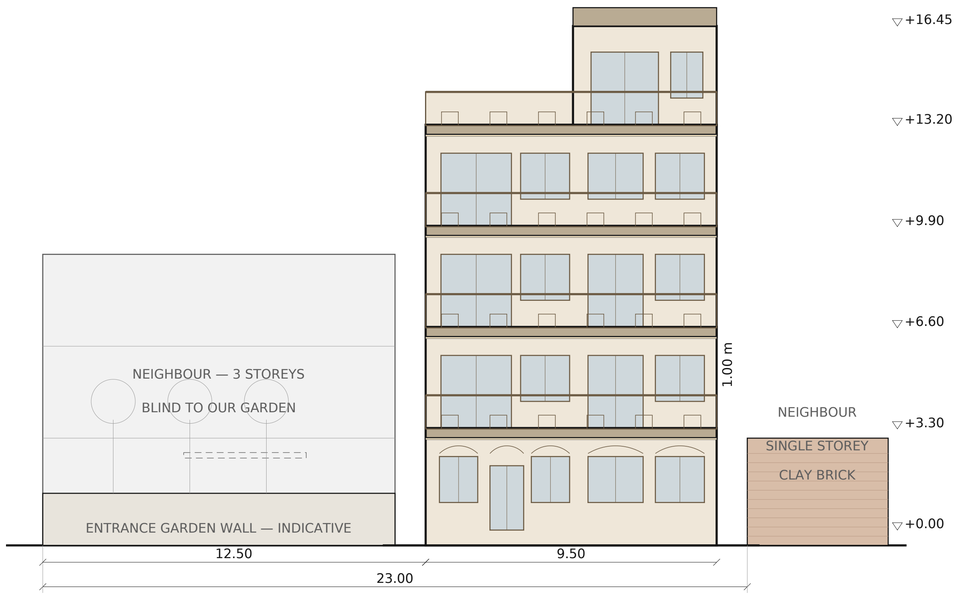
SIDE & REAR ELEVATIONS — SCHEME A “WAVE”

DRAWING No	REVISION	SCALE	DATE
A-401	**P21**	**1:75 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SCHEME A — WAVE, ADOPTED

STREET ELEVATION IN CONTEXT

LEVEL scheme A, as surveyed · 1:100 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

AS SURVEYED

PLOT WIDTH	23.00 m
ENTRANCE GARDEN, WEST	11.50 m
VILLA FRONTAGE	9.50 m
SIDE PASSAGE, EAST	1.00 m
WEST NEIGHBOUR	3 storeys, blind
EAST NEIGHBOUR	1 storey, clay brick
NORTH NEIGHBOUR	1 storey
OUR TOP OF BUILDING	+17.05
WE STAND OVER THE EAST	14.05 m

GENERAL NOTES

1. Preliminary — the exterior only, and the first drawing in the set to show the villa against the houses it actually stands between.

2. The house is not in the middle of its plot. It sits hard against the east boundary with a 1.00 m paved side passage, and the whole 11.50 m entrance garden is on the west, gated from the west boundary with a walk running east to the front door.

3. East: a single storey in clay brick, 1.00 m away. That is what lets the east wall be glazed at all — our lowest east sill at +4.20 clears its roof by 1.20 m. We stand 14.05 m over it, which is worth seeing before the elevations are agreed.

4. West: 3 storeys and blind toward us, drawn in thin line because it stands back behind the garden. Nothing of ours opens that way in any case — the west wall takes louvred ventilation only.

5. North: a single storey as well, confirmed by the client and not drawn on this south-facing view. The north wall carries windows on all five levels; only the ground floor looks at anything, and from +3.30 up they clear its roof as the east windows clear the east neighbour's.

6. The garden wall height is indicative and awaits the client's choice. Boundaries to be confirmed by measured survey.

GRAPHIC SCALE 1:100

0 2 4 6 8 10 m

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

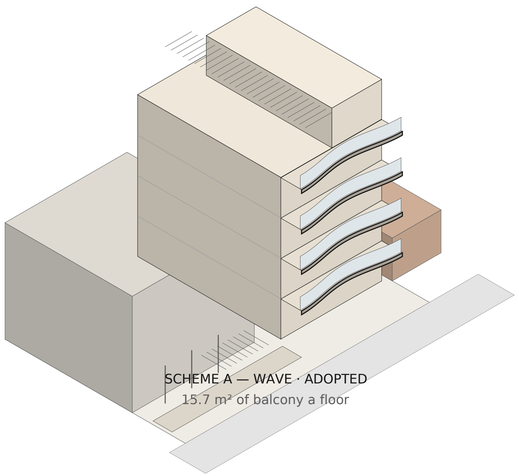
STREET ELEVATION IN CONTEXT

DRAWING No	REVISION	SCALE	DATE
A-402	**P21**	**1:100 @ A1**	**2026-10-07**

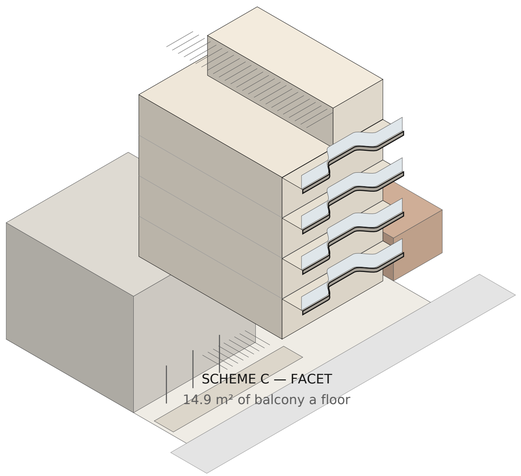
Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

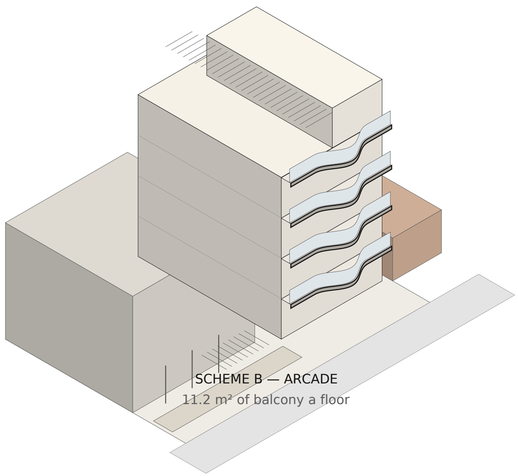
P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SCHEME A — WAVE · ADOPTED
15.7 m² of balcony a floor



SCHEME C — FACET
14.9 m² of balcony a floor



SCHEME B — ARCADE
11.2 m² of balcony a floor

AXONOMETRIC — THREE BALCONY OPTIONS

LEVEL from the south-west · 1:250 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 Finished floor level datum

THE FOUR CURVES OFFERED

A — SERPENTINE, ADOPTED	15.7 m² a floor
C — FACET	14.9 m² a floor
B — BOWED CENTRE BAY	11.2 m² a floor
SUPERSEDED CURVE	17.4 m² a floor
DEEPEST ALLOWED	2.00 m
SHALLOWEST USABLE	0.60 m
WEST NEIGHBOUR	3 storeys, blind
EAST NEIGHBOUR	1 storey, clay brick

GENERAL NOTES

1. Preliminary. Isometric from the south-west, which is the one corner that shows the street frontage, the blind west wall, the roof pergola and the entrance garden at once.

2. This is a projection, not a render: every line is a transform of the constants the plans are drawn from, so it cannot show a balcony, a garden or a neighbour that the drawings do not have.

3. All three sit on the real plot. The villa is hard against the east boundary with a 1.00 m side passage and the 11.50 m entrance garden on the west, the three-storey blind neighbour standing back behind it and the single-storey clay-brick one a metre to the east.

4. Scheme A is the one adopted, and the plans are drawn to its curve. B and C are kept here as the record of what was offered; their elevation sheets are stamped not selected and are not bound into the issue.

5. The balconies stop at the corner on all three: the 1.00 m setbacks forbid anything projecting west, north or east.

6. Faces are painted back to front rather than depth-sorted, so a detail may overlap where two surfaces nearly touch. The measured elevations A-400 and A-401 govern.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

AXONOMETRIC — THREE BALCONY OPTIONS

DRAWING No	REVISION	SCALE	DATE
A-430	**P21**	**1:250 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

DOOR & WINDOW SCHEDULE

95 OPENINGS · 24 TYPES · MEASURED FROM THE PLANS

MARK	1:100	W × H mm	SILL	FACE	L0	L1	L2	L3	L4	No.
D01 Door		2300 × 2400	floor	S	·	1	1	1	·	**3**
D02 Door		2200 × 2400	floor	S	·	·	·	·	1	**1**
D03 Door		1800 × 2400	floor	S	·	1	1	·	·	**2**
D04 Door		1500 × 2400	floor	N	·	·	·	·	1	**1**
D05 Door		1100 × 2100	floor	S	1	·	·	·	·	**1**
D06 Door		800 × 2400	floor	W	·	·	·	·	1	**1**
W01 Window		1800 × 1500	+900	SE	2	2	2	2	·	**8**
W02 Window		1700 × 1500	+900	E	·	·	·	1	1	**2**
W03 Window		1600 × 1500	+900	SE	1	2	2	2	1	**8**
W04 Window		1500 × 1500	+900	N	1	1	1	1	·	**4**
W05 Window		1400 × 1500	+900	ENW	2	3	3	2	1	**11**
W06 Window		1300 × 1500	+900	E	·	1	1	·	1	**3**
W07 Window		1250 × 1500	+900	S	2	·	·	·	·	**2**
W08 Window		1200 × 1500	+900	EN	4	·	·	·	·	**4**
W09 Window		1100 × 1500	+900	N	1	1	1	1	·	**4**
W10 Window		1050 × 1500	+900	S	·	·	·	·	1	**1**
W11 Window		1000 × 1500	+900	E	1	·	·	2	·	**3**
W12 Window		900 × 1500	+900	E	·	·	·	1	·	**1**
W13 Window		800 × 1500	+900	ENW	1	3	3	1	2	**10**
W14 Window		700 × 1500	+900	E	·	·	·	·	1	**1**
W15 Window		600 × 1500	+900	EW	·	1	1	·	3	**5**
W16 Window		500 × 1500	+900	E	1	·	·	1	·	**2**
L01 Louvre		800 × 500	+2100	W	4	4	4	4	·	**16**
L02 Louvre		600 × 500	+2100	W	·	·	1	·	·	**1**

DOOR & WINDOW SCHEDULE

SCHEDULE · ALL FIVE LEVELS · MILLIMETRES

THE OPENINGS

DOORS	9
WINDOWS	69
LOUVRES	17
TYPES	24
DOOR AREA	38.3 m²
GLAZED AREA	127.9 m²
LOUVRE AREA	6.7 m²
WINDOW HEAD, EVERY LEVEL	+2.40 above floor
WINDOW SILL, EVERY LEVEL	+0.90 above floor

GENERAL NOTES

1. Measured from the plans, not compiled beside them. Every opening is a pair of coordinates the plan cuts its wall with and the elevation draws its frame in, so this schedule and those drawings cannot disagree about a width.

2. Marks run largest first within each kind, so D01 is the widest door and W01 the widest window. L marks are the west wall's louvres: that face is blind below the penthouse and ventilates only.

3. Heads and sills are constant through the building — sill +0.90, head +2.40 — so a type is distinguished by its width alone except at the doors.

4. Operation, frame material, glazing specification, ironmongery and finishes are not set by the geometry and are not invented here. They belong to ARC-600 and the specification.

5. Dimensions are structural opening sizes. Frame and tolerance allowances to be taken off by the supplier against the construction drawings.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

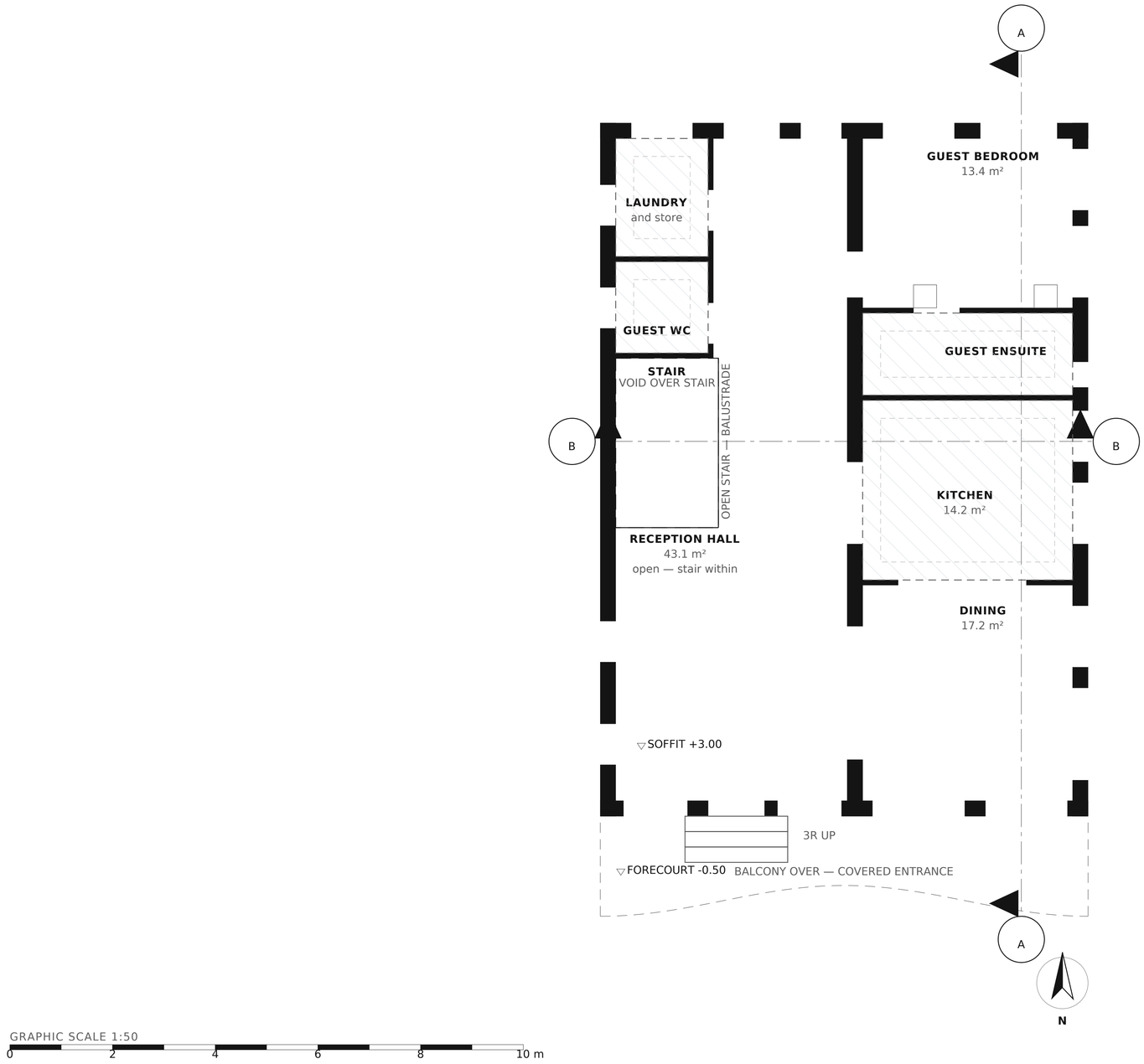
DOOR & WINDOW SCHEDULE

DRAWING No	REVISION	SCALE	DATE
ARC-500	**P21**	**1:100 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



REFLECTED CEILING PLAN — GROUND FLOOR

LEVEL +0.00 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE CEILING

FINISHED FLOOR LEVEL	+0.00
SOFFIT, GENERALLY	+3.00
CLEAR HEIGHT	3.00 m
STRUCTURAL ZONE OVER	300 mm
FLOOR TO FLOOR	3.30 m
BULKHEADS SHOWN	4 wet rooms
BULKHEAD DEPTH	by the services engineer
LIGHT FITTINGS	see EL-200 series

GENERAL NOTES

1. The ceiling of the ground floor, seen from below. Drawn by running the floor plan with its sanitaryware, furniture, door swings and dimension chains switched off, so it cannot stop agreeing with A-100 when a partition moves.

2. Soffit is +3.00 throughout, 3.00 m clear, under a 300 mm structural zone. Any local drop below that is a bulkhead and is shown as one.

3. A bulkhead is drawn over every room the plan calls wet, because that is where the drainage runs. The outline is the room; the depth, the exact extent and everything inside it are the services engineer's and are set on the ME and PL series.

4. No light fitting is drawn. Fittings, switching and emergency provision are the EL-200 series; a layout invented here would only have to be thrown away.

5. Ceiling finishes, board type, moisture resistance in the wet rooms and access panels are specification, not geometry, and are not set on this sheet.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

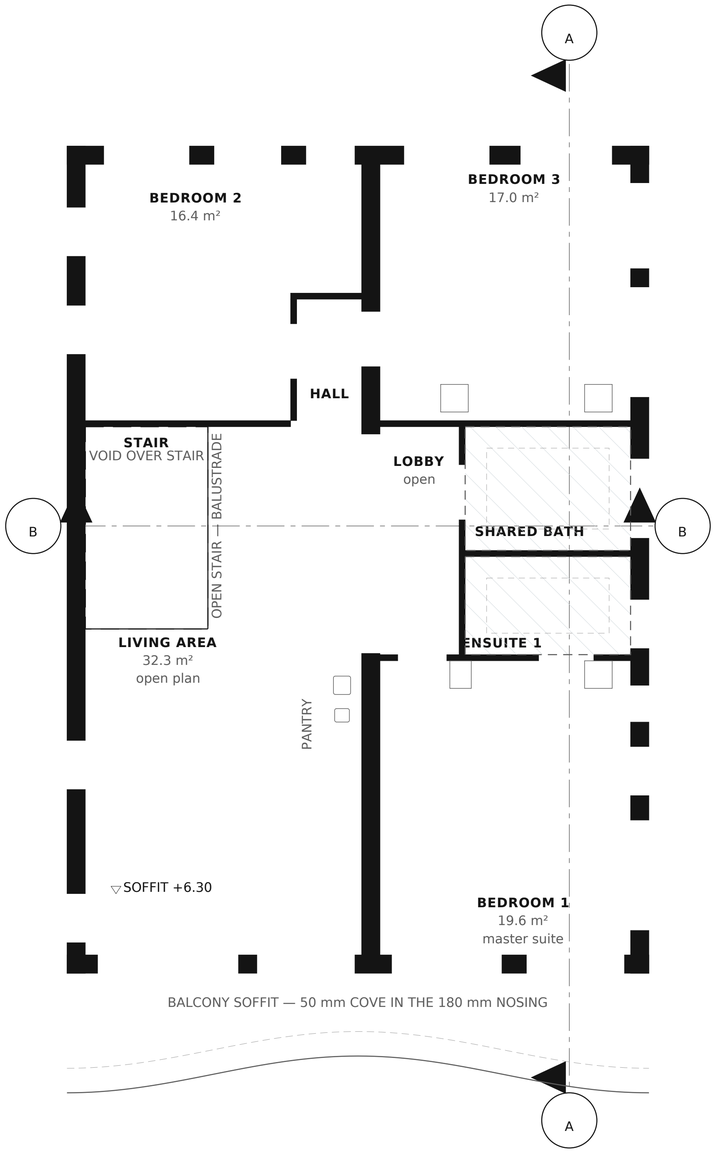
REFLECTED CEILING PLAN — GROUND FLOOR

DRAWING No	REVISION	SCALE	DATE
ARC-400	**P21**	**1:50 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

REFLECTED CEILING PLAN — FIRST FLOOR

LEVEL +3.30 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE CEILING

FINISHED FLOOR LEVEL	+3.30
SOFFIT, GENERALLY	+6.30
CLEAR HEIGHT	3.00 m
STRUCTURAL ZONE OVER	300 mm
FLOOR TO FLOOR	3.30 m
BULKHEADS SHOWN	2 wet rooms
BULKHEAD DEPTH	by the services engineer
LIGHT FITTINGS	see EL-200 series

GENERAL NOTES

1. The ceiling of the first floor, seen from below. Drawn by running the floor plan with its sanitaryware, furniture, door swings and dimension chains switched off, so it cannot stop agreeing with A-101 when a partition moves.

2. Soffit is +6.30 throughout, 3.00 m clear, under a 300 mm structural zone. Any local drop below that is a bulkhead and is shown as one.

3. A bulkhead is drawn over every room the plan calls wet, because that is where the drainage runs. The outline is the room; the depth, the exact extent and everything inside it are the services engineer's and are set on the ME and PL series.

4. No light fitting is drawn. Fittings, switching and emergency provision are the EL-200 series; a layout invented here would only have to be thrown away.

5. Ceiling finishes, board type, moisture resistance in the wet rooms and access panels are specification, not geometry, and are not set on this sheet.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

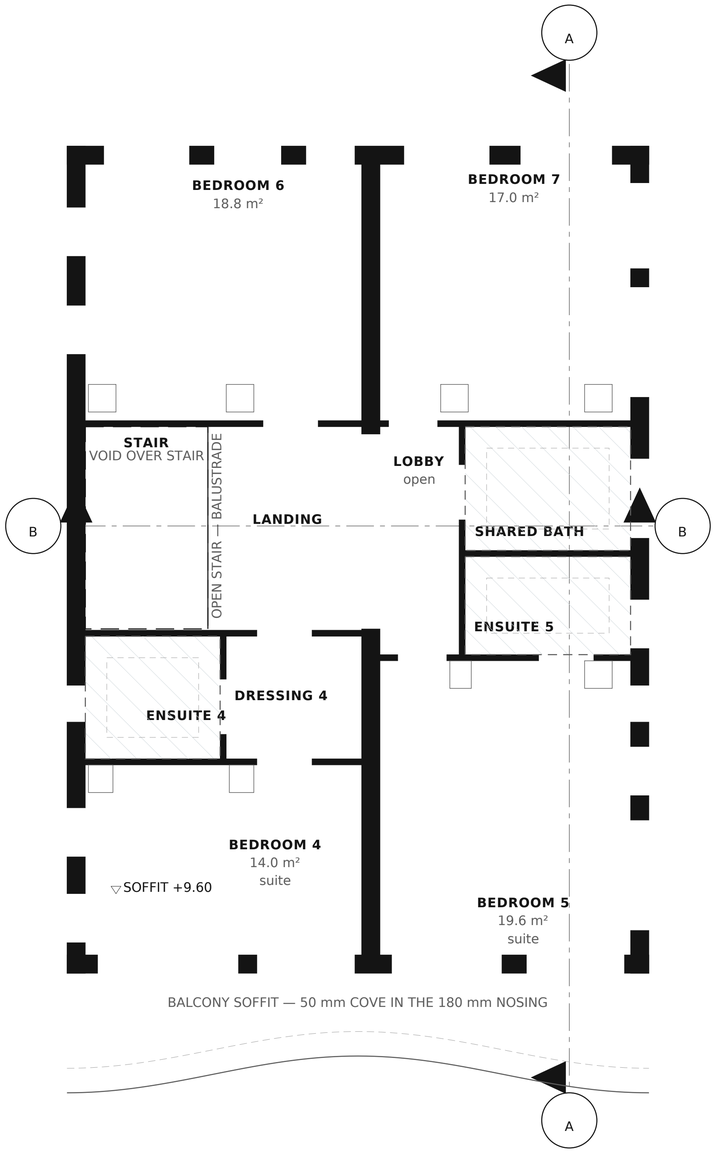
REFLECTED CEILING PLAN — FIRST FLOOR

DRAWING No	REVISION	SCALE	DATE
ARC-401	**P21**	**1:50 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

REFLECTED CEILING PLAN — SECOND FLOOR

LEVEL +6.60 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE CEILING

FINISHED FLOOR LEVEL	+6.60
SOFFIT, GENERALLY	+9.60
CLEAR HEIGHT	3.00 m
STRUCTURAL ZONE OVER	300 mm
FLOOR TO FLOOR	3.30 m
BULKHEADS SHOWN	3 wet rooms
BULKHEAD DEPTH	by the services engineer
LIGHT FITTINGS	see EL-200 series

GENERAL NOTES

1. The ceiling of the second floor, seen from below. Drawn by running the floor plan with its sanitaryware, furniture, door swings and dimension chains switched off, so it cannot stop agreeing with A-102 when a partition moves.

2. Soffit is +9.60 throughout, 3.00 m clear, under a 300 mm structural zone. Any local drop below that is a bulkhead and is shown as one.

3. A bulkhead is drawn over every room the plan calls wet, because that is where the drainage runs. The outline is the room; the depth, the exact extent and everything inside it are the services engineer's and are set on the ME and PL series.

4. No light fitting is drawn. Fittings, switching and emergency provision are the EL-200 series; a layout invented here would only have to be thrown away.

5. Ceiling finishes, board type, moisture resistance in the wet rooms and access panels are specification, not geometry, and are not set on this sheet.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

REFLECTED CEILING PLAN — SECOND FLOOR

DRAWING No	REVISION	SCALE	DATE
ARC-402	**P21**	**1:50 @ A1**	**2026-10-07**

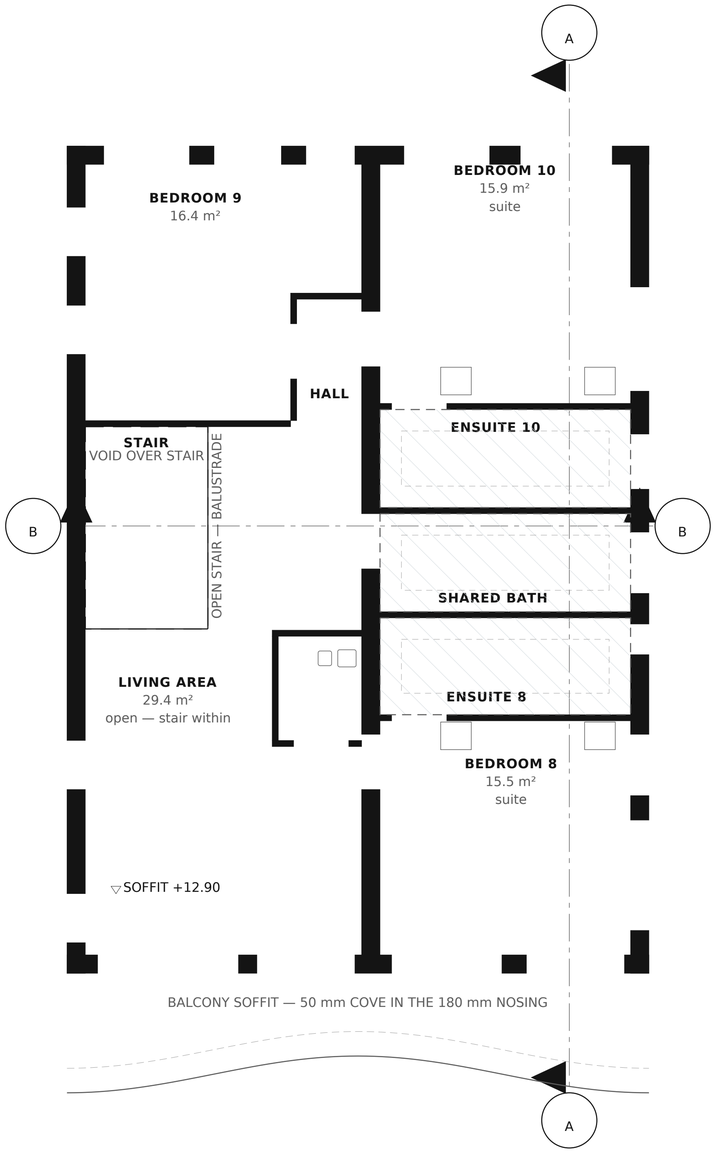
Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

REFLECTED CEILING PLAN — THIRD FLOOR

LEVEL +9.90 · 1:50 @ A1 · METRES



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

KEY

▬	External / party wall cut by section, 300 mm — brick, colour unrestricted
▬	Internal partition, 100 mm — red brick
▬	Terrace parapet, 250 mm × 1100 mm high
▬	Single leaf door, swing shown to opening side
▬	Glazed opening — arched head where noted
┄	Element above the cut plane (pergola, slab over)
▭	Void / open to sky
▽ +0.00	Finished floor level datum

THE CEILING

FINISHED FLOOR LEVEL	+9.90
SOFFIT, GENERALLY	+12.90
CLEAR HEIGHT	3.00 m
STRUCTURAL ZONE OVER	300 mm
FLOOR TO FLOOR	3.30 m
BULKHEADS SHOWN	3 wet rooms
BULKHEAD DEPTH	by the services engineer
LIGHT FITTINGS	see EL-200 series

GENERAL NOTES

1. The ceiling of the third floor, seen from below. Drawn by running the floor plan with its sanitaryware, furniture, door swings and dimension chains switched off, so it cannot stop agreeing with A-103 when a partition moves.

2. Soffit is +12.90 throughout, 3.00 m clear, under a 300 mm structural zone. Any local drop below that is a bulkhead and is shown as one.

3. A bulkhead is drawn over every room the plan calls wet, because that is where the drainage runs. The outline is the room; the depth, the exact extent and everything inside it are the services engineer's and are set on the ME and PL series.

4. No light fitting is drawn. Fittings, switching and emergency provision are the EL-200 series; a layout invented here would only have to be thrown away.

5. Ceiling finishes, board type, moisture resistance in the wet rooms and access panels are specification, not geometry, and are not set on this sheet.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

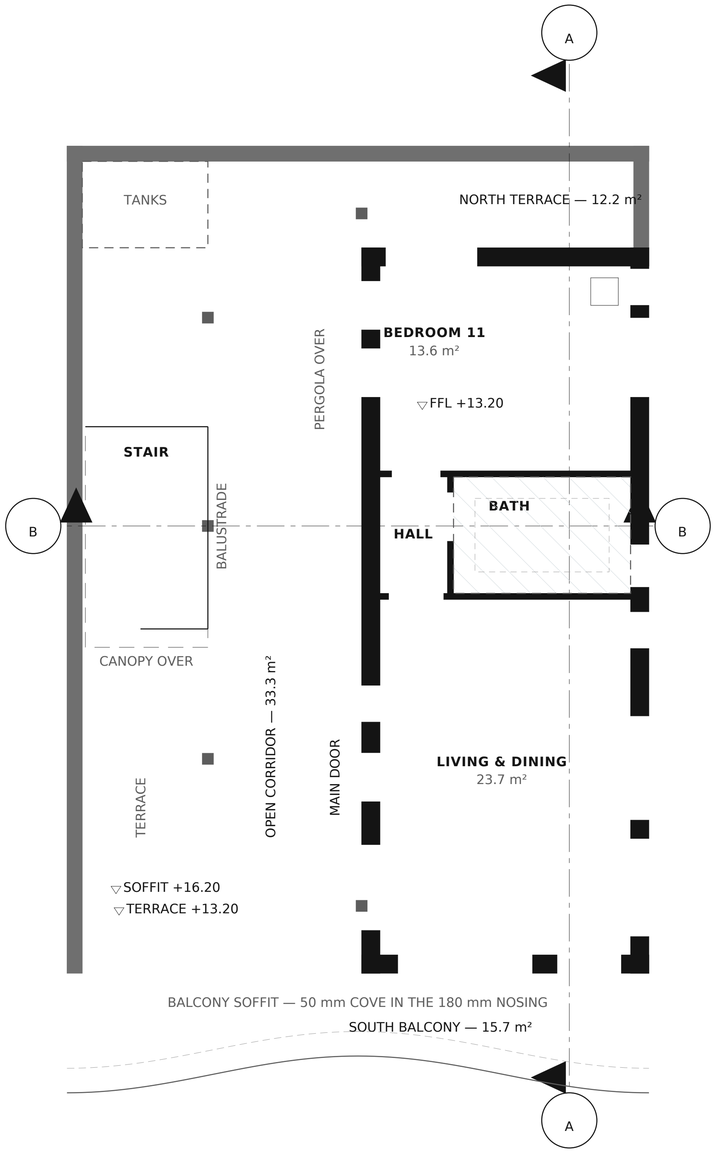
REFLECTED CEILING PLAN — THIRD FLOOR

DRAWING No	REVISION	SCALE	DATE
ARC-403	**P21**	**1:50 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

REFLECTED CEILING PLAN — PENTHOUSE FLOOR

LEVEL +13.20 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE CEILING

FINISHED FLOOR LEVEL	+13.20
SOFFIT, GENERALLY	+16.20
CLEAR HEIGHT	3.00 m
STRUCTURAL ZONE OVER	300 mm
FLOOR TO FLOOR	3.30 m
BULKHEADS SHOWN	1 wet rooms
BULKHEAD DEPTH	by the services engineer
LIGHT FITTINGS	see EL-200 series

GENERAL NOTES

1. The ceiling of the penthouse floor, seen from below. Drawn by running the floor plan with its sanitaryware, furniture, door swings and dimension chains switched off, so it cannot stop agreeing with A-104 when a partition moves.

2. Soffit is +16.20 throughout, 3.00 m clear, under a 300 mm structural zone. Any local drop below that is a bulkhead and is shown as one.

3. A bulkhead is drawn over every room the plan calls wet, because that is where the drainage runs. The outline is the room; the depth, the exact extent and everything inside it are the services engineer's and are set on the ME and PL series.

4. No light fitting is drawn. Fittings, switching and emergency provision are the EL-200 series; a layout invented here would only have to be thrown away.

5. Ceiling finishes, board type, moisture resistance in the wet rooms and access panels are specification, not geometry, and are not set on this sheet.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

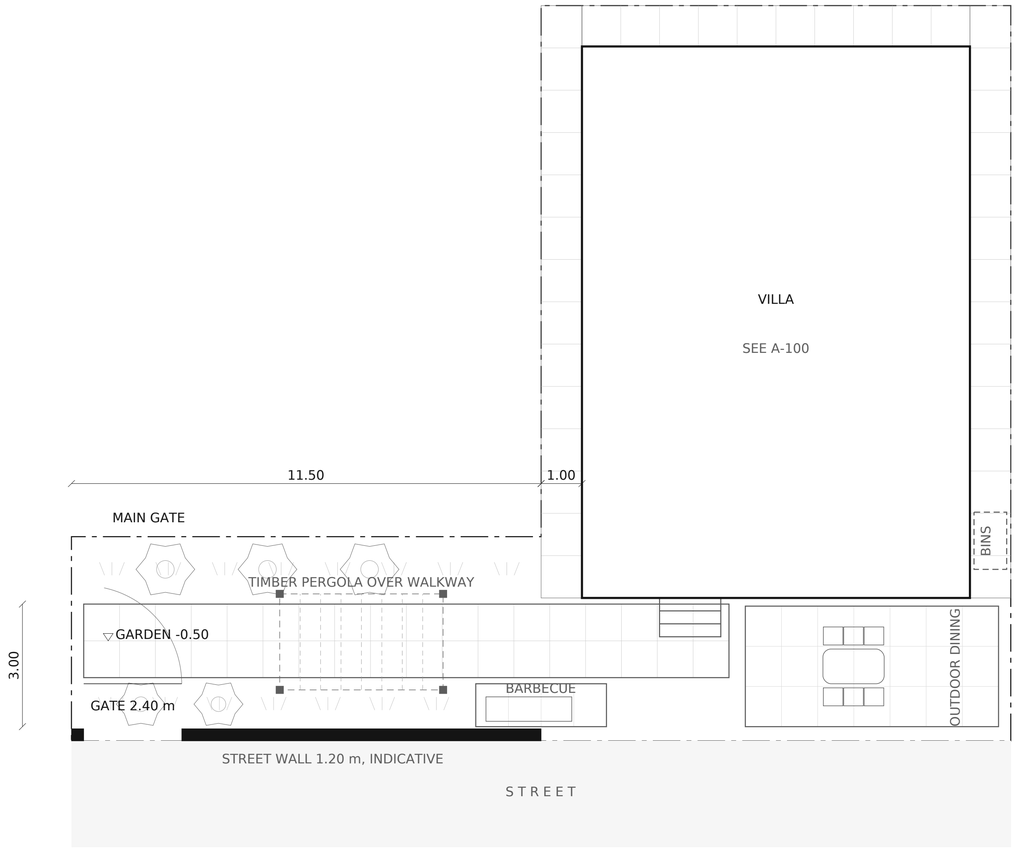
REFLECTED CEILING PLAN — PENTHOUSE FLOOR

DRAWING No	REVISION	SCALE	DATE
ARC-404	**P21**	**1:50 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:75

0 2 4 6 8 10 m

N

EXTERNAL WORKS & LANDSCAPE

LEVEL -0.50 garden, +0.00 villa · 1:75 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 Finished floor level datum

AREA SCHEDULE

PAVED WALK	28.4 m²
FORECOURT	18.3 m²
BARBECUE TERRACE	3.4 m²
SIDE PASSAGES	38.5 m²
TURF	24.7 m²
TOTAL	**113.3 m²**
PERGOLA OVER THE WALK	9.4 m² timber
TREES	5 nr
STREET WALL	9.10 m at 1.20 m, indicative
GATE	2.40 m leaf
STEPS TO THE FRONT DOOR	3 risers
PLOT LESS THE VILLA	136.2 m²
NOT MEASURED HERE	23.0 m²

GENERAL NOTES

1. The garden as a trade package rather than as context. Every area is read from the constants A-000 draws from, so this sheet and the site plan cannot disagree about a width.

2. The three 1.00 m setbacks are our own land and are paved as side passages, reached from the forecourt on the east and the garden on the west. They are the largest paved item on the sheet.

3. The street wall is 1.20 m high and INDICATIVE: its height, its coping and its finish await the client, and its line awaits the measured survey. Nothing else on the sheet depends on it.

4. Paving materials, bedding, falls, sub-base, turf specification and irrigation are not set by the geometry and are not invented here. The quantities are ours; the specification is not.

5. The forecourt is overhung by the balcony for up to 1.95 m, which is what covers the front door without a porch. See A-000 and A-400.

6. The schedule measures ground cover, so the pergola is listed beside it rather than in it - it stands over paving already counted. 23.0 m² of the plot outside the villa is not measured here: margins, the bin standing and the entrance steps. It is shown rather than absorbed.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

EXTERNAL WORKS & LANDSCAPE

DRAWING No	REVISION	SCALE	DATE
ARC-700	**P21**	**1:75 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

STRUCTURAL NOTES & DESIGN CRITERIA

SHAMANDEEL, QUESNA, MENOUFIA, EGYPT · ECP 201 — Egyptian Code for Loads · ECP 203 — Egyptian Code for Reinforced Concrete

WHAT THIS PACKAGE IS

The frame set out: a grid, twelve column coordinates, six slab outlines, the spans between them and the lines where a beam is required. All of it falls out of the architecture and all of it is exact.

It exists so the structural engineer starts from a coordinated layout rather than from nothing, and so the architecture cannot quietly move underneath the structure.

WHAT THIS PACKAGE IS NOT

It is not a structural design. No member size, no slab or footing depth, no reinforcement, no cover, no load and no analysis appears anywhere in it.

Design criteria — dead and imposed loads, wind, the seismic zone and its design ground acceleration — are set and signed by the structural engineer to ECP 201, and the concrete to ECP 203. ECP 201 has five seismic zones; which governs Shamandeel is named by the engineer, not assumed here.

WHAT IT WAITS ON

GEO-001, the soil investigation. The ground is central Nile Delta and well characterised as a region, which is enough to assume a foundation type and lay it out, and not enough to size one. Allowable bearing pressure, the water table and sulphate content are plot figures.

SUR-001, the measured survey, for the boundary STR-400 sits on.

THE ONE RULE THE ARCHITECTURE ALREADY SETTLED

A 300 mm wall needs a beam under it, and a beam is a column line, a wall below, or a secondary beam landing on primary beams. Every 300 mm wall in this building stands on a grid line, with one exception: the penthouse's north wall, which is a line load on the +13.20 slab spanning B to C. STR-305 carries a secondary beam under it for that reason.

The 100 mm partitions do not stand on grid lines and do not need to; they are carried by the slab.

STRUCTURAL NOTES & DESIGN CRITERIA

LEVEL the whole frame · NTS · METRES

KEY

▬▬	External / party wall cut by section, 300 mm — brick, colour unrestricted
▬▬	Internal partition, 100 mm — red brick
▬▬	Terrace parapet, 250 mm × 1100 mm high
▬◠	Single leaf door, swing shown to opening side
▬▭▬	Glazed opening — arched head where noted
┄┄┄	Element above the cut plane (pergola, slab over)
▭▭	Void / open to sky
▽ +0.00	Finished floor level datum

THE FRAME

GRID	3 × 4, A–C / 1–4
SPAN A–B	4.81 m
SPAN B–C	4.39 m
SPAN 1–2	4.81 m
SPAN 2–3	4.10 m
SPAN 3–4	4.29 m
COLUMNS	12
COLUMN SECTION	300 × 300 mm
STRUCTURAL ZONE	300 mm
STOREY HEIGHT	3.30 m

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. Design criteria to ECP 201 and concrete to ECP 203, set and signed by the structural engineer. No value is assumed here.

4. The package waits on GEO-001 for anything below ground and on SUR-001 for the boundary.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

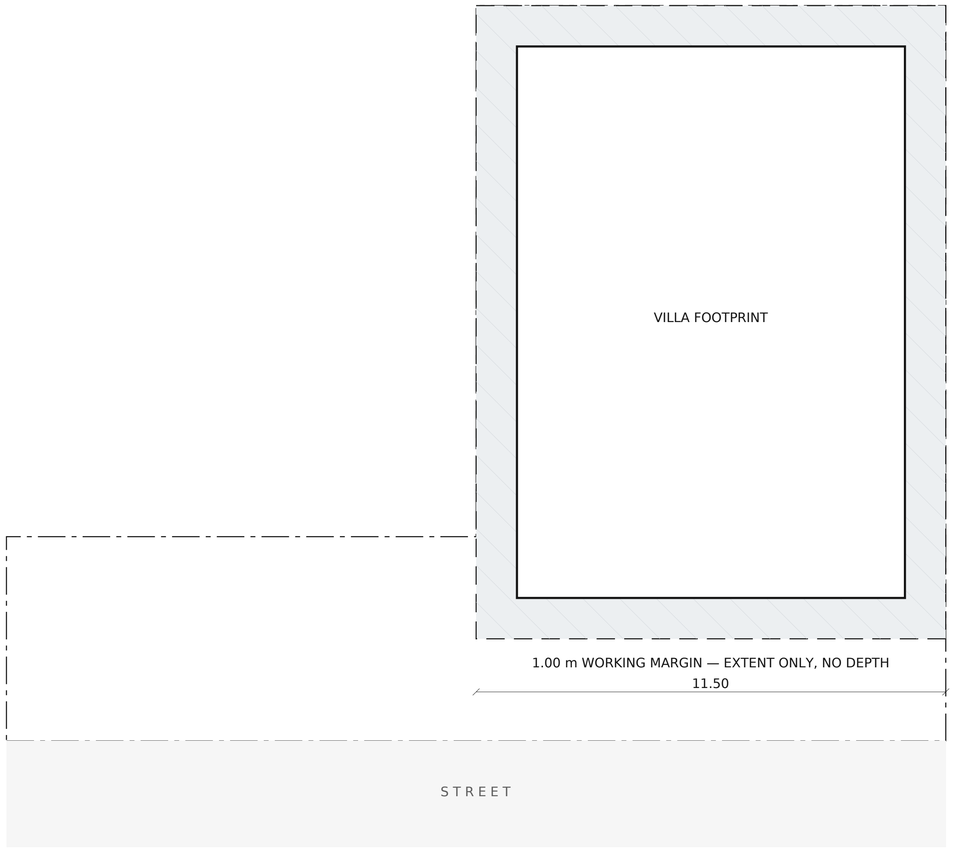
STRUCTURAL NOTES & DESIGN CRITERIA

DRAWING No	REVISION	SCALE	DATE
STR-001	**P21**	**NTS**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



EXCAVATION, GRADING & SHORING

LEVEL -0.50 garden, +0.00 villa · 1:75 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE FRAME

FOOTPRINT	128.2 m²
WORKING MARGIN	1.00 m
DIG EXTENT	178.2 m²
DIG DEPTH	by GEO-001 and STR-101
FORMATION LEVEL	by GEO-001
SHORING	by the engineer
WATER TABLE	by GEO-001
NEIGHBOUR, EAST	1 storey, 1.00 m away

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. The extent of the dig is a layout decision and is drawn. Its depth is not: formation level follows the foundation, the foundation follows the bearing pressure, and that is GEO-001.

4. Shoring, dewatering and the sequence of excavation are the engineer's and the contractor's. The east neighbour stands 1.00 m from the dig and is a single storey in clay brick; its protection is to be designed before anything is excavated against it.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

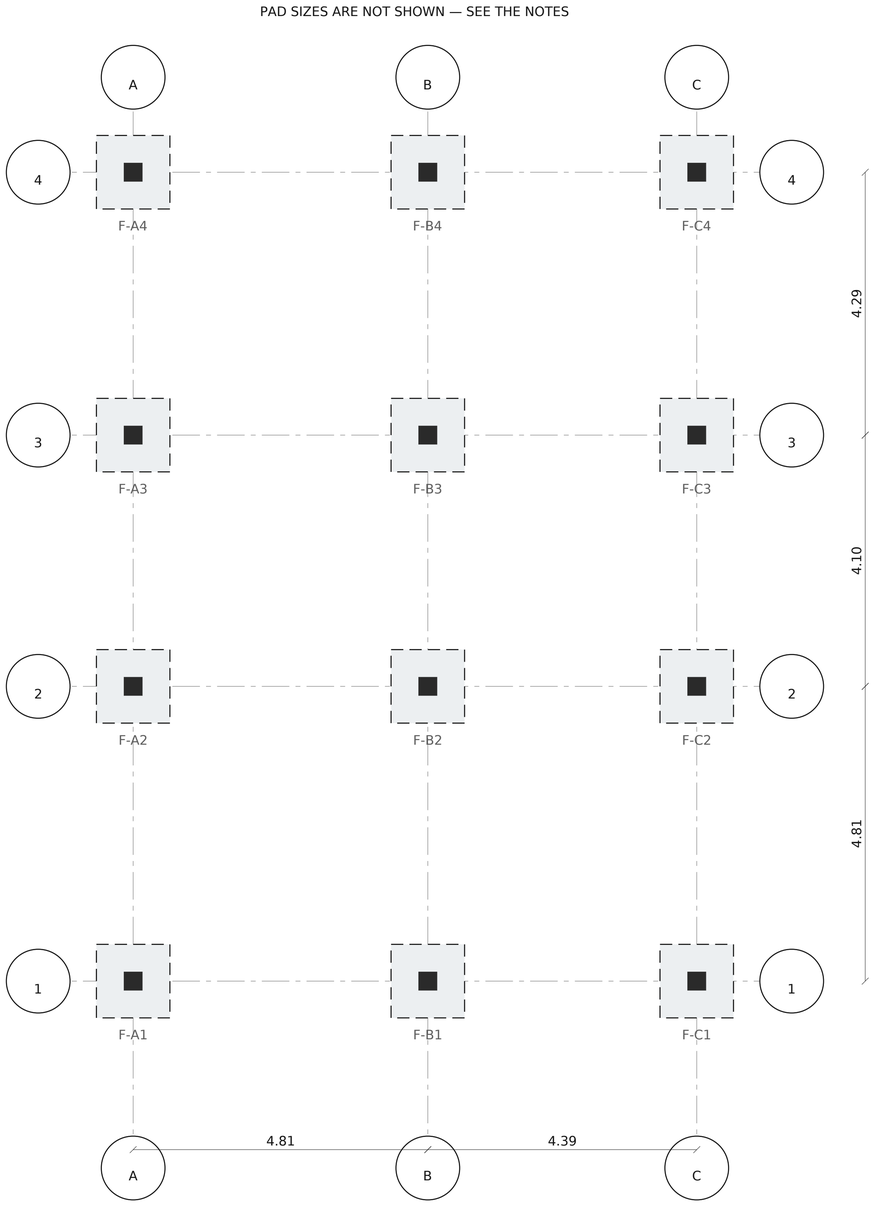
EXCAVATION, GRADING & SHORING

DRAWING No	REVISION	SCALE	DATE
STR-100	**P21**	**1:75 @ A1**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



FOUNDATION LAYOUT

LEVEL formation by GEO-001 · 1:50 @ A1 · METRES

KEY

External / party wall cut by section, 300 mm — brick, colour unrestricted	
Internal partition, 100 mm — red brick	
Terrace parapet, 250 mm × 1100 mm high	
Single leaf door, swing shown to opening side	
Glazed opening — arched head where noted	
Element above the cut plane (pergola, slab over)	
Void / open to sky	
+0.00	Finished floor level datum

THE FRAME

GRID	3 × 4, A–C / 1–4
SPAN A–B	4.81 m
SPAN B–C	4.39 m
SPAN 1–2	4.81 m
SPAN 2–3	4.10 m
SPAN 3–4	4.29 m
COLUMNS	12
COLUMN SECTION	300 × 300 mm
STRUCTURAL ZONE	300 mm
STOREY HEIGHT	3.30 m
PADS	12
PAD SIZE	by the engineer
FOUNDING DEPTH	by GEO-001

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. A pad is drawn under every column at the coordinate the column stands on. The square shown is a nominal 1.20 m so the layout reads; it is not a size and must not be scaled.

4. Pad sizes, founding depth, blinding and reinforcement follow the allowable bearing pressure, the water table and the sulphate content, which come from GEO-001 and do not exist yet.

5. A raft or piles may prove more appropriate than pads. That is the engineer's call once the ground is known; the coordinates on this sheet hold either way.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

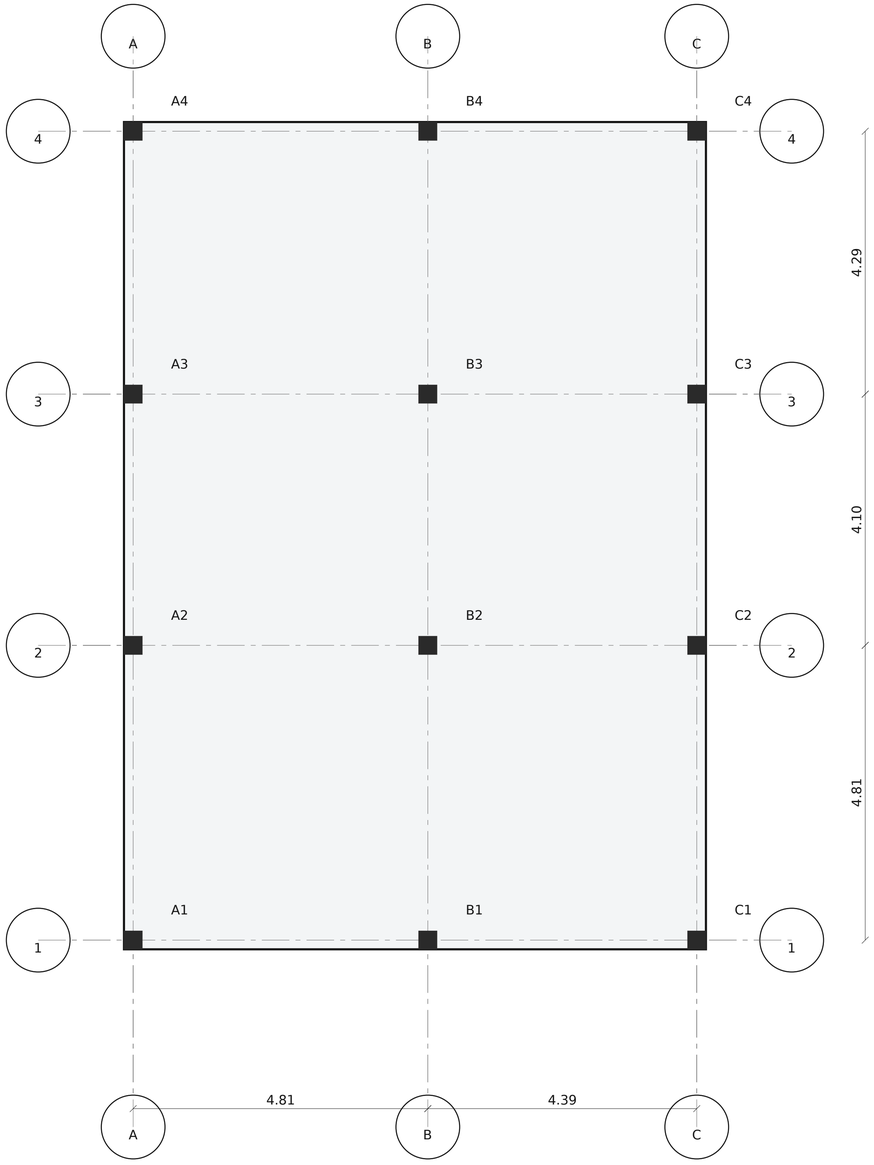
FOUNDATION LAYOUT

DRAWING No	REVISION	SCALE	DATE
STR-101	**P21**	**1:50 @ A1**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



COLUMN SCHEDULE & GRID

LEVEL all levels · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- +0.00 Finished floor level datum

THE FRAME

GRID	3 × 4, A–C / 1–4
SPAN A–B	4.81 m
SPAN B–C	4.39 m
SPAN 1–2	4.81 m
SPAN 2–3	4.10 m
SPAN 3–4	4.29 m
COLUMNS	12
COLUMN SECTION	300 × 300 mm
STRUCTURAL ZONE	300 mm
STOREY HEIGHT	3.30 m
COLUMNS TO +9.90	8
COLUMNS TO +13.20	4
SECTION	by the engineer

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. Every column is marked by its grid intersection and stands at the coordinate printed beside it. Eight run from the foundation to +13.20; four — B2, B3, C2 and C3 — carry on to the roof over the penthouse.

4. Sections and reinforcement are the engineer's. The marks and coordinates are what this sheet issues.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

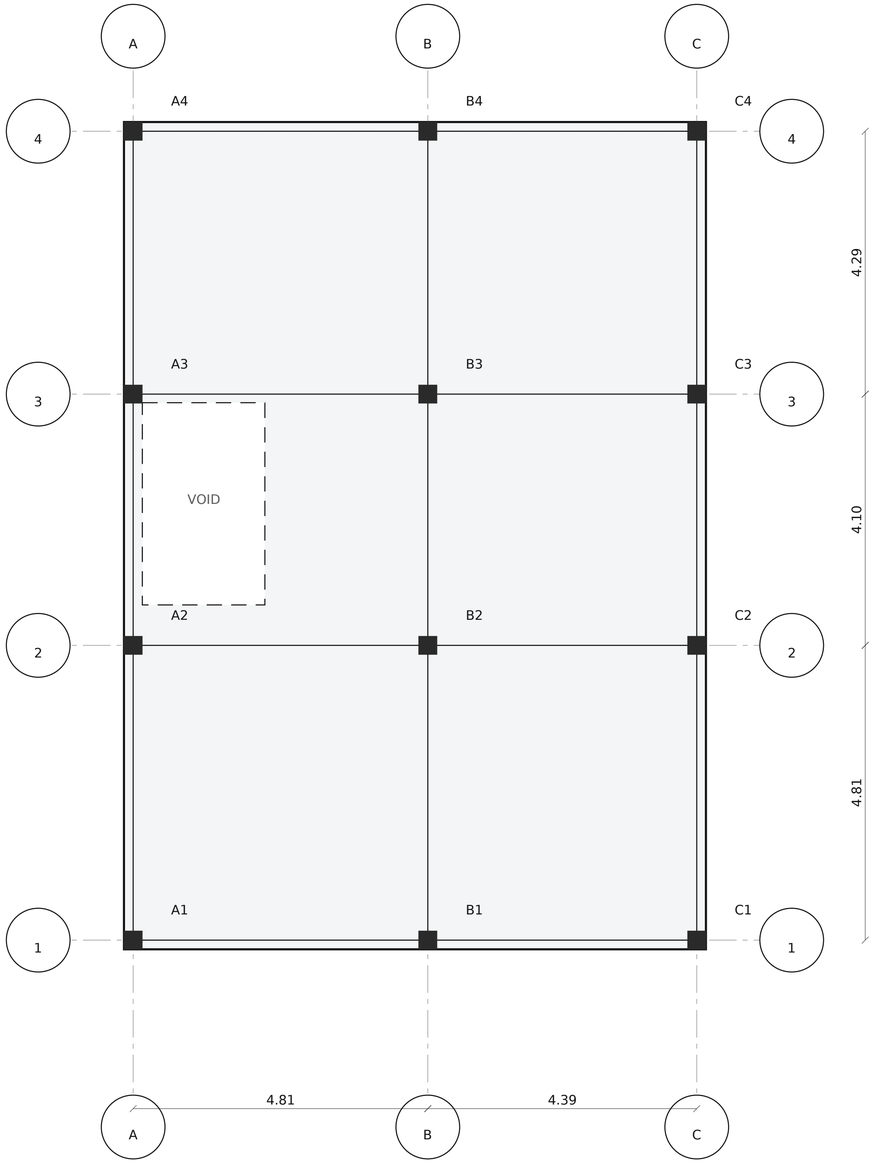
COLUMN SCHEDULE & GRID

DRAWING No	REVISION	SCALE	DATE
STR-200	**P21**	**1:50 @ A1**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SLAB FRAMING LAYOUT AT +0.00

LEVEL +0.00 · 1:50 @ A1 · METRES

KEY

External / party wall cut by section, 300 mm — brick, colour unrestricted	
Internal partition, 100 mm — red brick	
Terrace parapet, 250 mm × 1100 mm high	
Single leaf door, swing shown to opening side	
Glazed opening — arched head where noted	
Element above the cut plane (pergola, slab over)	
Void / open to sky	
+0.00	Finished floor level datum

THE FRAME

GRID	3 × 4, A–C / 1–4
SPAN A–B	4.81 m
SPAN B–C	4.39 m
SPAN 1–2	4.81 m
SPAN 2–3	4.10 m
SPAN 3–4	4.29 m
COLUMNS	12
COLUMN SECTION	300 × 300 mm
STRUCTURAL ZONE	300 mm
STOREY HEIGHT	3.30 m
THIS SLAB	+0.00
SLAB THICKNESS	by the engineer

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. Primary beams run on the grid, because every 300 mm wall stands on a grid line. The 100 mm partitions do not and do not need to: they are carried by the slab.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

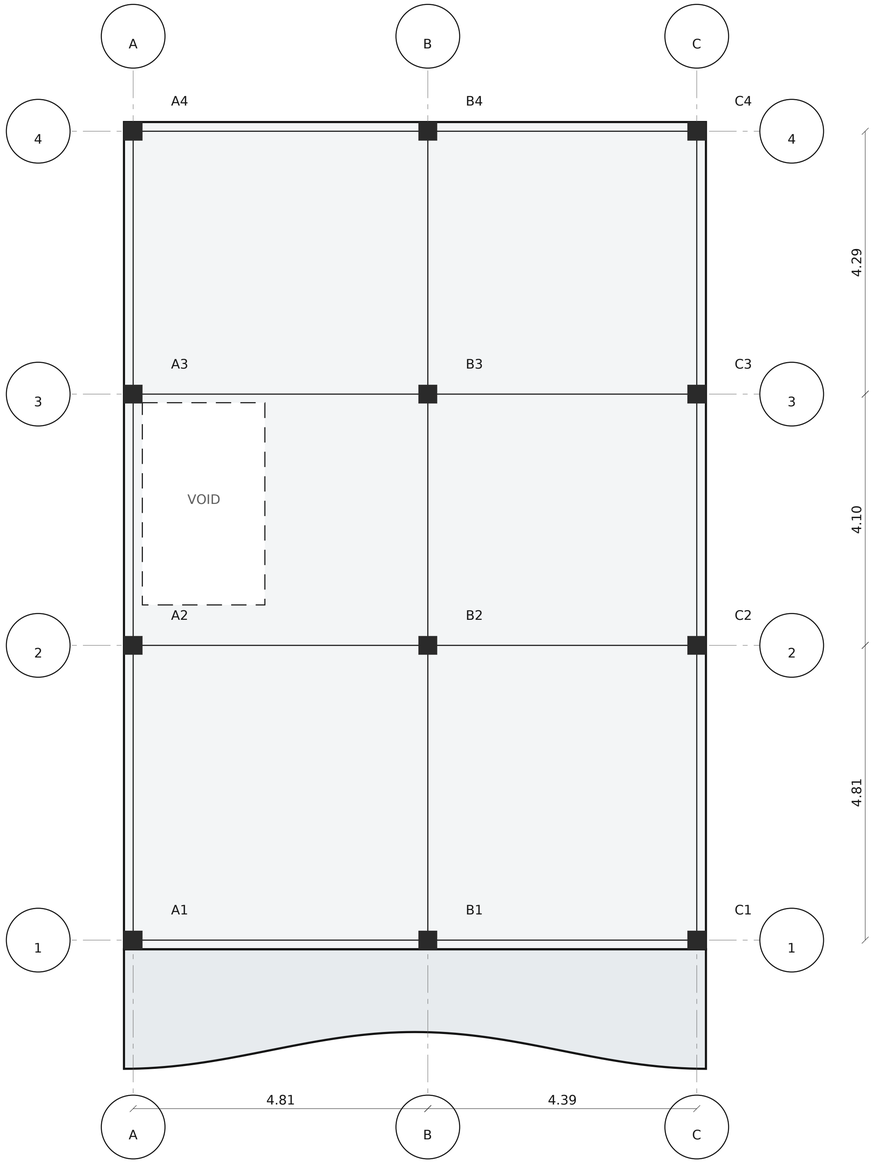
SLAB FRAMING LAYOUT AT +0.00

DRAWING No	REVISION	SCALE	DATE
STR-301	**P21**	**1:50 @ A1**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SLAB FRAMING LAYOUT AT +3.30

LEVEL +3.30 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE FRAME

GRID	3 × 4, A–C / 1–4
SPAN A–B	4.81 m
SPAN B–C	4.39 m
SPAN 1–2	4.81 m
SPAN 2–3	4.10 m
SPAN 3–4	4.29 m
COLUMNS	12
COLUMN SECTION	300 × 300 mm
STRUCTURAL ZONE	300 mm
STOREY HEIGHT	3.30 m
THIS SLAB	+3.30
SLAB THICKNESS	by the engineer

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. The cantilever beyond grid 1 is the balcony: 15.7 m² a floor, 1.95 m at its deepest. It is the most engineering-sensitive element here and has its own sheet, STR-500.

4. Primary beams run on the grid, because every 300 mm wall stands on a grid line. The 100 mm partitions do not and do not need to: they are carried by the slab.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

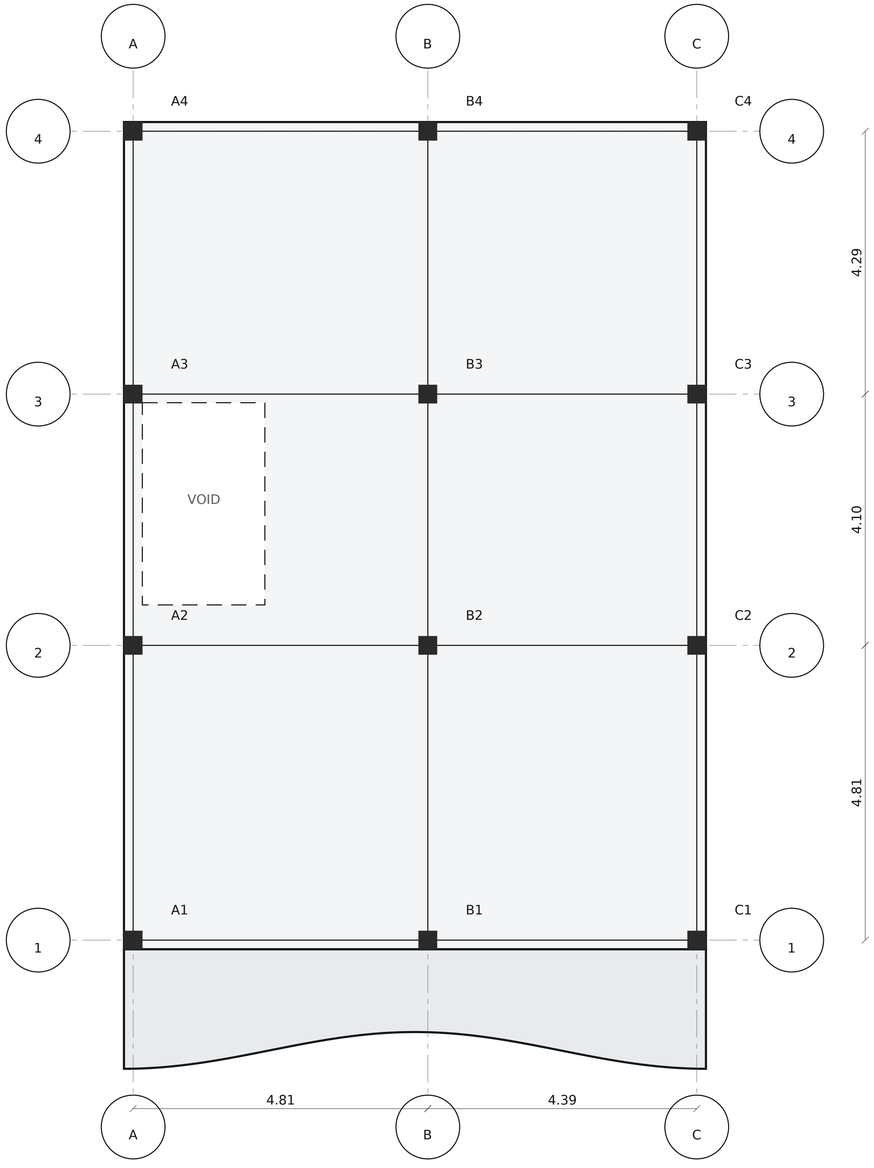
SLAB FRAMING LAYOUT AT +3.30

DRAWING No	REVISION	SCALE	DATE
STR-302	**P21**	**1:50 @ A1**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SLAB FRAMING LAYOUT AT +6.60

LEVEL +6.60 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE FRAME

GRID	3 × 4, A–C / 1–4
SPAN A–B	4.81 m
SPAN B–C	4.39 m
SPAN 1–2	4.81 m
SPAN 2–3	4.10 m
SPAN 3–4	4.29 m
COLUMNS	12
COLUMN SECTION	300 × 300 mm
STRUCTURAL ZONE	300 mm
STOREY HEIGHT	3.30 m
THIS SLAB	+6.60
SLAB THICKNESS	by the engineer

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. The cantilever beyond grid 1 is the balcony: 15.7 m² a floor, 1.95 m at its deepest. It is the most engineering-sensitive element here and has its own sheet, STR-500.

4. Primary beams run on the grid, because every 300 mm wall stands on a grid line. The 100 mm partitions do not and do not need to: they are carried by the slab.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

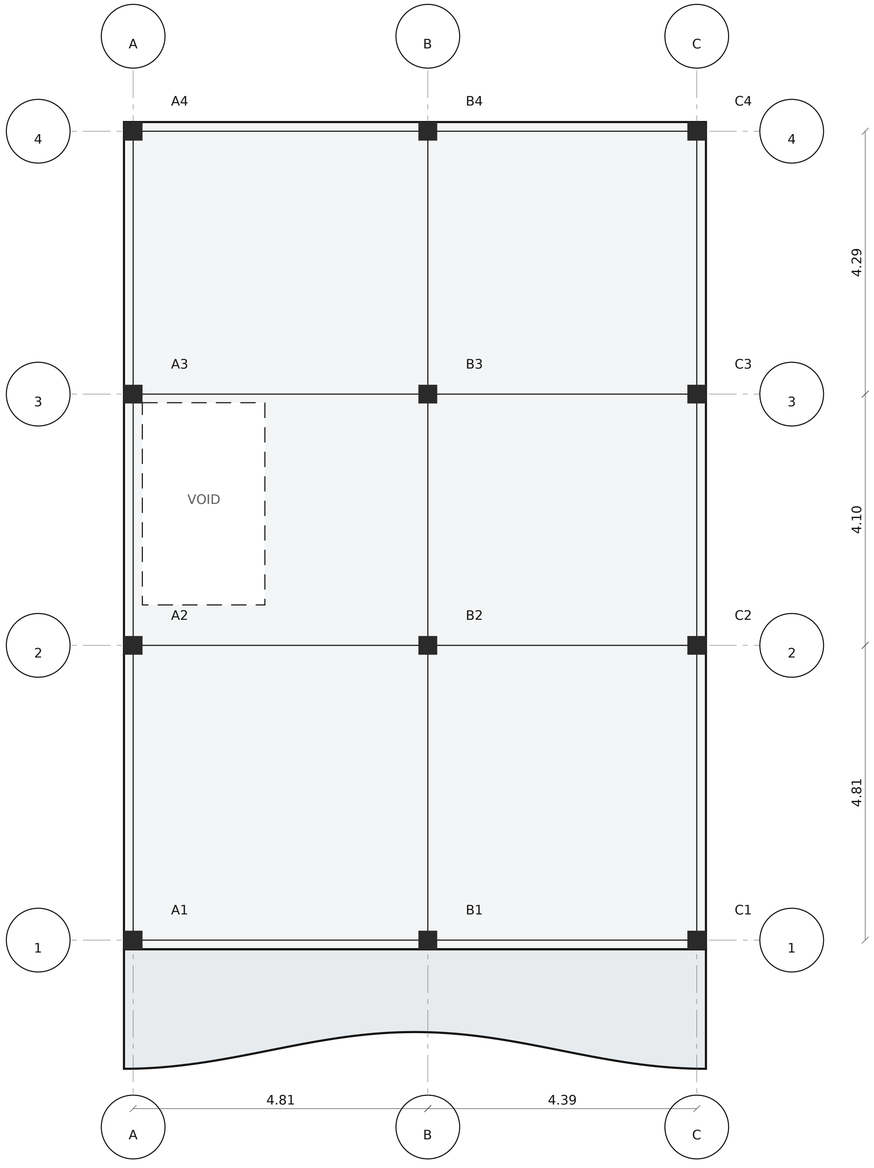
SLAB FRAMING LAYOUT AT +6.60

DRAWING No	REVISION	SCALE	DATE
STR-303	**P21**	**1:50 @ A1**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SLAB FRAMING LAYOUT AT +9.90

LEVEL +9.90 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE FRAME

GRID	3 × 4, A–C / 1–4
SPAN A–B	4.81 m
SPAN B–C	4.39 m
SPAN 1–2	4.81 m
SPAN 2–3	4.10 m
SPAN 3–4	4.29 m
COLUMNS	12
COLUMN SECTION	300 × 300 mm
STRUCTURAL ZONE	300 mm
STOREY HEIGHT	3.30 m
THIS SLAB	+9.90
SLAB THICKNESS	by the engineer

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. The cantilever beyond grid 1 is the balcony: 15.7 m² a floor, 1.95 m at its deepest. It is the most engineering-sensitive element here and has its own sheet, STR-500.

4. Primary beams run on the grid, because every 300 mm wall stands on a grid line. The 100 mm partitions do not and do not need to: they are carried by the slab.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

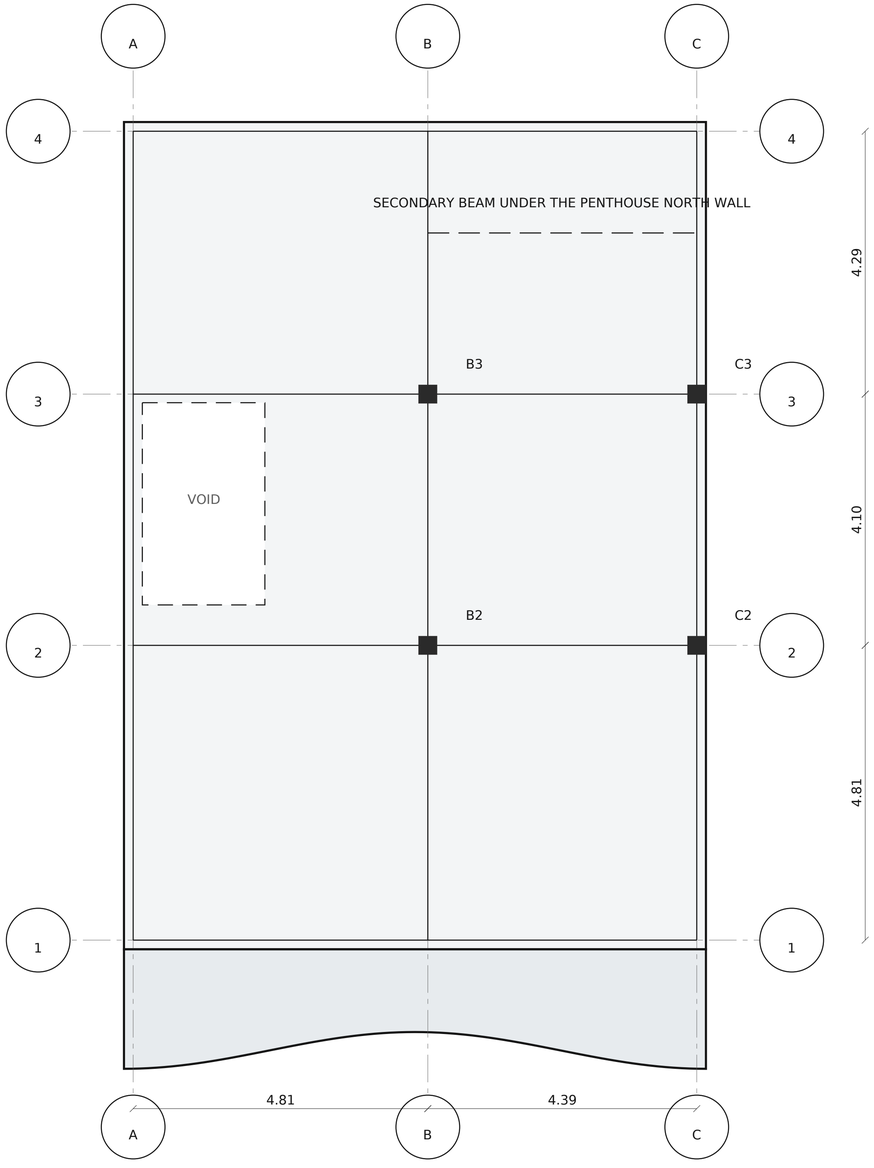
SLAB FRAMING LAYOUT AT +9.90

DRAWING No	REVISION	SCALE	DATE
STR-304	**P21**	**1:50 @ A1**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SLAB FRAMING LAYOUT AT +13.20

LEVEL +13.20 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- +0.00 Finished floor level datum

THE FRAME

GRID	3 × 4, A–C / 1–4
SPAN A–B	4.81 m
SPAN B–C	4.39 m
SPAN 1–2	4.81 m
SPAN 2–3	4.10 m
SPAN 3–4	4.29 m
COLUMNS	12
COLUMN SECTION	300 × 300 mm
STRUCTURAL ZONE	300 mm
STOREY HEIGHT	3.30 m
THIS SLAB	+13.20
SLAB THICKNESS	by the engineer

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. This slab carries the one wall in the building that does not stand on a grid line: the penthouse's north wall, a line load spanning B to C. The secondary beam under it is drawn dashed; its size is the engineer's.

4. The cantilever beyond grid 1 is the balcony: 15.7 m² a floor, 1.95 m at its deepest. It is the most engineering-sensitive element here and has its own sheet, STR-500.

5. Primary beams run on the grid, because every 300 mm wall stands on a grid line. The 100 mm partitions do not and do not need to: they are carried by the slab.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

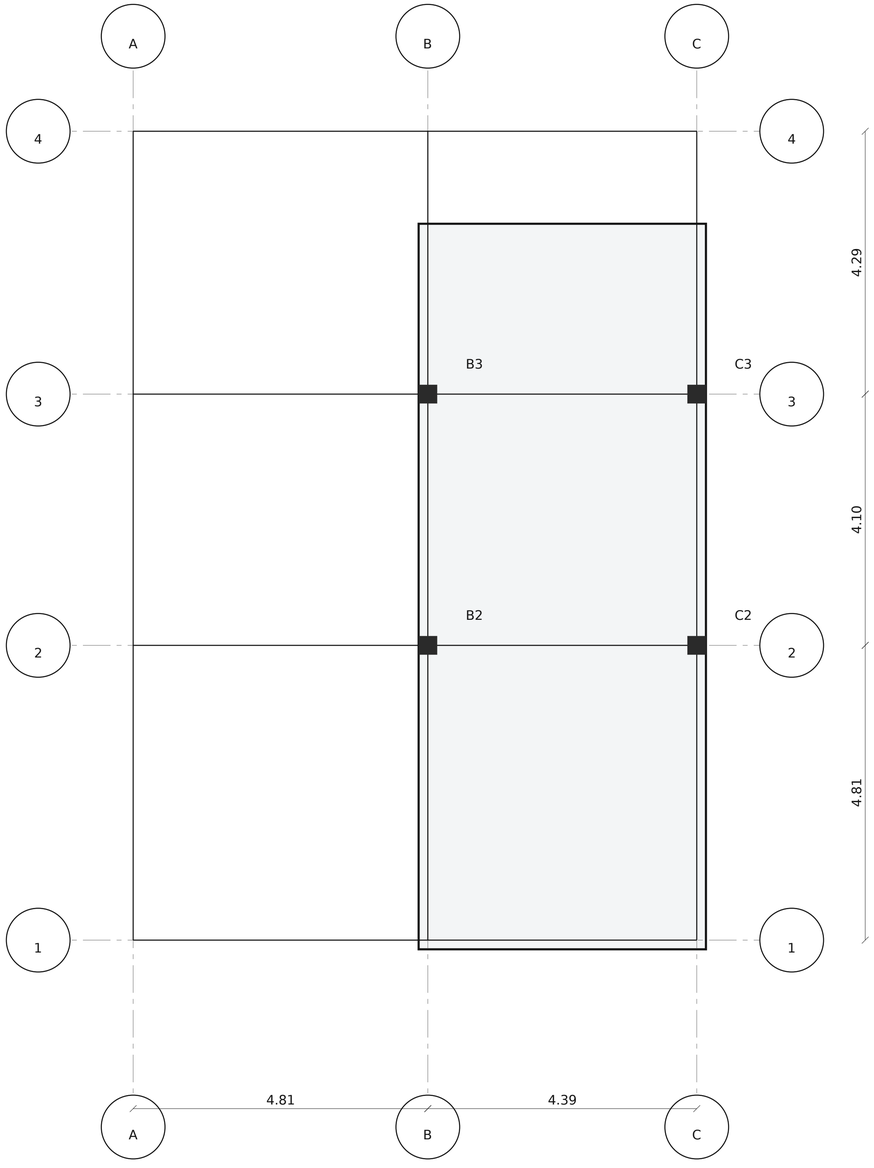
SLAB FRAMING LAYOUT AT +13.20

DRAWING No	REVISION	SCALE	DATE
STR-305	**P21**	**1:50 @ A1**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



SLAB FRAMING LAYOUT AT +16.45

LEVEL +16.45 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- +0.00 Finished floor level datum

THE FRAME

GRID	3 × 4, A–C / 1–4
SPAN A–B	4.81 m
SPAN B–C	4.39 m
SPAN 1–2	4.81 m
SPAN 2–3	4.10 m
SPAN 3–4	4.29 m
COLUMNS	12
COLUMN SECTION	300 × 300 mm
STRUCTURAL ZONE	300 mm
STOREY HEIGHT	3.30 m
THIS SLAB	+16.45
SLAB THICKNESS	by the engineer

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. Primary beams run on the grid, because every 300 mm wall stands on a grid line. The 100 mm partitions do not and do not need to: they are carried by the slab.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

SLAB FRAMING LAYOUT AT +16.45

DRAWING No	REVISION	SCALE	DATE
STR-306	**P21**	**1:50 @ A1**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



RETAINING & BOUNDARY WALLS

LEVEL -0.50 to +0.00 · 1:75 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE FRAME

STREET WALL	9.10 m
STREET WALL HEIGHT	1.20 m, indicative
GATE	2.40 m
GARDEN LEVEL	-0.50
VILLA LEVEL	+0.00
GROUND RETAINED	0.50 m
WALL DESIGN	by the engineer
BOUNDARY LINE	by SUR-001

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. The garden sits 0.50 m below the villa, so the change in level is retained somewhere along the west face and around the forecourt. Where is drawn; how is not.

4. The street wall's height is the client's choice and indicative at 1.20 m; its line awaits the measured survey. Nothing else on this sheet depends on either.

5. Retaining design, drainage behind the wall, movement joints and foundations are the engineer's and wait on GEO-001.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

RETAINING & BOUNDARY WALLS

DRAWING No	REVISION	SCALE	DATE
STR-400	**P21**	**1:75 @ A1**	**2026-10-07**

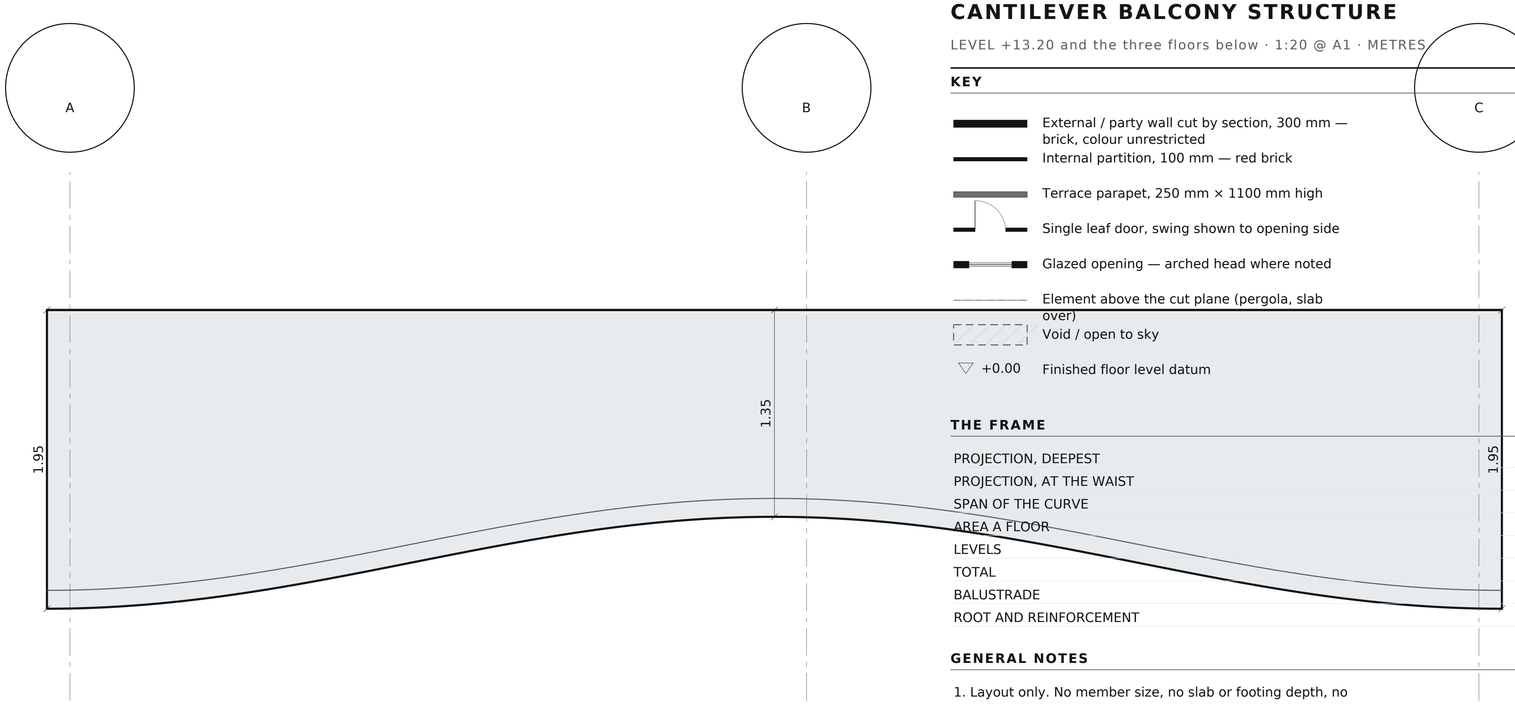
Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

CANTILEVER BALCONY STRUCTURE

LEVEL +13.20 and the three floors below · 1:20 @ A1 · METRES



SLAB EDGE — THICKNESS, ROOT DEPTH AND REINFORCEMENT BY THE ENGINEER

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- +0.00 Finished floor level datum

THE FRAME

PROJECTION, DEEPEST	1.95 m
PROJECTION, AT THE WAIST	1.35 m
SPAN OF THE CURVE	9.50 m
AREA A FLOOR	15.7 m²
LEVELS	4
TOTAL	62.7 m²
BALUSTRADE	glass, 120 mm inboard
ROOT AND REINFORCEMENT	by the engineer

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. A 1.95 m cantilever on a curve, repeated on four levels, is the single most engineering-sensitive element in this building. It is drawn at 1:20 for that reason and for no other.

4. The curve, the span, the projection at every point and the balustrade line are exact. The slab thickness, the root depth into the floor behind, the top steel and the deflection and vibration checks are entirely the engineer's.

5. Grid line 1 runs along the face the cantilever springs from. The balcony projects south of it and is unsupported at its far edge on all four levels.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

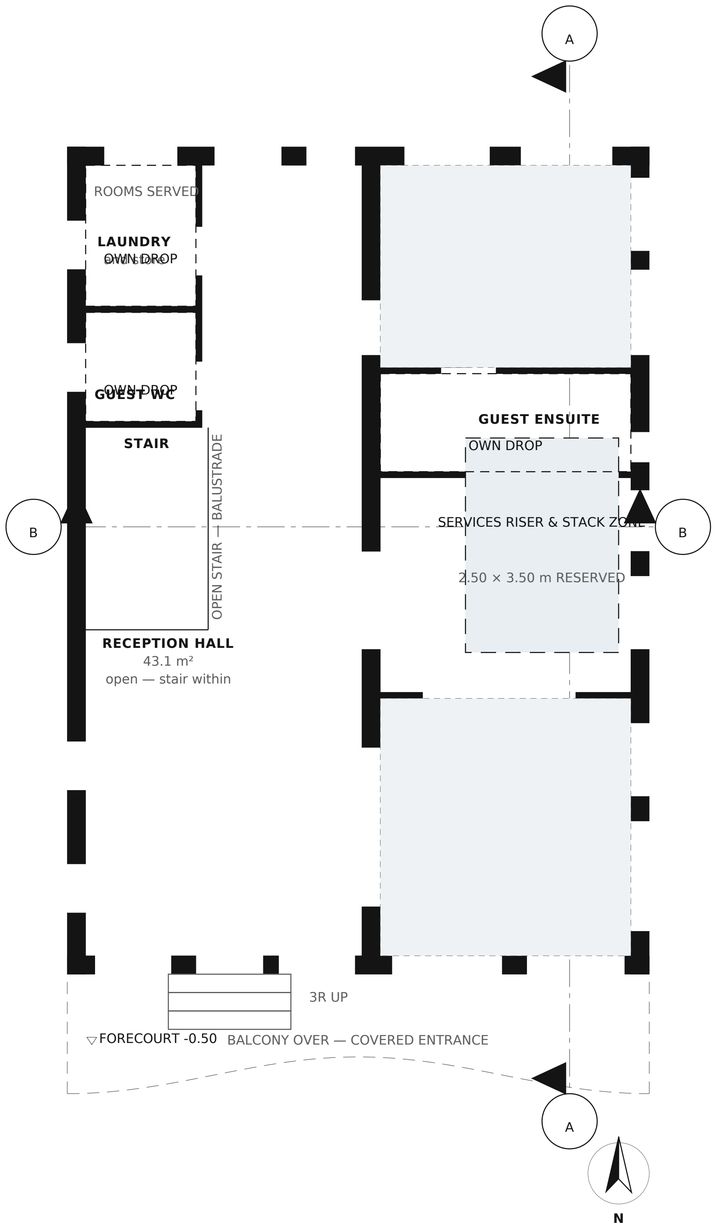
CANTILEVER BALCONY STRUCTURE

DRAWING No	REVISION	SCALE	DATE
STR-500	**P21**	**1:20 @ A1**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

POWER & SMALL POWER LAYOUT — GROUND FLOOR

LEVEL +0.00 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+0.00
DISCIPLINE	Electrical
ROOMS MARKED	2
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	3
SOFFIT	+3.00
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Reserved riser and the distribution route. Outlets, circuits, board schedules and loads are the electrical engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-100 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

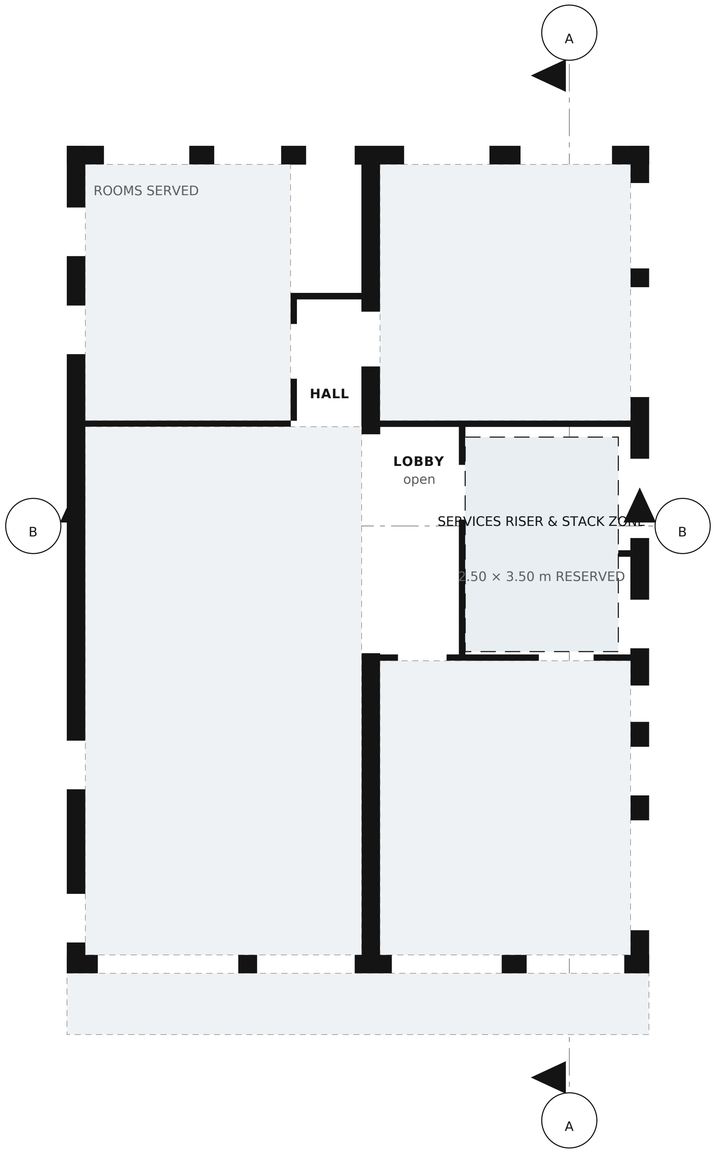
POWER & SMALL POWER LAYOUT — GROUND FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-100	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

POWER & SMALL POWER LAYOUT — FIRST FLOOR

LEVEL +3.30 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+3.30
DISCIPLINE	Electrical
ROOMS MARKED	5
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+6.30
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Reserved riser and the distribution route. Outlets, circuits, board schedules and loads are the electrical engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-101 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

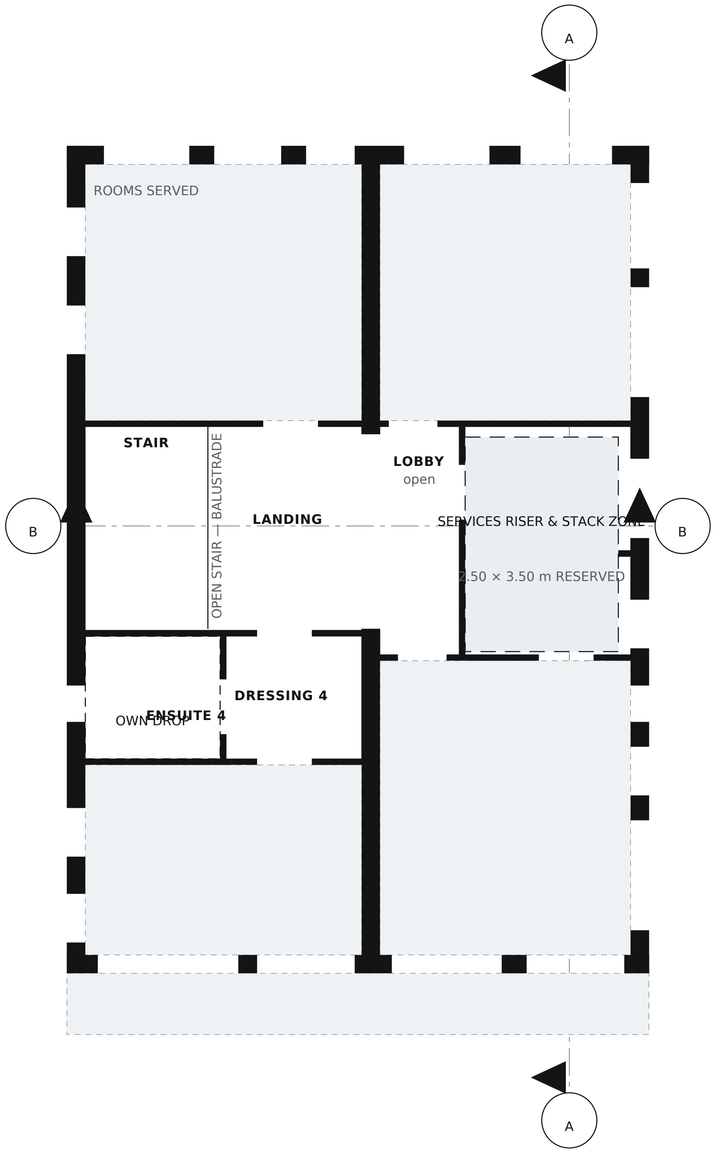
POWER & SMALL POWER LAYOUT — FIRST FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-101	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

POWER & SMALL POWER LAYOUT — SECOND FLOOR

LEVEL +6.60 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+6.60
DISCIPLINE	Electrical
ROOMS MARKED	5
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+9.60
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Reserved riser and the distribution route. Outlets, circuits, board schedules and loads are the electrical engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-102 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

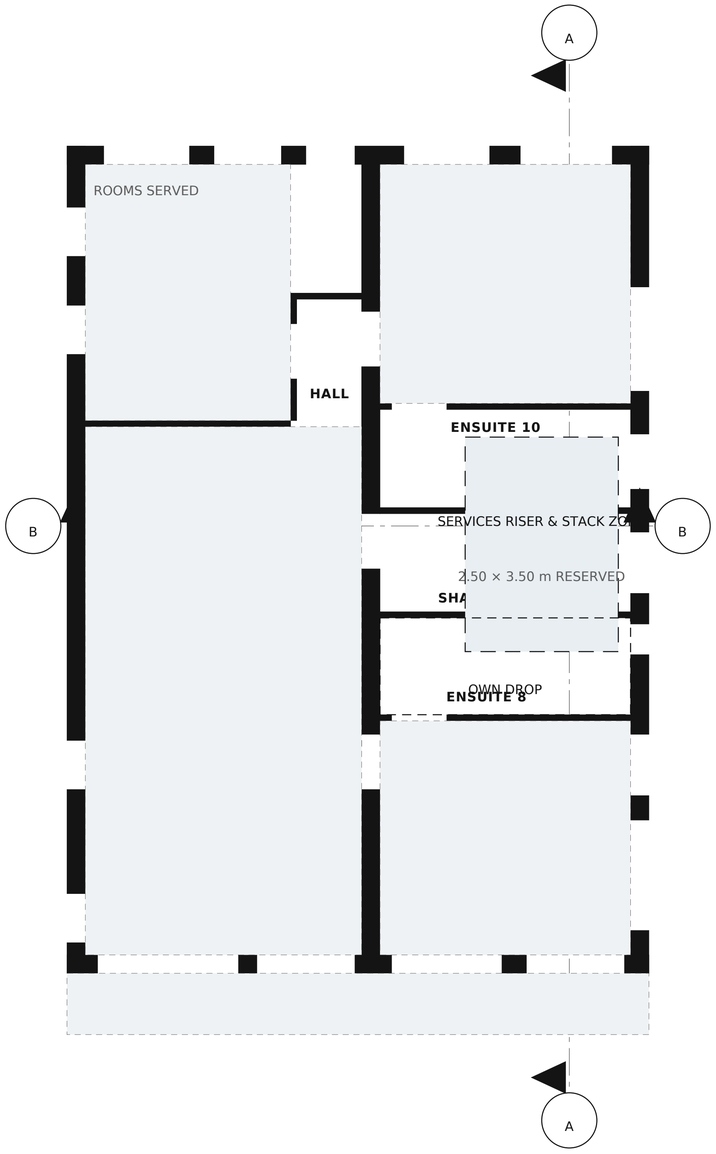
POWER & SMALL POWER LAYOUT — SECOND FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-102	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



N

GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

POWER & SMALL POWER LAYOUT — THIRD FLOOR

LEVEL +9.90 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+9.90
DISCIPLINE	Electrical
ROOMS MARKED	5
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+12.90
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Reserved riser and the distribution route. Outlets, circuits, board schedules and loads are the electrical engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-103 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

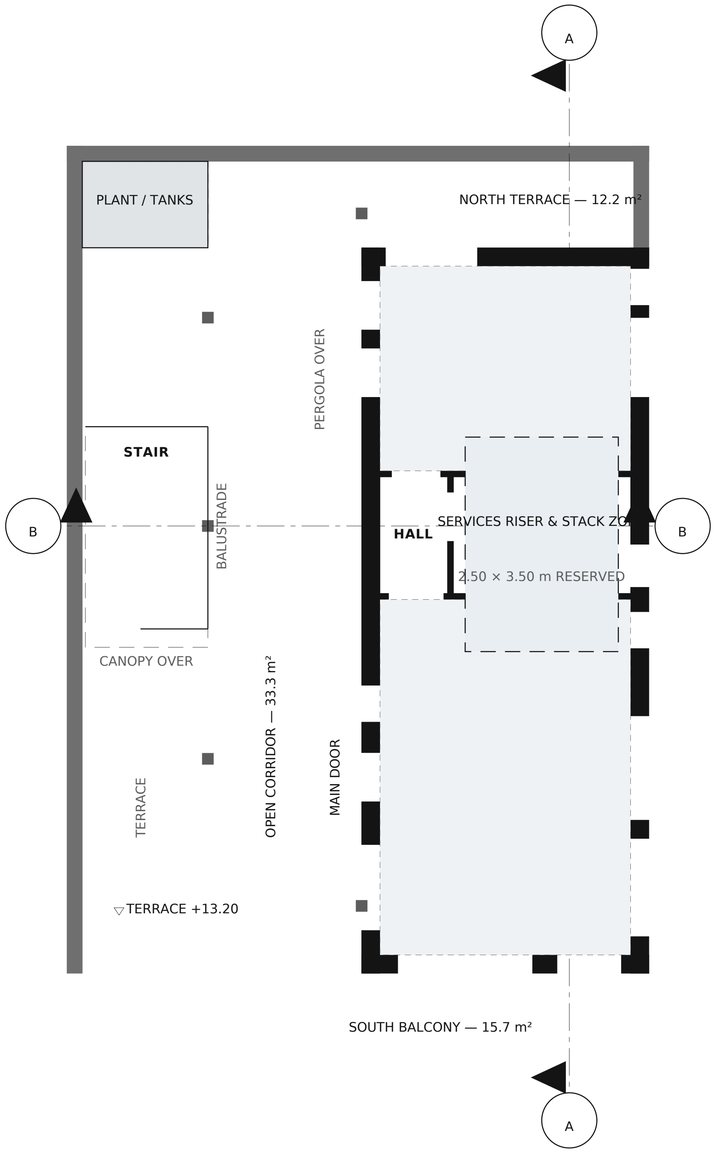
POWER & SMALL POWER LAYOUT — THIRD FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-103	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

POWER & SMALL POWER LAYOUT — PENTHOUSE FLOOR

LEVEL +13.20 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+13.20
DISCIPLINE	Electrical
ROOMS MARKED	2
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+16.20
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Reserved riser and the distribution route. Outlets, circuits, board schedules and loads are the electrical engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-104 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

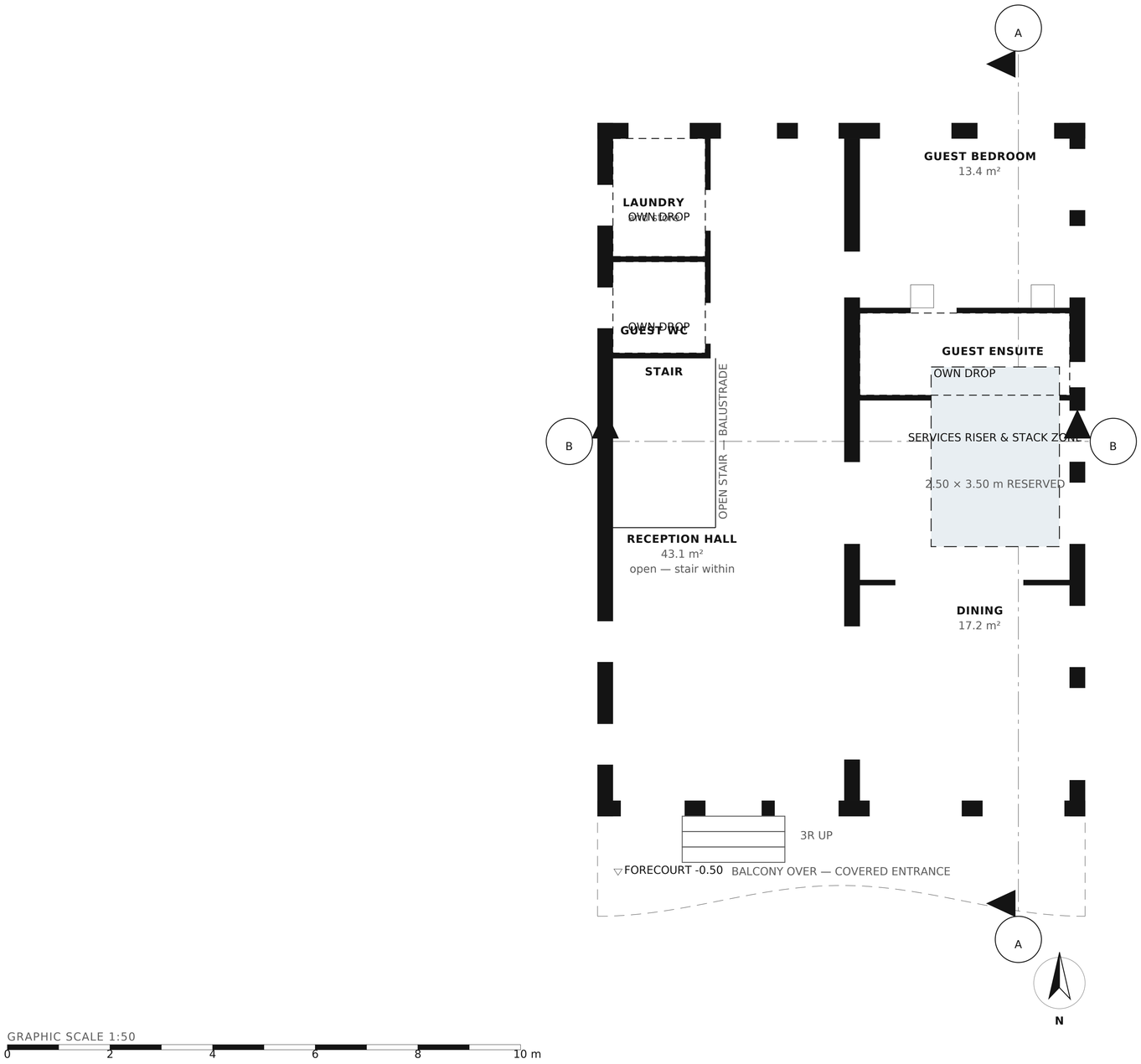
POWER & SMALL POWER LAYOUT — PENTHOUSE FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-104	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



LIGHTING & SWITCHING LAYOUT — GROUND FLOOR

LEVEL +0.00 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+0.00
DISCIPLINE	Electrical
ROOMS MARKED	8
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	3
SOFFIT	+3.00
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. The ceiling it has to work with, and the soffit level. No fitting is drawn; see the ARC-400 series for the bulkheads.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-100 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

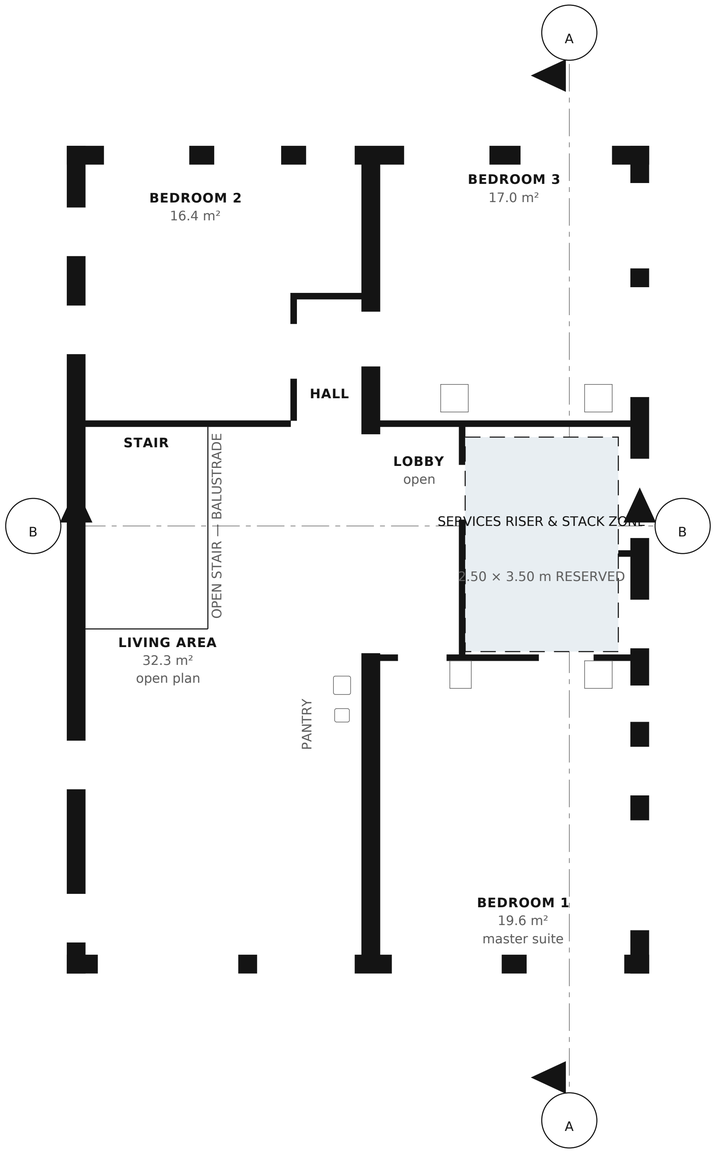
LIGHTING & SWITCHING LAYOUT — GROUND FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-200	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

LIGHTING & SWITCHING LAYOUT — FIRST FLOOR

LEVEL +3.30 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+3.30
DISCIPLINE	Electrical
ROOMS MARKED	10
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+6.30
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. The ceiling it has to work with, and the soffit level. No fitting is drawn; see the ARC-400 series for the bulkheads.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-101 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

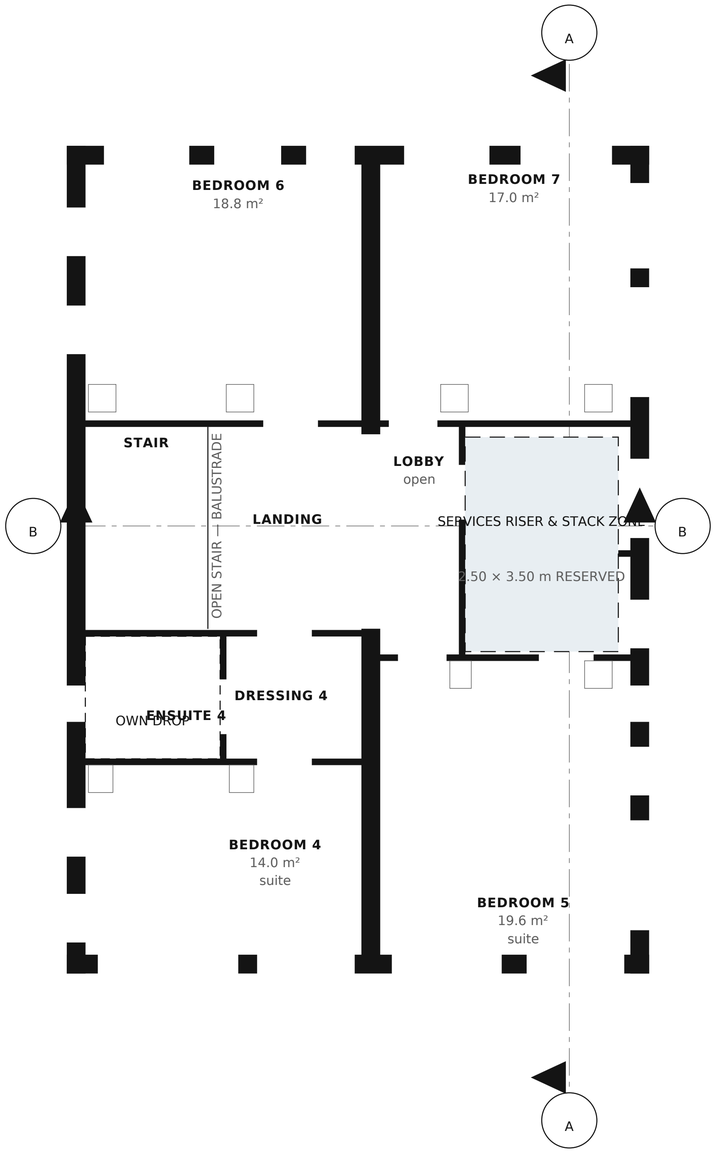
LIGHTING & SWITCHING LAYOUT — FIRST FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-201	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

LIGHTING & SWITCHING LAYOUT — SECOND FLOOR

LEVEL +6.60 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+6.60
DISCIPLINE	Electrical
ROOMS MARKED	12
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+9.60
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. The ceiling it has to work with, and the soffit level. No fitting is drawn; see the ARC-400 series for the bulkheads.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-102 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

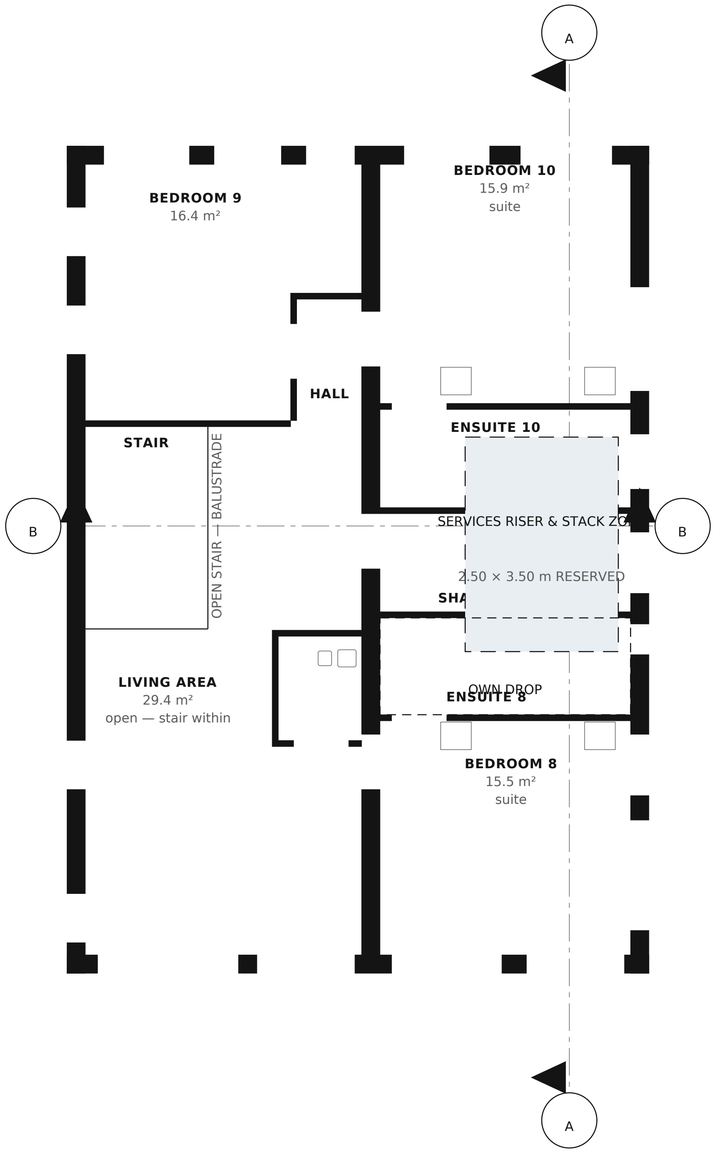
LIGHTING & SWITCHING LAYOUT — SECOND FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-202	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

LIGHTING & SWITCHING LAYOUT — THIRD FLOOR

LEVEL +9.90 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+9.90
DISCIPLINE	Electrical
ROOMS MARKED	10
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+12.90
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. The ceiling it has to work with, and the soffit level. No fitting is drawn; see the ARC-400 series for the bulkheads.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-103 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

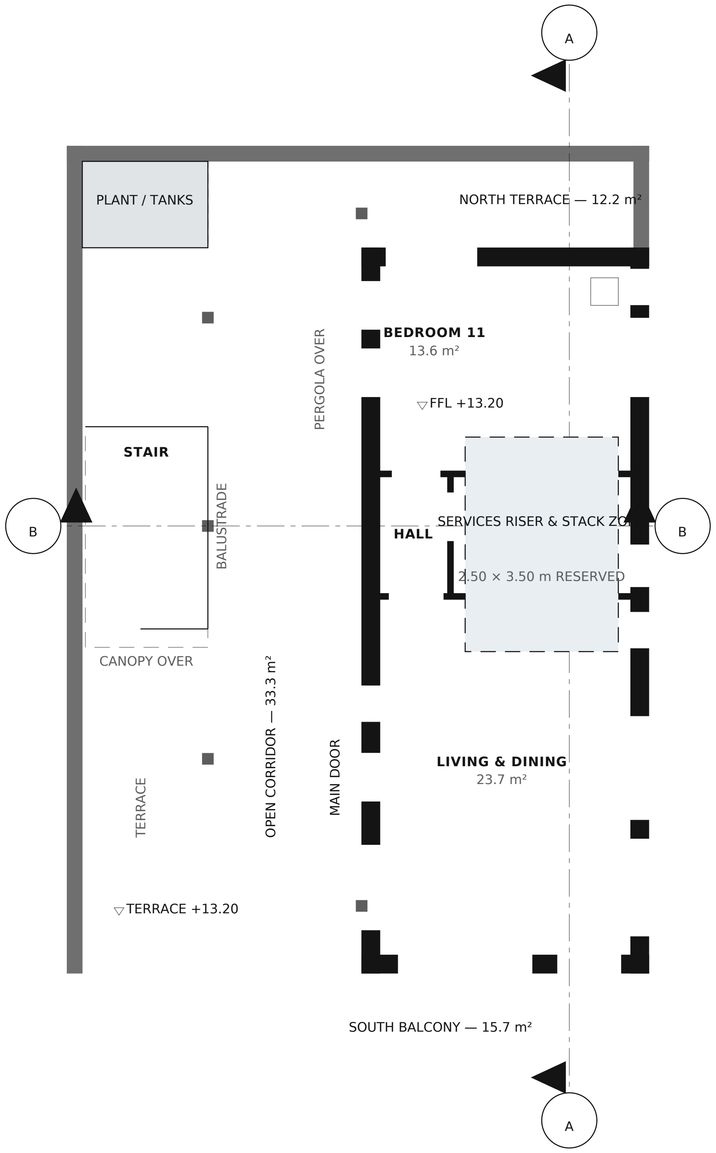
LIGHTING & SWITCHING LAYOUT — THIRD FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-203	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

LIGHTING & SWITCHING LAYOUT — PENTHOUSE FLOOR

LEVEL +13.20 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 Finished floor level datum

COORDINATION

LEVEL	+13.20
DISCIPLINE	Electrical
ROOMS MARKED	5
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+16.20
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. The ceiling it has to work with, and the soffit level. No fitting is drawn; see the ARC-400 series for the bulkheads.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-104 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

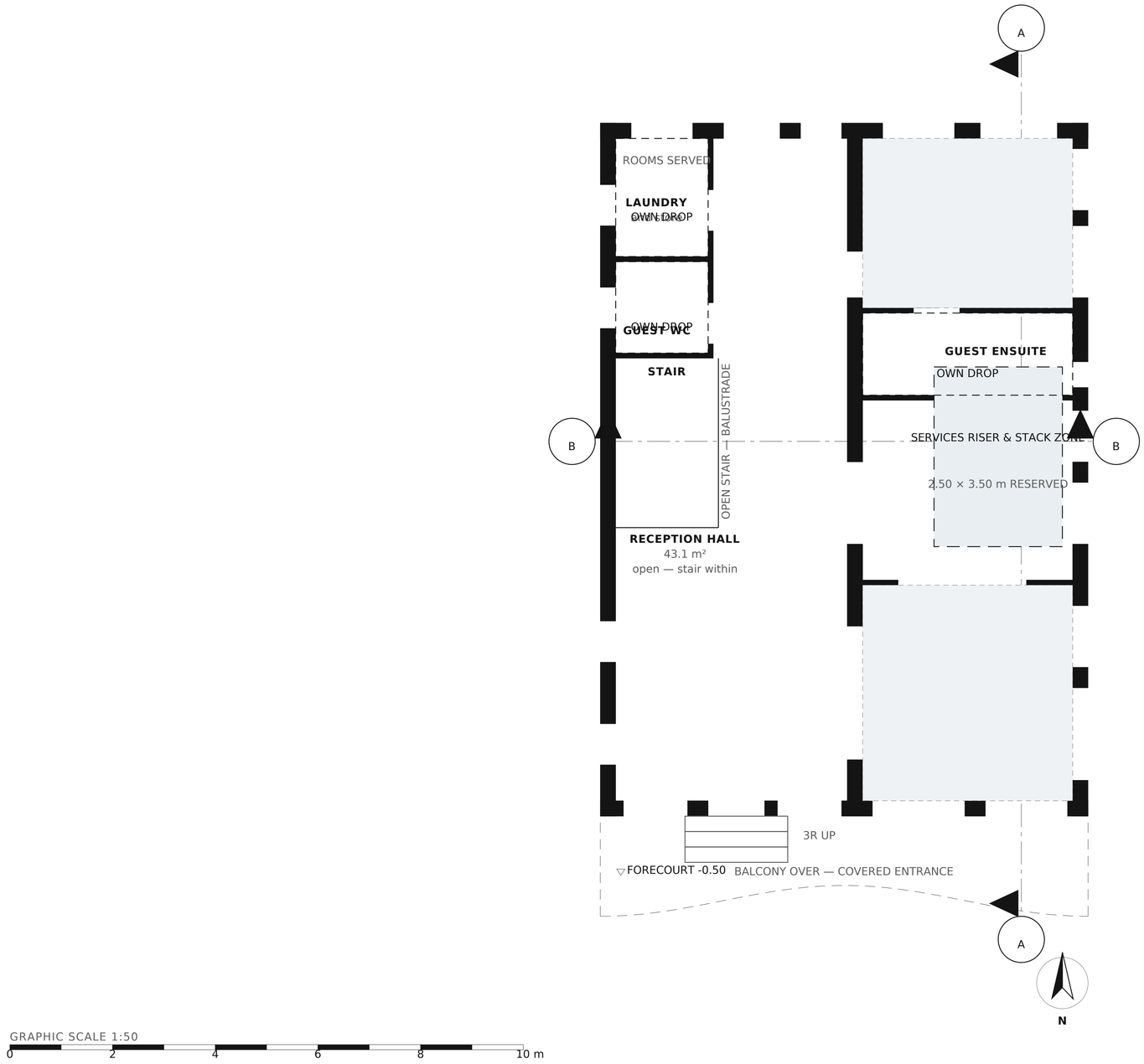
LIGHTING & SWITCHING LAYOUT — PENTHOUSE FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-204	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



LOW VOLTAGE, DATA & SECURITY LAYOUT — GROUND FLOOR

LEVEL +0.00 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+0.00
DISCIPLINE	Electrical
ROOMS MARKED	2
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	3
SOFFIT	+3.00
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Entry point and riser. Devices, containment and head-end are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-100 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

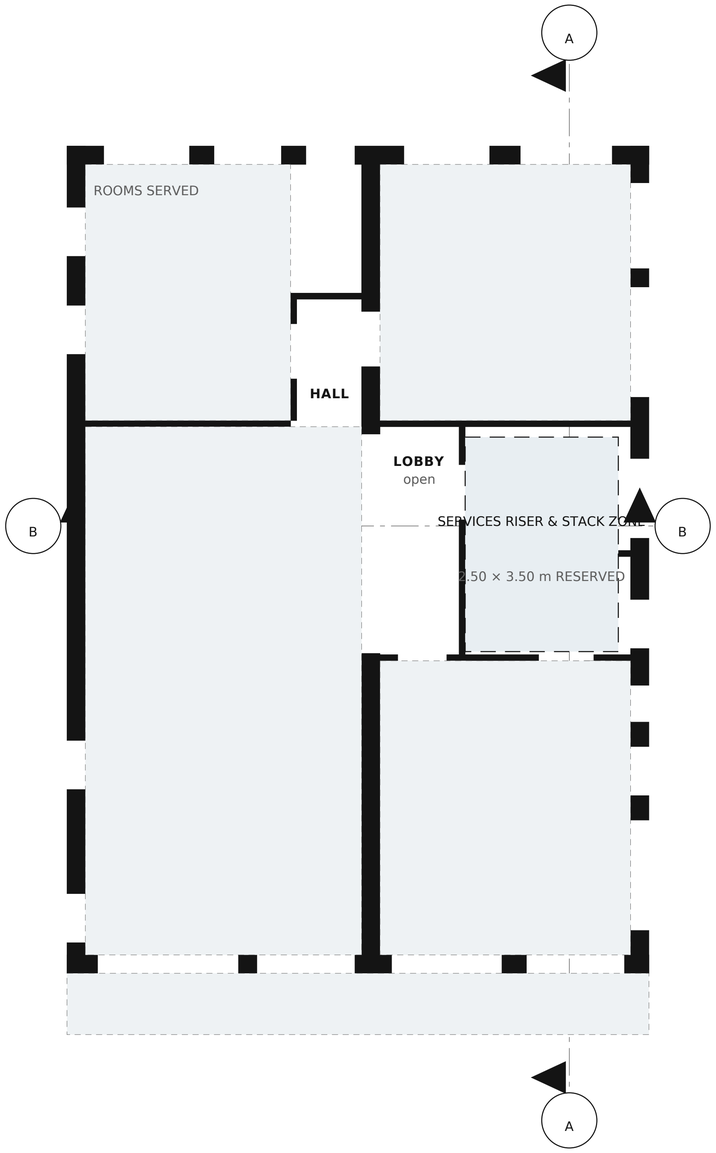
LOW VOLTAGE, DATA & SECURITY LAYOUT — GROUND FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-300	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

LOW VOLTAGE, DATA & SECURITY LAYOUT — FIRST FLOOR

LEVEL +3.30 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+3.30
DISCIPLINE	Electrical
ROOMS MARKED	5
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+6.30
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Entry point and riser. Devices, containment and head-end are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-101 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

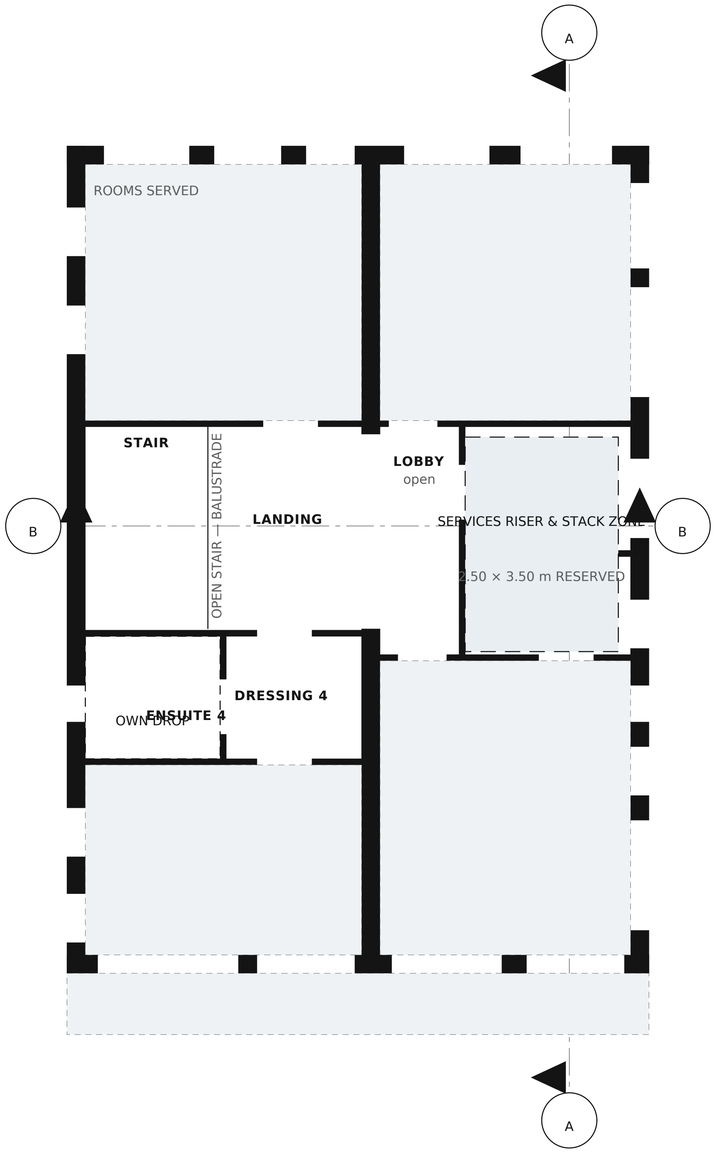
LOW VOLTAGE, DATA & SECURITY LAYOUT — FIRST FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-301	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

LOW VOLTAGE, DATA & SECURITY LAYOUT — SECOND FLOOR

LEVEL +6.60 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+6.60
DISCIPLINE	Electrical
ROOMS MARKED	5
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+9.60
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Entry point and riser. Devices, containment and head-end are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-102 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

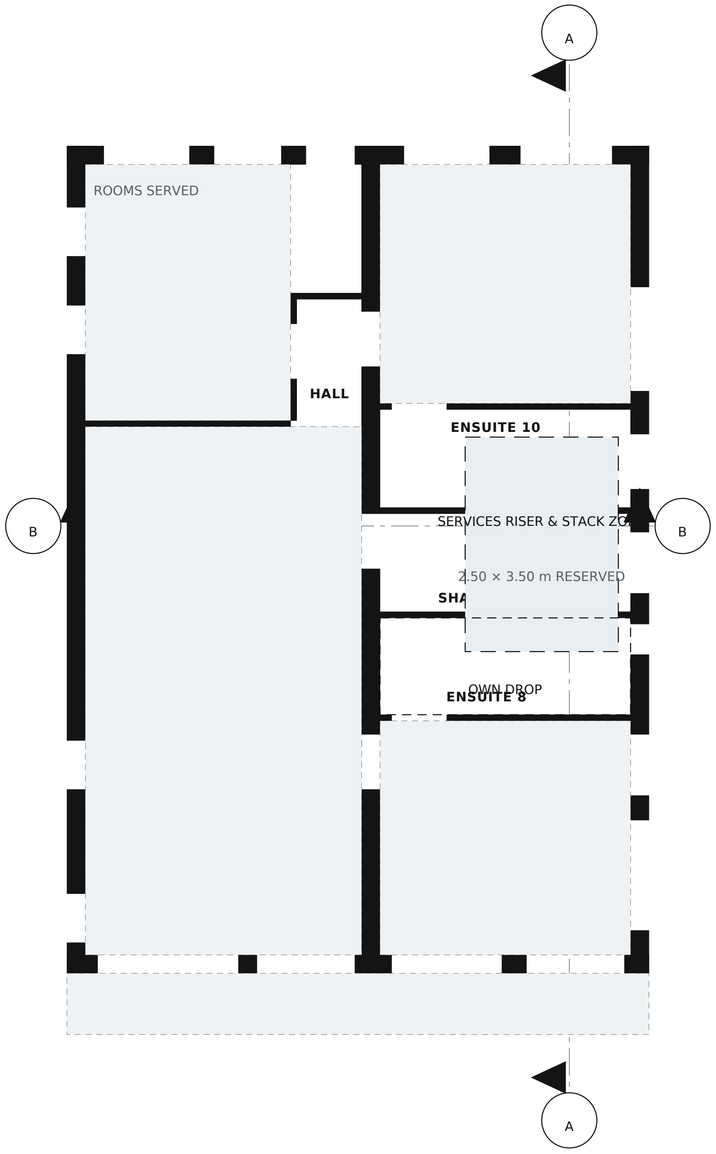
LOW VOLTAGE, DATA & SECURITY LAYOUT — SECOND FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-302	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

LOW VOLTAGE, DATA & SECURITY LAYOUT — THIRD FLOOR

LEVEL +9.90 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+9.90
DISCIPLINE	Electrical
ROOMS MARKED	5
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+12.90
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Entry point and riser. Devices, containment and head-end are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-103 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

LOW VOLTAGE, DATA & SECURITY LAYOUT — THIRD FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-303	**P21**	**1:50 @ A1**	**2026-10-07**

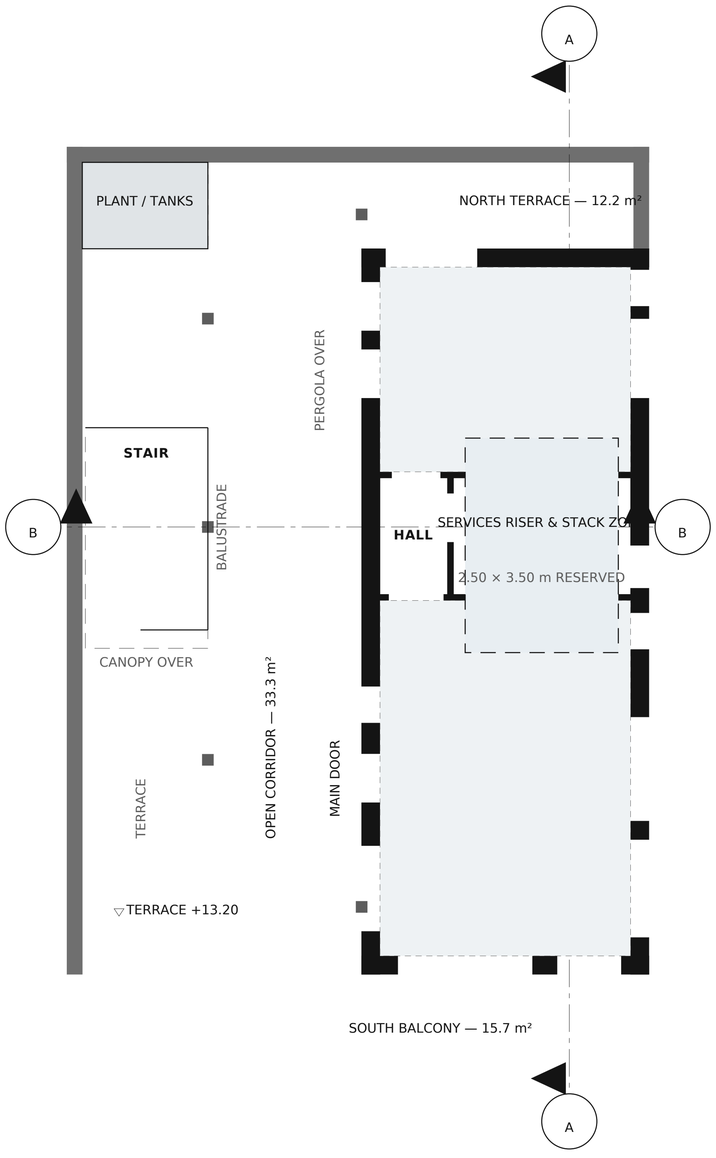
Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

LOW VOLTAGE, DATA & SECURITY LAYOUT — PENTHOUSE FLO

LEVEL +13.20 · 1:50 @ A1 · METRES



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+13.20
DISCIPLINE	Electrical
ROOMS MARKED	2
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+16.20
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Entry point and riser. Devices, containment and head-end are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-104 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

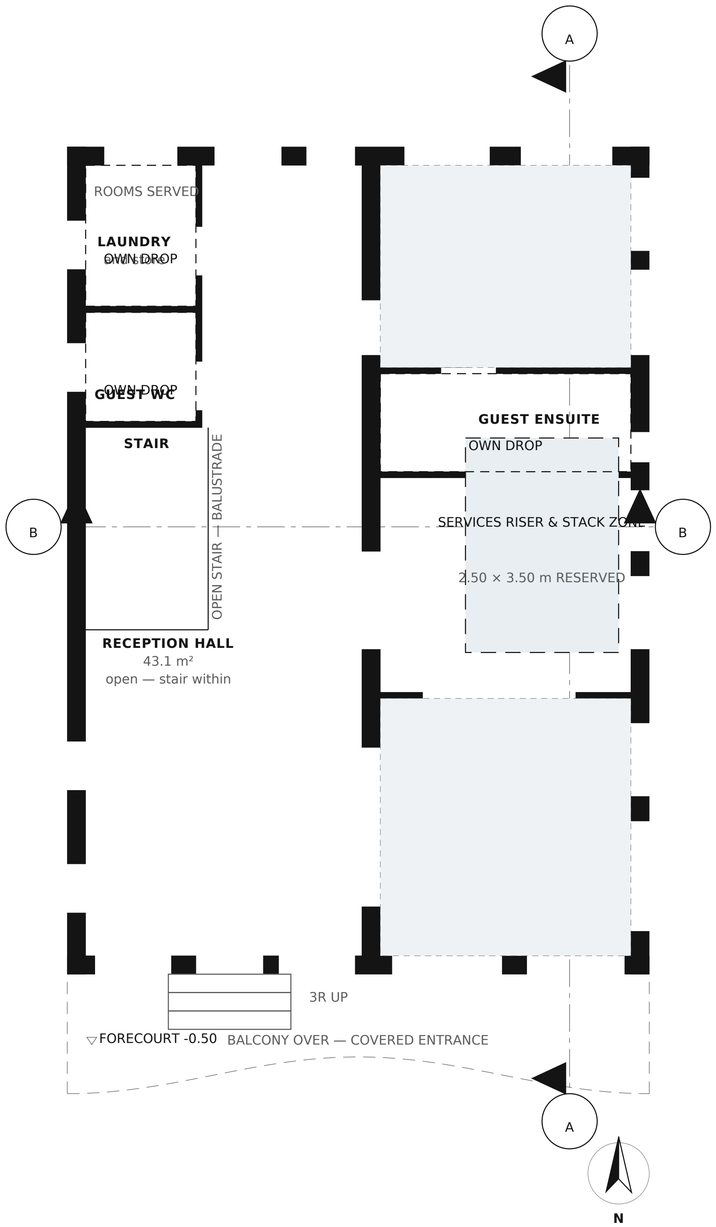
LOW VOLTAGE, DATA & SECURITY LAYOUT — PENTHOUSE FLOOR

DRAWING No	REVISION	SCALE	DATE
EL-304	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

HVAC LAYOUT — GROUND FLOOR

LEVEL +0.00 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+0.00
DISCIPLINE	Mechanical
ROOMS MARKED	2
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	3
SOFFIT	+3.00
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Which rooms are conditioned, where the plant stands and where the risers run. Capacities, duct sizes and unit selection are the mechanical engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-100 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

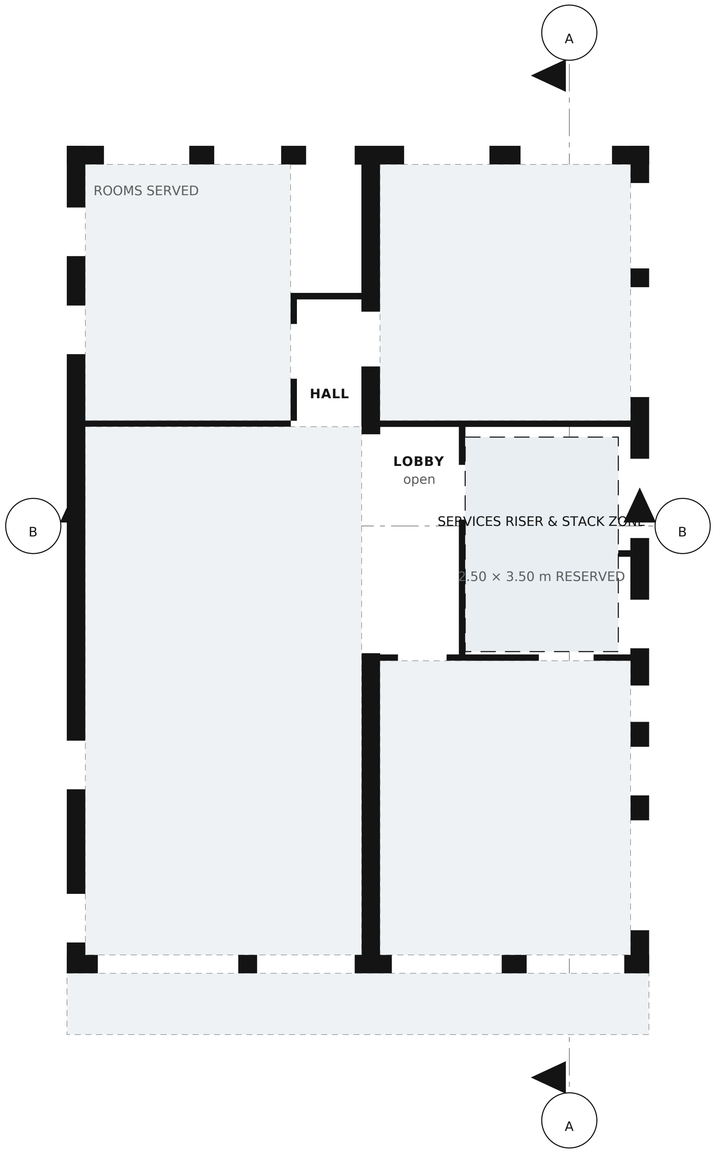
HVAC LAYOUT — GROUND FLOOR

DRAWING No	REVISION	SCALE	DATE
ME-100	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

HVAC LAYOUT — FIRST FLOOR

LEVEL +3.30 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+3.30
DISCIPLINE	Mechanical
ROOMS MARKED	5
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+6.30
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Which rooms are conditioned, where the plant stands and where the risers run. Capacities, duct sizes and unit selection are the mechanical engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-101 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

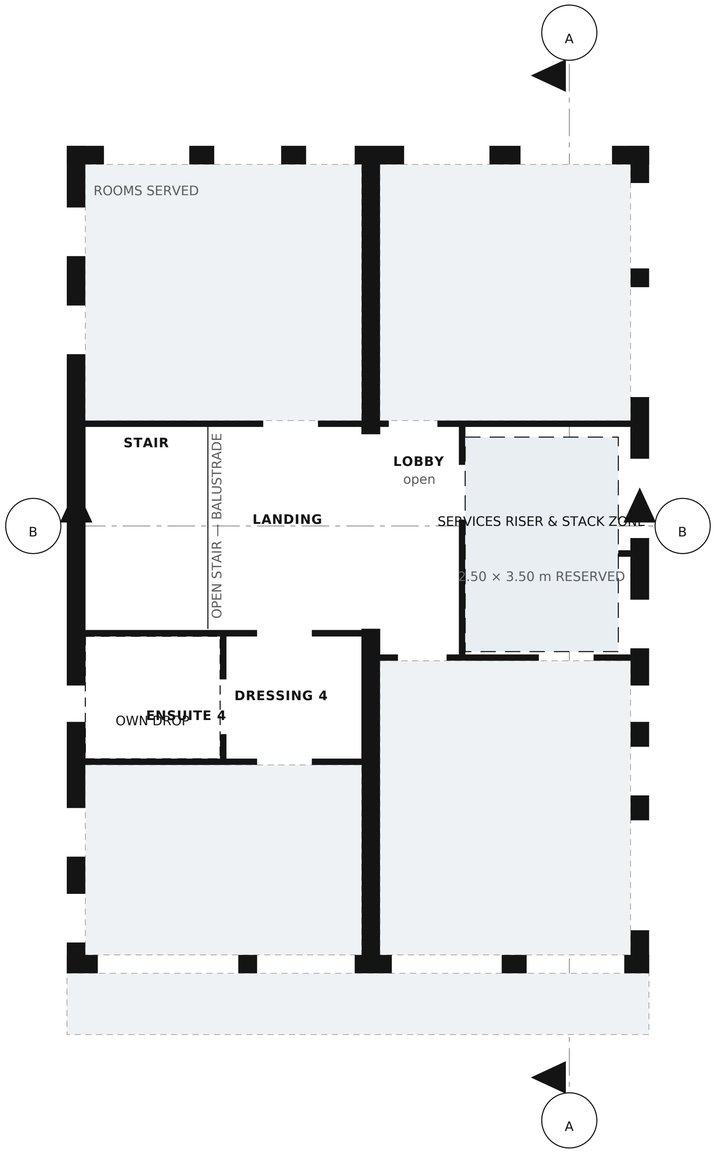
HVAC LAYOUT — FIRST FLOOR

DRAWING No	REVISION	SCALE	DATE
ME-101	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

HVAC LAYOUT — SECOND FLOOR

LEVEL +6.60 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+6.60
DISCIPLINE	Mechanical
ROOMS MARKED	5
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+9.60
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Which rooms are conditioned, where the plant stands and where the risers run. Capacities, duct sizes and unit selection are the mechanical engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-102 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

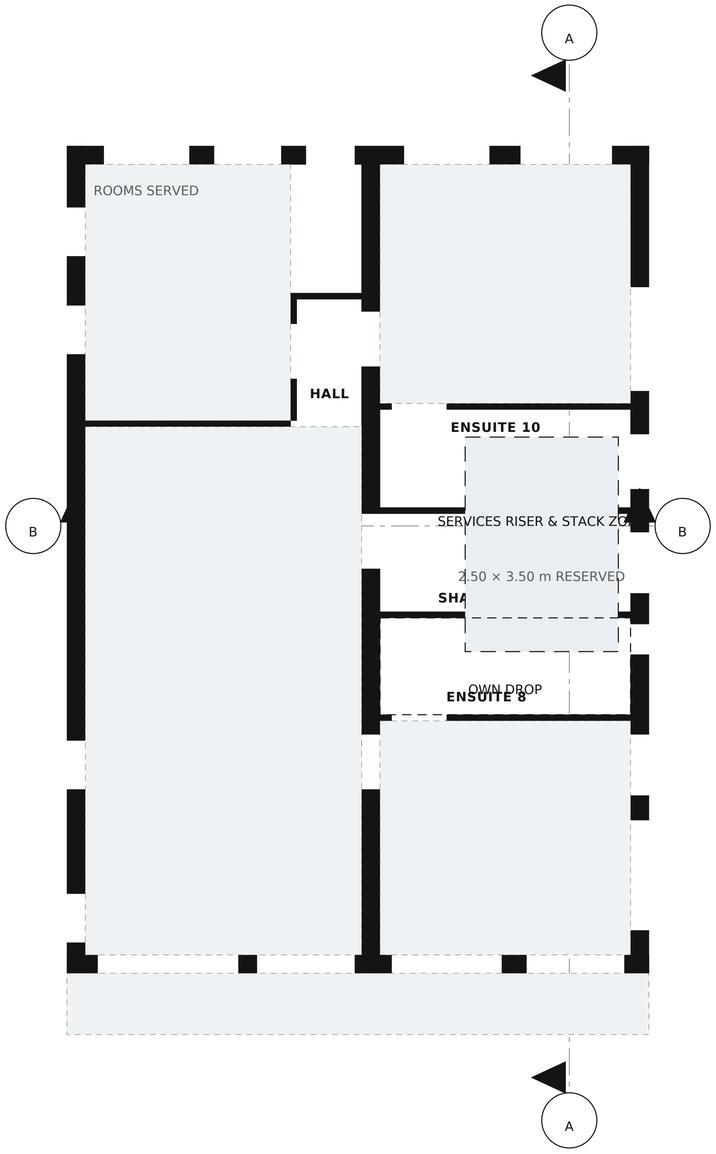
HVAC LAYOUT — SECOND FLOOR

DRAWING No	REVISION	SCALE	DATE
ME-102	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

HVAC LAYOUT — THIRD FLOOR

LEVEL +9.90 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+9.90
DISCIPLINE	Mechanical
ROOMS MARKED	5
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+12.90
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Which rooms are conditioned, where the plant stands and where the risers run. Capacities, duct sizes and unit selection are the mechanical engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-103 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

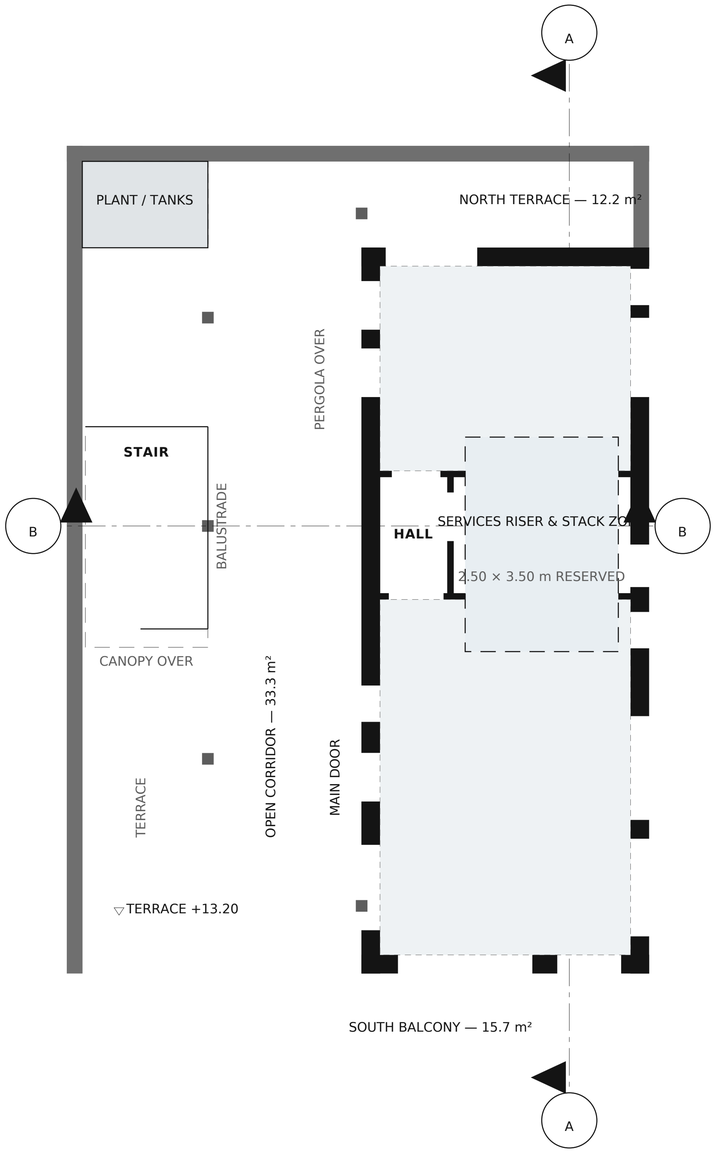
HVAC LAYOUT — THIRD FLOOR

DRAWING No	REVISION	SCALE	DATE
ME-103	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

HVAC LAYOUT — PENTHOUSE FLOOR

LEVEL +13.20 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+13.20
DISCIPLINE	Mechanical
ROOMS MARKED	2
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+16.20
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Which rooms are conditioned, where the plant stands and where the risers run. Capacities, duct sizes and unit selection are the mechanical engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-104 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

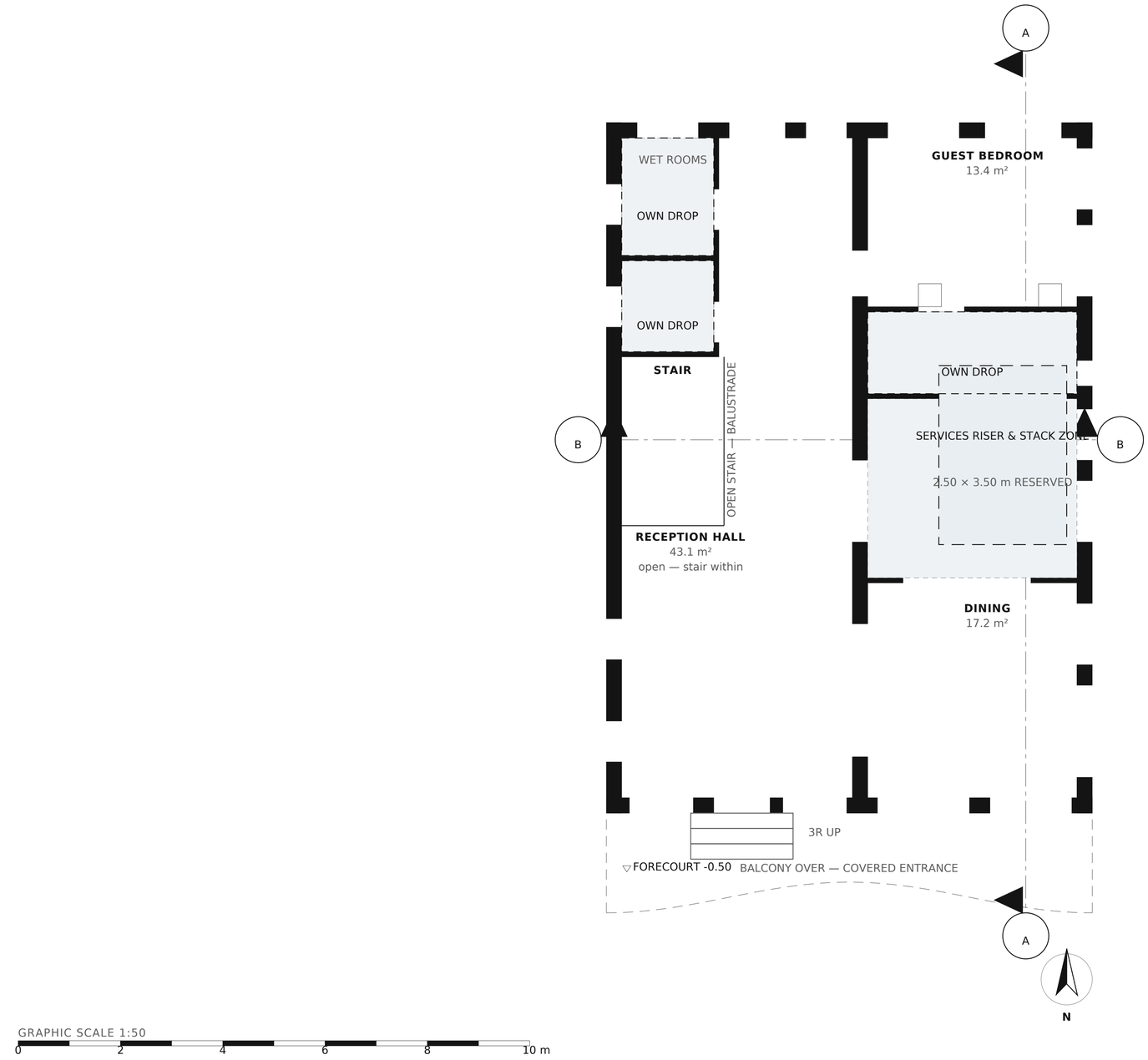
HVAC LAYOUT — PENTHOUSE FLOOR

DRAWING No	REVISION	SCALE	DATE
ME-104	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



WATER SUPPLY LAYOUT — GROUND FLOOR

LEVEL +0.00 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+0.00
DISCIPLINE	Plumbing
ROOMS MARKED	4
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	3
SOFFIT	+3.00
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Tanks over, riser down, wet rooms served. Pipe sizes, pressures and hot water generation are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-100 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

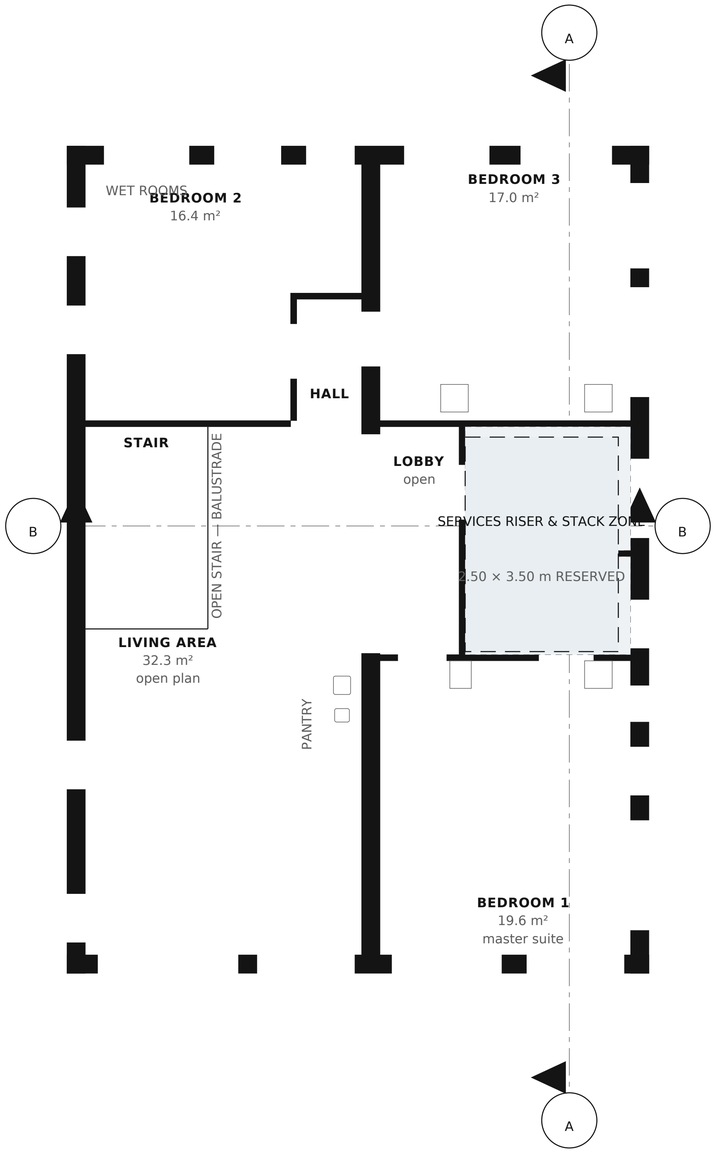
WATER SUPPLY LAYOUT — GROUND FLOOR

DRAWING No	REVISION	SCALE	DATE
PL-100	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

WATER SUPPLY LAYOUT — FIRST FLOOR

LEVEL +3.30 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+3.30
DISCIPLINE	Plumbing
ROOMS MARKED	2
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+6.30
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Tanks over, riser down, wet rooms served. Pipe sizes, pressures and hot water generation are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-101 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

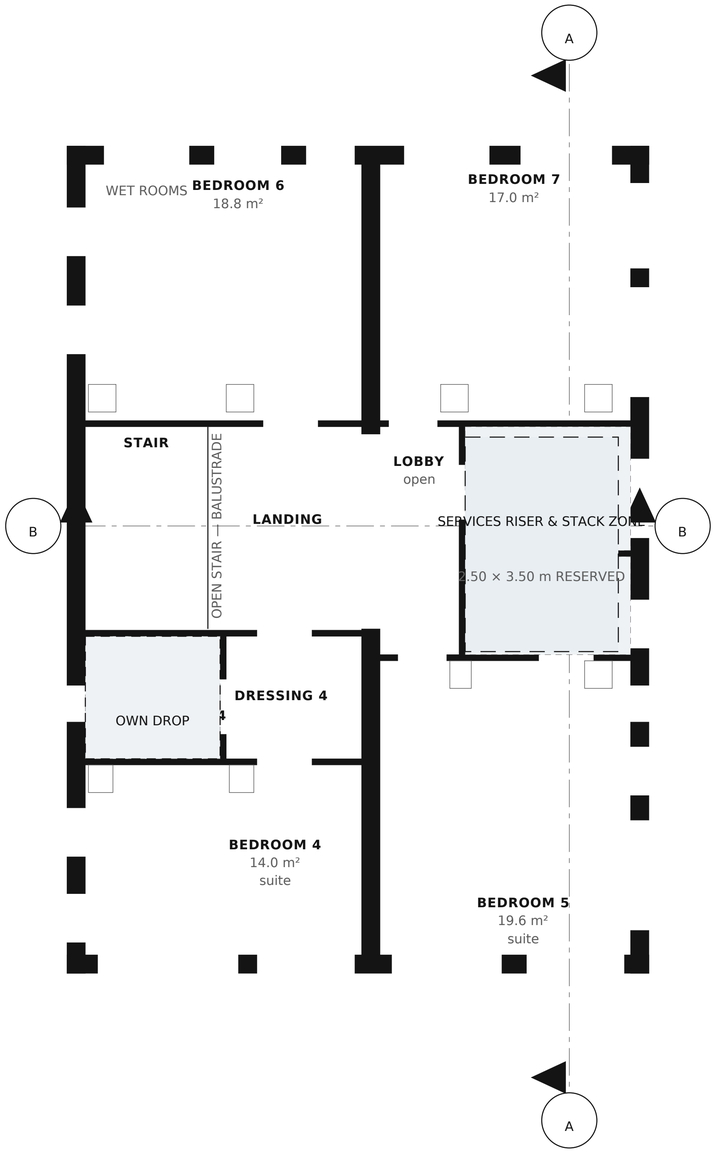
WATER SUPPLY LAYOUT — FIRST FLOOR

DRAWING No	REVISION	SCALE	DATE
PL-101	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

WATER SUPPLY LAYOUT — SECOND FLOOR

LEVEL +6.60 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+6.60
DISCIPLINE	Plumbing
ROOMS MARKED	3
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+9.60
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Tanks over, riser down, wet rooms served. Pipe sizes, pressures and hot water generation are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-102 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

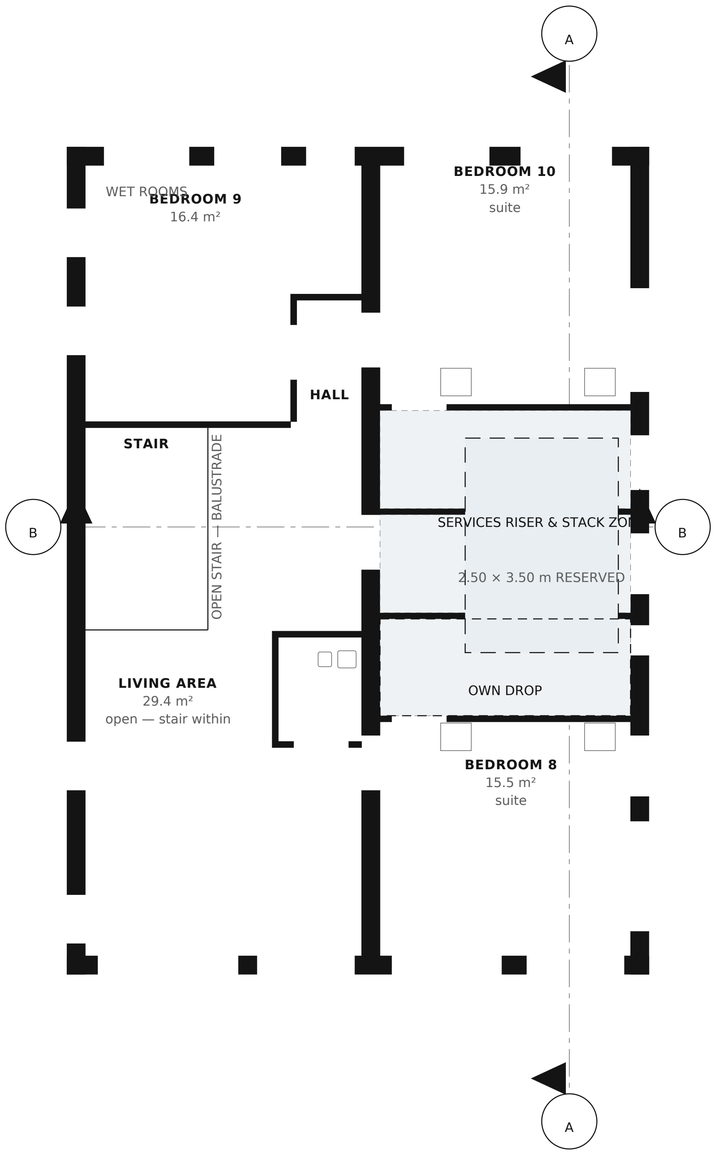
WATER SUPPLY LAYOUT — SECOND FLOOR

DRAWING No	REVISION	SCALE	DATE
PL-102	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

WATER SUPPLY LAYOUT — THIRD FLOOR

LEVEL +9.90 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+9.90
DISCIPLINE	Plumbing
ROOMS MARKED	3
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+12.90
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Tanks over, riser down, wet rooms served. Pipe sizes, pressures and hot water generation are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-103 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

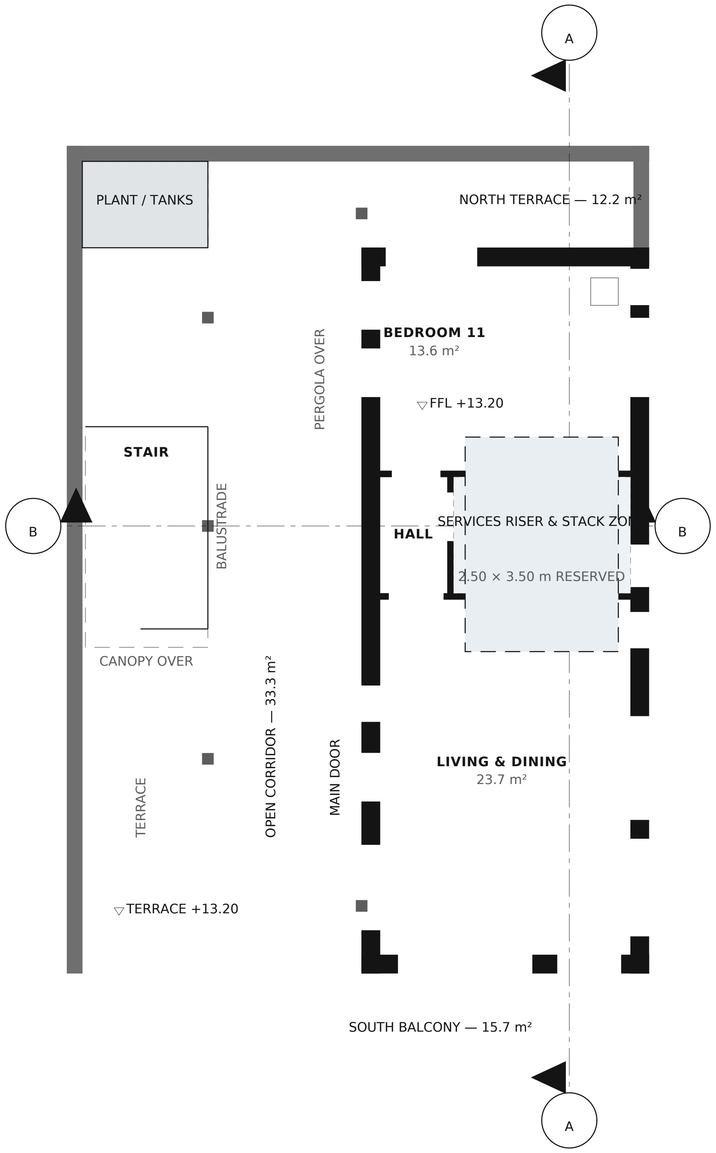
WATER SUPPLY LAYOUT — THIRD FLOOR

DRAWING No	REVISION	SCALE	DATE
PL-103	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted





GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

WATER SUPPLY LAYOUT — PENTHOUSE FLOOR

LEVEL +13.20 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+13.20
DISCIPLINE	Plumbing
ROOMS MARKED	1
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+16.20
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Tanks over, riser down, wet rooms served. Pipe sizes, pressures and hot water generation are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-104 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

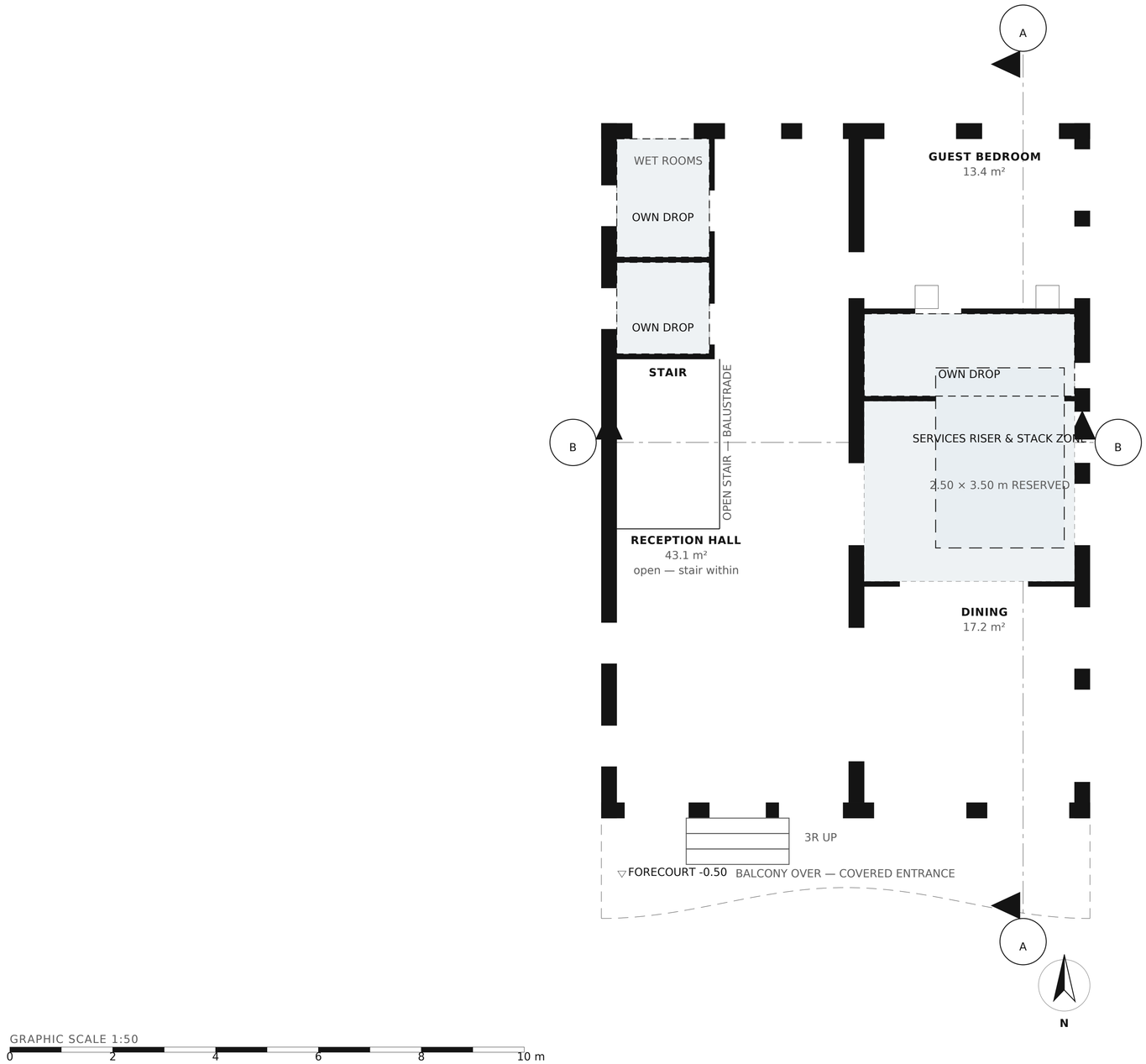
WATER SUPPLY LAYOUT — PENTHOUSE FLOOR

DRAWING No	REVISION	SCALE	DATE
PL-104	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



DRAINAGE LAYOUT — GROUND FLOOR

LEVEL +0.00 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+0.00
DISCIPLINE	Plumbing
ROOMS MARKED	4
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	3
SOFFIT	+3.00
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Where the stacks drop, measured from where the wet rooms actually stack. Falls, pipe sizes, venting and the below-ground drainage are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-100 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

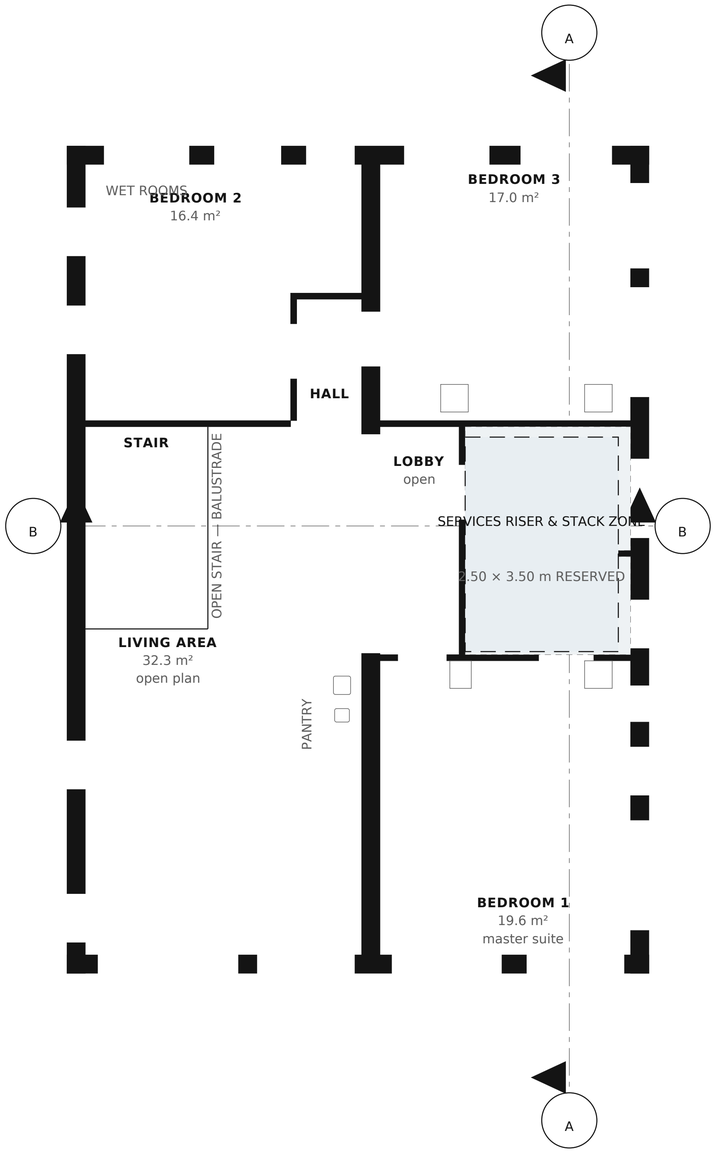
DRAINAGE LAYOUT — GROUND FLOOR

DRAWING No	REVISION	SCALE	DATE
PL-200	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

DRAINAGE LAYOUT — FIRST FLOOR

LEVEL +3.30 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+3.30
DISCIPLINE	Plumbing
ROOMS MARKED	2
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+6.30
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Where the stacks drop, measured from where the wet rooms actually stack. Falls, pipe sizes, venting and the below-ground drainage are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-101 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

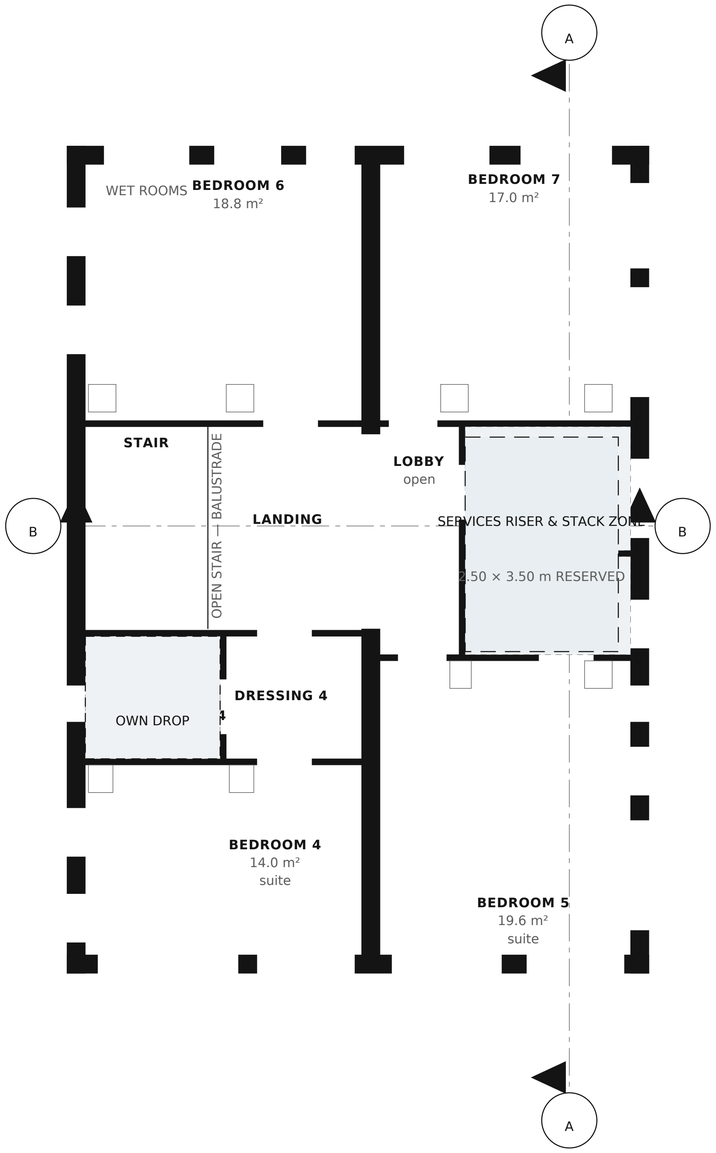
DRAINAGE LAYOUT — FIRST FLOOR

DRAWING No	REVISION	SCALE	DATE
PL-201	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

DRAINAGE LAYOUT — SECOND FLOOR

LEVEL +6.60 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+6.60
DISCIPLINE	Plumbing
ROOMS MARKED	3
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+9.60
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Where the stacks drop, measured from where the wet rooms actually stack. Falls, pipe sizes, venting and the below-ground drainage are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-102 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

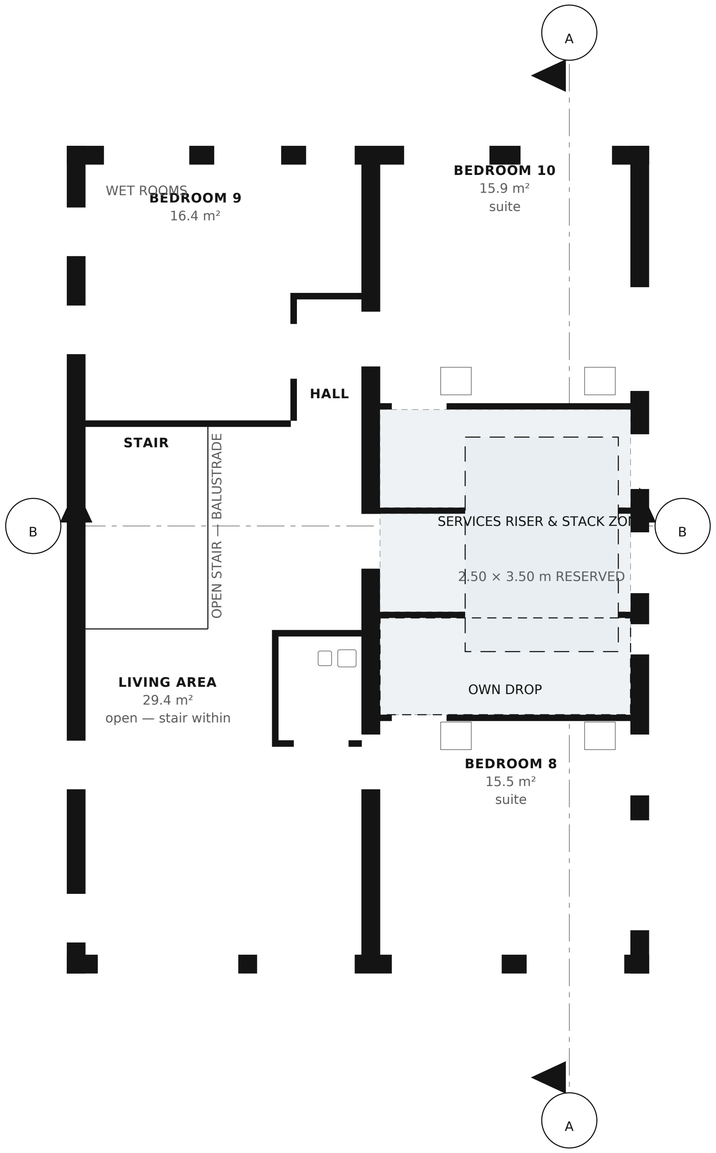
DRAINAGE LAYOUT — SECOND FLOOR

DRAWING No	REVISION	SCALE	DATE
PL-202	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

DRAINAGE LAYOUT — THIRD FLOOR

LEVEL +9.90 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+9.90
DISCIPLINE	Plumbing
ROOMS MARKED	3
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	1
SOFFIT	+12.90
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Where the stacks drop, measured from where the wet rooms actually stack. Falls, pipe sizes, venting and the below-ground drainage are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-103 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

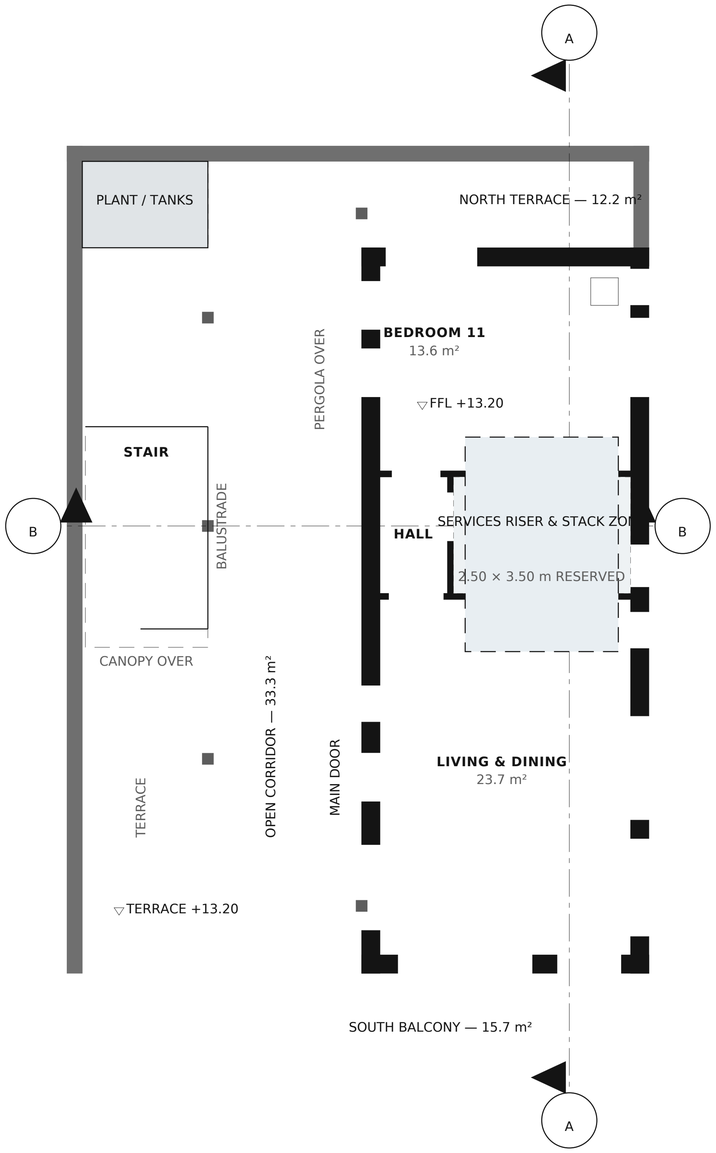
DRAINAGE LAYOUT — THIRD FLOOR

DRAWING No	REVISION	SCALE	DATE
PL-203	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



GRAPHIC SCALE 1:50

0 2 4 6 8 10 m

N

DRAINAGE LAYOUT — PENTHOUSE FLOOR

LEVEL +13.20 · 1:50 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

COORDINATION

LEVEL	+13.20
DISCIPLINE	Plumbing
ROOMS MARKED	1
RISER RESERVED	2.50 × 3.50 m
ISOLATED WET ROOMS	0
SOFFIT	+16.20
STRUCTURAL ZONE	300 mm
SIZES AND LOADS	by the engineer

GENERAL NOTES

1. Where the stacks drop, measured from where the wet rooms actually stack. Falls, pipe sizes, venting and the below-ground drainage are the engineer's.

2. Coordination only. No fitting, outlet, diffuser, pipe or cable is drawn, and no size, load or capacity appears. Those are the services engineer's design; a layout invented here would be thrown away, and in the meantime somebody might price it.

3. What this sheet does fix is space: the riser reserved through the building, the rooms that have to be served, and the plant position. Those are expensive to change later, which is why the architecture states them now.

4. The plan is run from the floor plan with its fittings switched off, so it cannot drift from A-104 when a partition moves.

5. Builder's work — holes, chases, upstands and bases — is to be agreed against this layout before any slab is poured.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

DRAINAGE LAYOUT — PENTHOUSE FLOOR

DRAWING No	REVISION	SCALE	DATE
PL-204	**P21**	**1:50 @ A1**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

ROOF

PENTHOUSE

THIRD

SECOND

FIRST

GROUND

ENGINEER

ELECTRICAL SINGLE LINE DIAGRAM

LEVEL the whole building · schematic · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE RISER

LEVELS SERVED	5
RISER RESERVED	2.50 × 3.50 m
WET ON	5 levels
PLANT	roof, north-west
HEIGHT SERVED	16.45 m
SIZES	by the engineer
CAPACITIES	by the engineer
BELOW GROUND	by the engineer

GENERAL NOTES

1. One diagram for the building, not one per level.

2. A diagram, not a drawing to scale, and it fixes one thing: the riser runs the full height of the building in the east strip, where the wet rooms were measured to stack on every level.

3. No size, capacity, rating or load appears. The engineer's design fills them in; this reserves the space it has to fit through.

4. The connection below ground waits on GEO-001 and the drainage design, and on whatever the authority requires at the boundary.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

ELECTRICAL SINGLE LINE DIAGRAM

DRAWING No	REVISION	SCALE	DATE
EL-400	**P21**	**schematic**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

ROOF

PENTHOUSE

THIRD

SECOND

FIRST

GROUND

ENGINEER

HVAC SCHEMATIC & PLANT ARRANGEMENT

LEVEL the whole building · schematic · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE RISER

LEVELS SERVED	5
RISER RESERVED	2.50 × 3.50 m
WET ON	5 levels
PLANT	roof, north-west
HEIGHT SERVED	16.45 m
SIZES	by the engineer
CAPACITIES	by the engineer
BELOW GROUND	by the engineer

GENERAL NOTES

1. Plant on the roof, risers down, which is all the architecture fixes.

2. A diagram, not a drawing to scale, and it fixes one thing: the riser runs the full height of the building in the east strip, where the wet rooms were measured to stack on every level.

3. No size, capacity, rating or load appears. The engineer's design fills them in; this reserves the space it has to fit through.

4. The connection below ground waits on GEO-001 and the drainage design, and on whatever the authority requires at the boundary.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

HVAC SCHEMATIC & PLANT ARRANGEMENT

DRAWING No	REVISION	SCALE	DATE
ME-200	**P21**	**schematic**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted

ROOF

PENTHOUSE

THIRD

SECOND

FIRST

GROUND

ENGINEER

DRAINAGE & WATER RISER DIAGRAM

LEVEL the whole building · schematic · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THE RISER

LEVELS SERVED	5
RISER RESERVED	2.50 × 3.50 m
WET ON	5 levels
PLANT	roof, north-west
HEIGHT SERVED	16.45 m
SIZES	by the engineer
CAPACITIES	by the engineer
BELOW GROUND	by the engineer

GENERAL NOTES

1. The stack, level by level, from the tanks to the ground.

2. A diagram, not a drawing to scale, and it fixes one thing: the riser runs the full height of the building in the east strip, where the wet rooms were measured to stack on every level.

3. No size, capacity, rating or load appears. The engineer's design fills them in; this reserves the space it has to fit through.

4. The connection below ground waits on GEO-001 and the drainage design, and on whatever the authority requires at the boundary.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

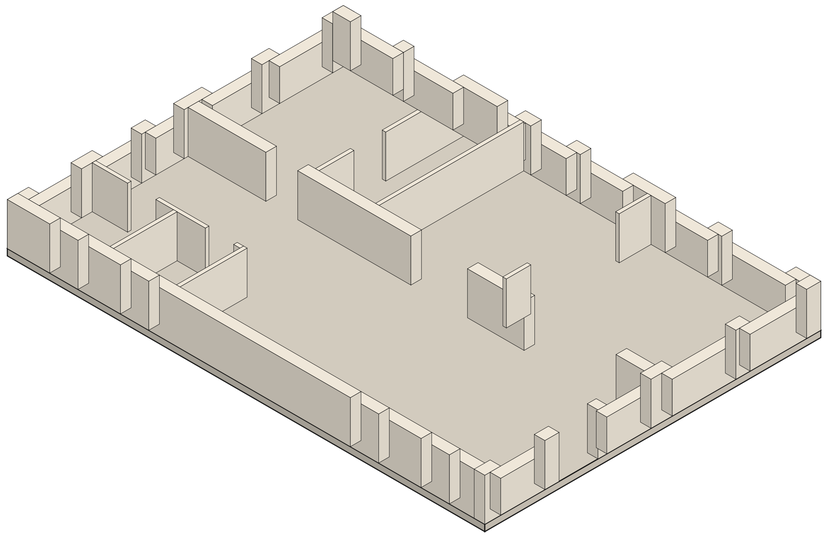
DRAINAGE & WATER RISER DIAGRAM

DRAWING No	REVISION	SCALE	DATE
PL-300	**P21**	**schematic**	**2026-10-07**

Coordination layout for the services engineer's design — no fitting, no size, no load. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



INTERIOR CUTAWAY — GROUND FLOOR

LEVEL +0.00 · 1:75 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THIS FLOOR

LEVEL	+0.00
ROOMS	8
NET INTERNAL	108.3 m²
WALLS CUT AT	1.20 m above the floor
VIEW	isometric from the south-west
SCHEME	A — Wave
BALCONY	—
DRAWN FROM	A-100

GENERAL NOTES

1. The ground floor as a doll's house, its walls cut 1.20 m above the floor and seen from the south-west, which is the corner A-430 uses.

2. Not a render, and not redrawn. The walls come back from the same probe that validates the plans, so this is A-100 in three dimensions and cannot be a different building.

3. A window leaves the wall under it standing to its sill; a door leaves a gap to the floor. Both come from the opening heights scheduled on ARC-500.

4. Faces are painted back to front rather than depth-sorted, so a detail may overlap where two surfaces nearly touch. The plan governs.

5. For looking at. Nothing is dimensioned here and nothing should be measured off it.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

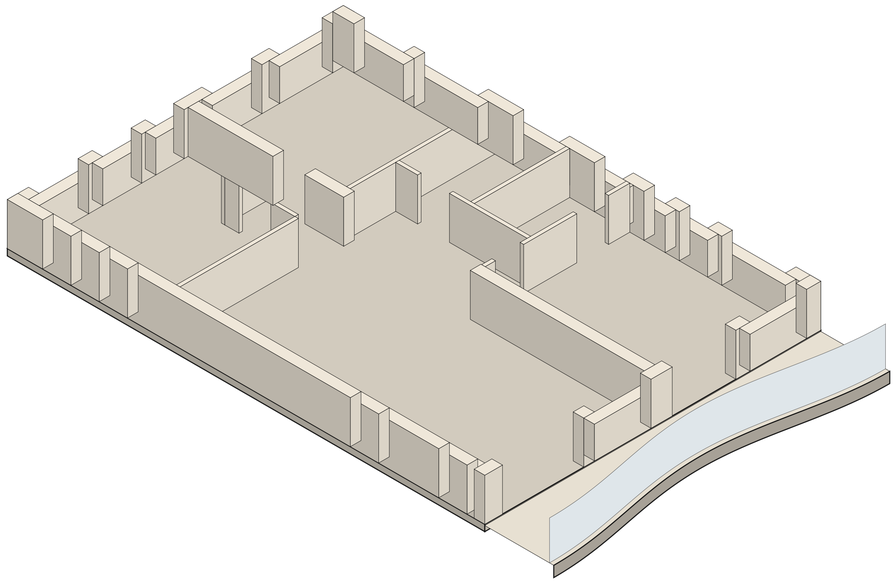
DRAWING

INTERIOR CUTAWAY — GROUND FLOOR

DRAWING No	REVISION	SCALE	DATE
A-02	**P21**	**1:75 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS
P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



INTERIOR CUTAWAY — FIRST FLOOR

LEVEL +3.30 · 1:75 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THIS FLOOR

LEVEL	+3.30
ROOMS	9
NET INTERNAL	108.3 m²
WALLS CUT AT	1.20 m above the floor
VIEW	isometric from the south-west
SCHEME	A — Wave
BALCONY	15.7 m²
DRAWN FROM	A-101

GENERAL NOTES

1. The first floor as a doll's house, its walls cut 1.20 m above the floor and seen from the south-west, which is the corner A-430 uses.

2. Not a render, and not redrawn. The walls come back from the same probe that validates the plans, so this is A-101 in three dimensions and cannot be a different building.

3. A window leaves the wall under it standing to its sill; a door leaves a gap to the floor. Both come from the opening heights scheduled on ARC-500.

4. Faces are painted back to front rather than depth-sorted, so a detail may overlap where two surfaces nearly touch. The plan governs.

5. For looking at. Nothing is dimensioned here and nothing should be measured off it.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

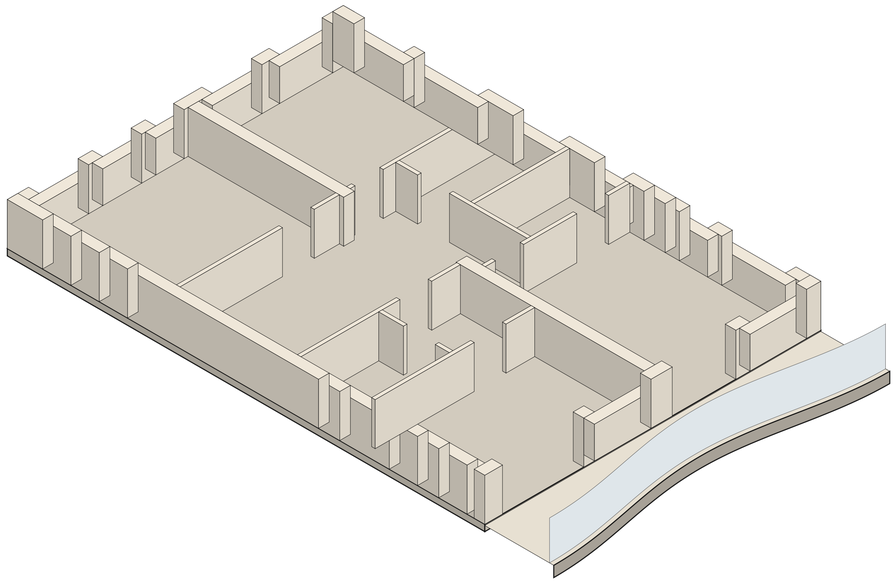
INTERIOR CUTAWAY — FIRST FLOOR

DRAWING No	REVISION	SCALE	DATE
A-03	**P21**	**1:75 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



INTERIOR CUTAWAY — SECOND FLOOR

LEVEL +6.60 · 1:75 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THIS FLOOR

LEVEL	+6.60
ROOMS	11
NET INTERNAL	107.0 m²
WALLS CUT AT	1.20 m above the floor
VIEW	isometric from the south-west
SCHEME	A — Wave
BALCONY	15.7 m²
DRAWN FROM	A-102

GENERAL NOTES

1. The second floor as a doll's house, its walls cut 1.20 m above the floor and seen from the south-west, which is the corner A-430 uses.

2. Not a render, and not redrawn. The walls come back from the same probe that validates the plans, so this is A-102 in three dimensions and cannot be a different building.

3. A window leaves the wall under it standing to its sill; a door leaves a gap to the floor. Both come from the opening heights scheduled on ARC-500.

4. Faces are painted back to front rather than depth-sorted, so a detail may overlap where two surfaces nearly touch. The plan governs.

5. For looking at. Nothing is dimensioned here and nothing should be measured off it.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

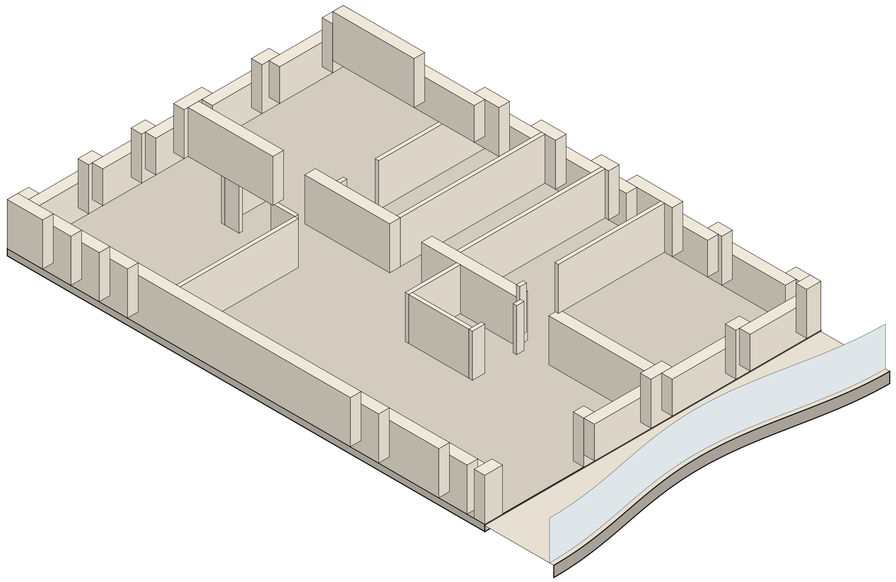
INTERIOR CUTAWAY — SECOND FLOOR

DRAWING No	REVISION	SCALE	DATE
A-04	**P21**	**1:75 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



INTERIOR CUTAWAY — THIRD FLOOR

LEVEL +9.90 · 1:75 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 — Finished floor level datum

THIS FLOOR

LEVEL	+9.90
ROOMS	9
NET INTERNAL	105.3 m²
WALLS CUT AT	1.20 m above the floor
VIEW	isometric from the south-west
SCHEME	A — Wave
BALCONY	15.7 m²
DRAWN FROM	A-103

GENERAL NOTES

1. The third floor as a doll's house, its walls cut 1.20 m above the floor and seen from the south-west, which is the corner A-430 uses.

2. Not a render, and not redrawn. The walls come back from the same probe that validates the plans, so this is A-103 in three dimensions and cannot be a different building.

3. A window leaves the wall under it standing to its sill; a door leaves a gap to the floor. Both come from the opening heights scheduled on ARC-500.

4. Faces are painted back to front rather than depth-sorted, so a detail may overlap where two surfaces nearly touch. The plan governs.

5. For looking at. Nothing is dimensioned here and nothing should be measured off it.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

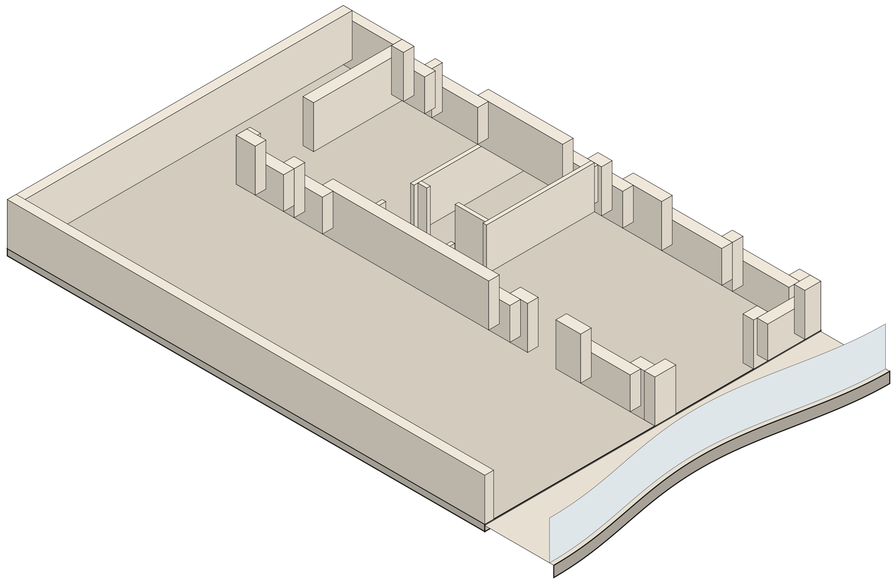
INTERIOR CUTAWAY — THIRD FLOOR

DRAWING No	REVISION	SCALE	DATE
A-05	**P21**	**1:75 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted



INTERIOR CUTAWAY — PENTHOUSE FLOOR

LEVEL +13.20 · 1:75 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- ▽ +0.00 Finished floor level datum

THIS FLOOR

LEVEL	+13.20
ROOMS	4
NET INTERNAL	44.8 m²
WALLS CUT AT	1.20 m above the floor
VIEW	isometric from the south-west
SCHEME	A — Wave
BALCONY	15.7 m²
DRAWN FROM	A-104

GENERAL NOTES

1. The penthouse floor as a doll's house, its walls cut 1.20 m above the floor and seen from the south-west, which is the corner A-430 uses.

2. Not a render, and not redrawn. The walls come back from the same probe that validates the plans, so this is A-104 in three dimensions and cannot be a different building.

3. A window leaves the wall under it standing to its sill; a door leaves a gap to the floor. Both come from the opening heights scheduled on ARC-500.

4. Faces are painted back to front rather than depth-sorted, so a detail may overlap where two surfaces nearly touch. The plan governs.

5. For looking at. Nothing is dimensioned here and nothing should be measured off it.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

INTERIOR CUTAWAY — PENTHOUSE FLOOR

DRAWING No	REVISION	SCALE	DATE
A-06	**P21**	**1:75 @ A1**	**2026-10-07**

Preliminary design issue — not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted